第一章　习题讲解

1-1 两球自重为G_1和G_2, 以绳悬挂如图1-33试画；①小球②大球③两球合在一起的受力图。

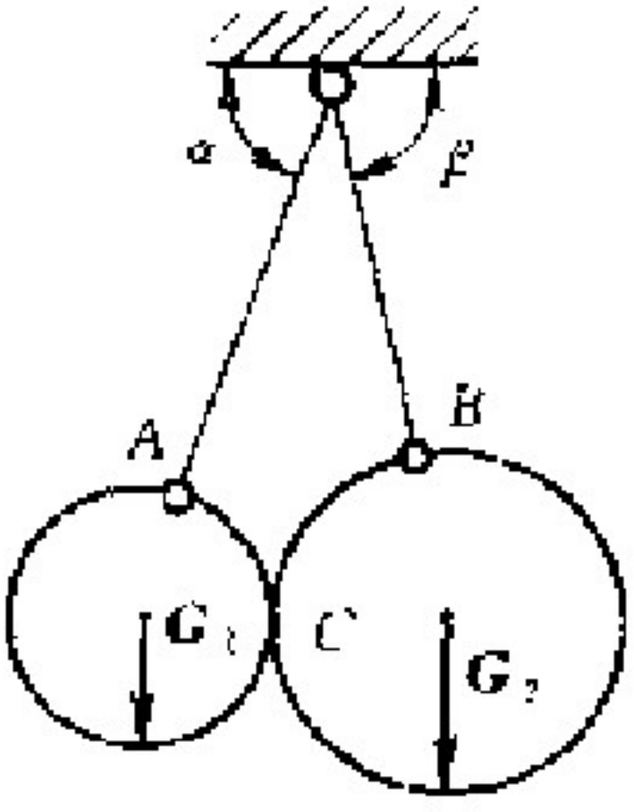
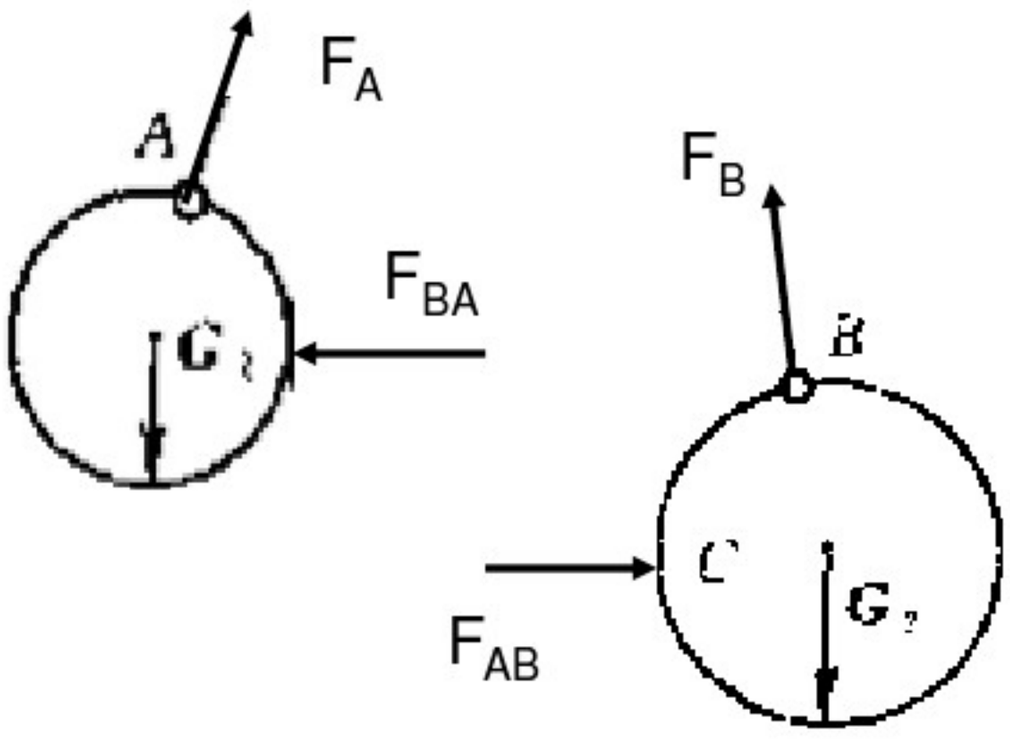
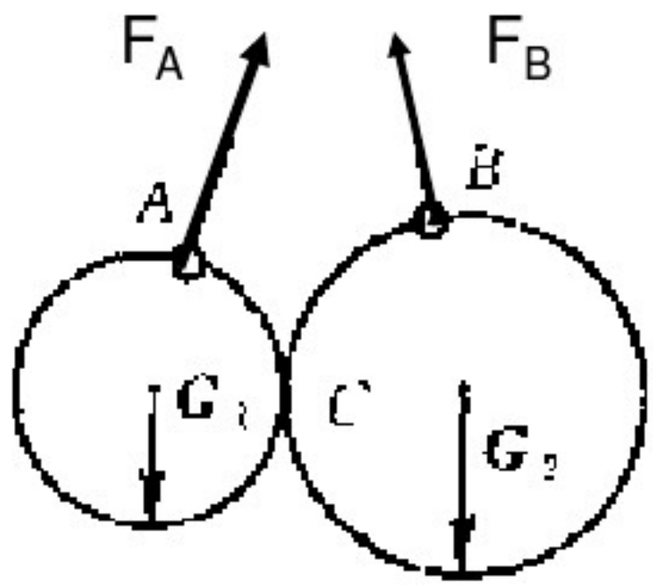
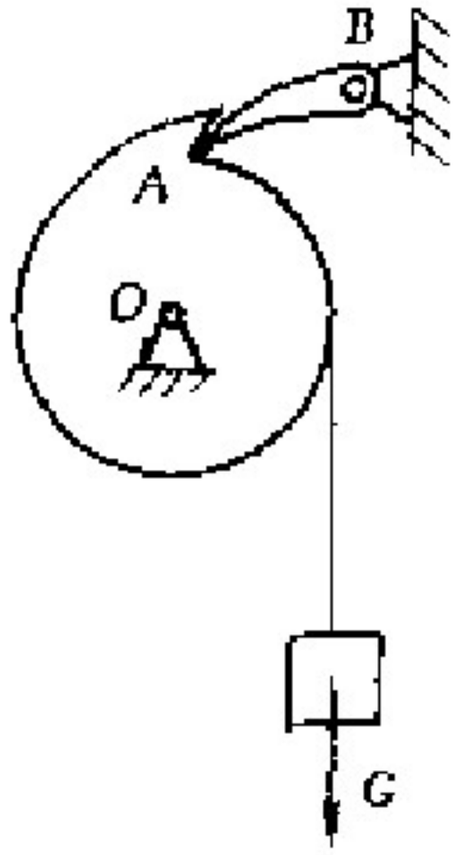


图 1-33　题 1-1 图




F_A
F_B
A
B
G_1
C
G_2

1-4 棘轮装置如图1-36所示。通过绳子悬挂重量为G的物体，AB为棘轮的止推爪，B处为平面铰链。试画出棘轮的受力图。



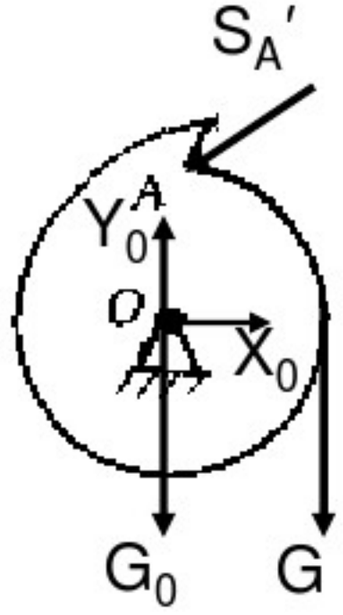
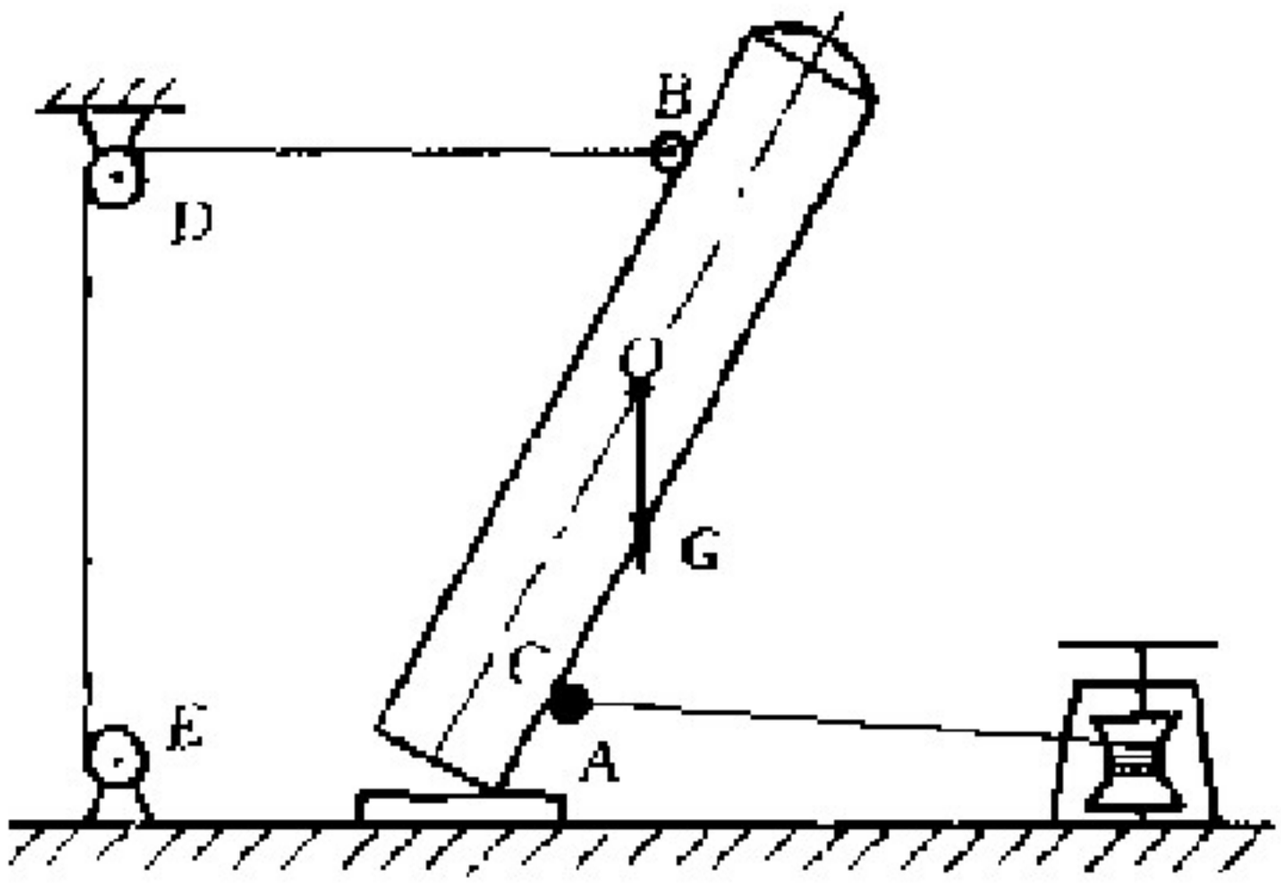


图 1-36　题 1-4 图

1-5 塔器竖起的过程如图1-37所示。下断搁在基础上，在C处系以钢绳并用绞盘拉住，上端在B处系以钢绳通过定滑轮D连接到卷扬机E。设塔重为G，试画出塔器的受力图。



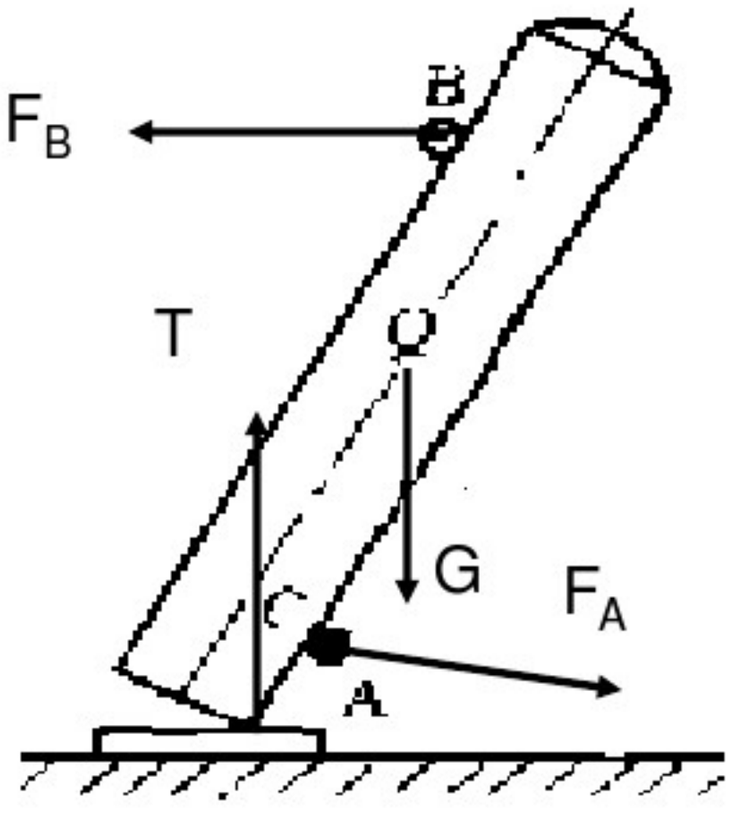


图 1-37　题 1-5 图

1-6 化工厂中起吊反应器时为了不致破坏栏杆，施加一水平力p，使反应器与栏杆相离开（图1-38）。已知此时牵引绳与铅垂线的夹角为30℃，反应器重量为30KN，试求水平力p的大小和绳子的拉力T。

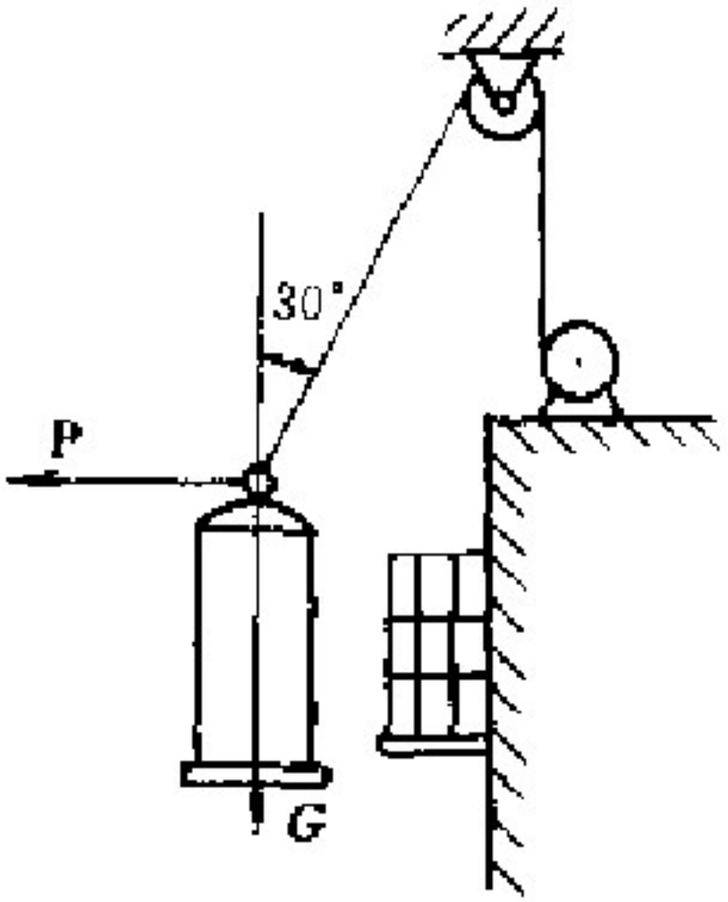
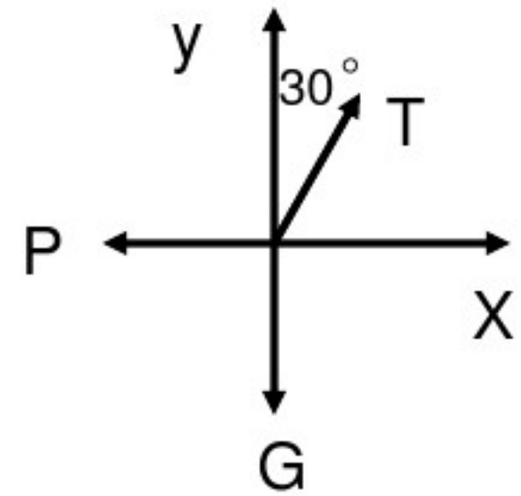


图 1-38　题 1-6 图

解：取反应器为研究对象，画受力图如图



$$\sum Y = 0, T\cos 30° = G, T = 20\sqrt{3}KN$$

$$\sum X = 0, T\sin 30° = P, P = 10\sqrt{3}KN$$

1-8 用三轴钻床在水平工件上钻孔时，每个钻头对工件施加一个力偶（图1-40）。已知三个力偶的矩分别为：m_1=1kN·m，m_2= 1.4kN·m，m_3=2kN·m，固定工件的两螺栓A和B与工件成光滑面接触，两螺栓的距离L=0.2m，求两螺栓受到的横向力。

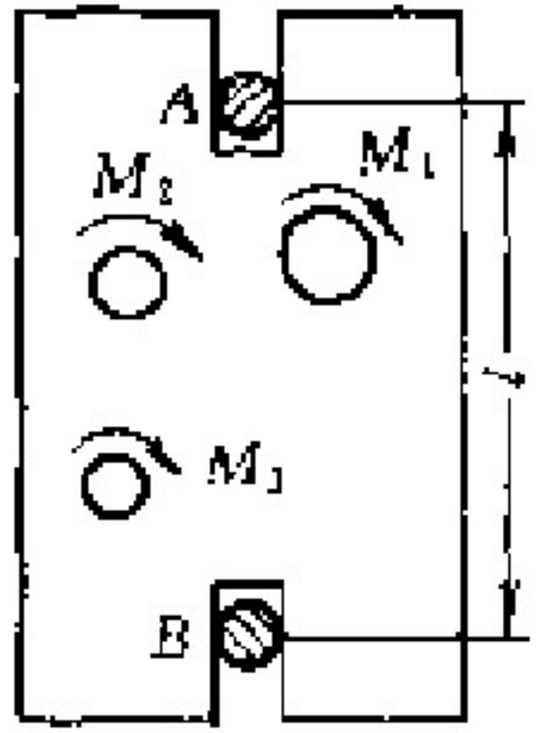


图 1-40　题 1-8 图

解 设两螺栓受到的横向力为R，
由合力矩原理

$$m_0(R) = \sum m_0(F)$$

$$m_1 + m_2 + m_3 = Rl$$

代入数值得 R=22kN

1-9 塔器的加热釜以侧塔的形式悬挂在主塔上，侧塔在A处搁在主塔的托架上，并用螺栓垂直固定；在B处顶在主塔的水平支杆上，并用水平螺栓做定位连接（图1-41）。已知侧塔G=20KN,尺寸如图。试求支座A、B对侧塔的约束反力。

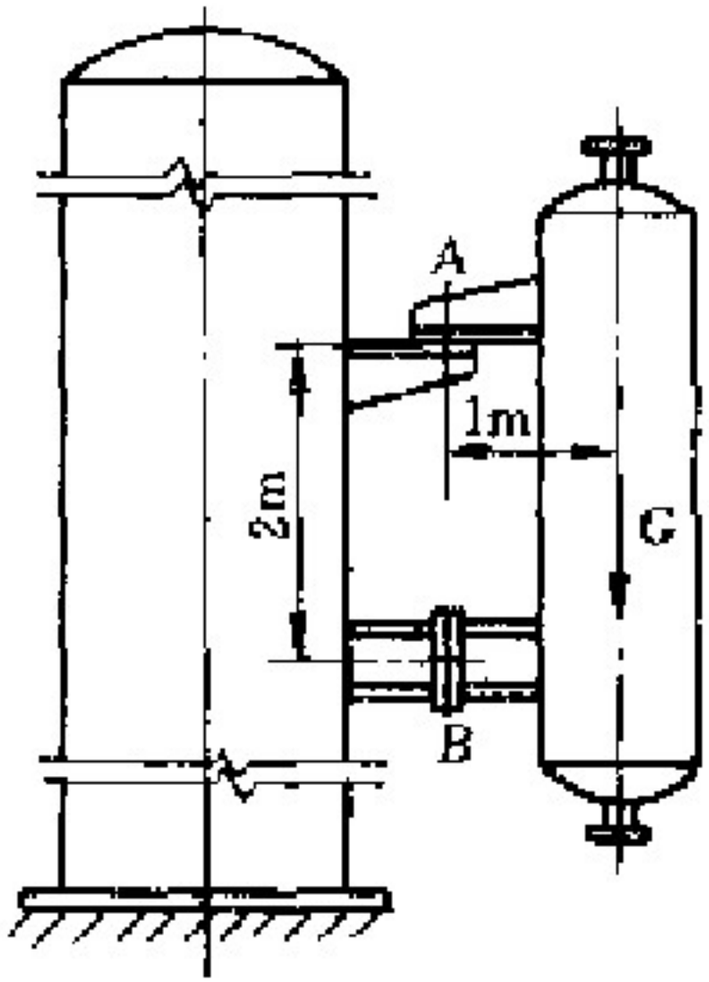
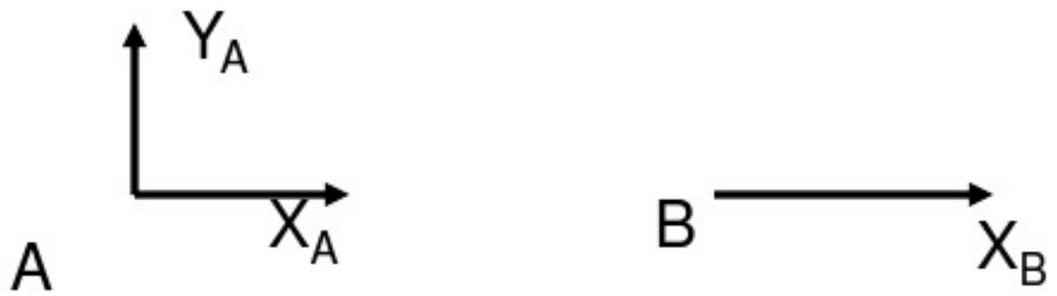


图 1-41　题 1-9 图

解 以侧塔为研究对象，由平衡方程

A、B点受力如图



$\sum M_A = 0$　$2X_B - G \times 1 = 0$，　$X_B = 10kN$

$\sum X = 0$　$X_A + X_B = 0$，　$X_A = -10kN$

$\sum Y = 0$　$Y_A - G = 0$，　$Y_A = 20kN$

1-10 如图1-42所示，有一管道支架ABC。A、B、C处均为理想的圆柱形铰链约束。已知该支架承受的两管道的重量均为G=4.5KN，图中尺寸均为mm。试求管架中梁AB和杆BC所受的力。

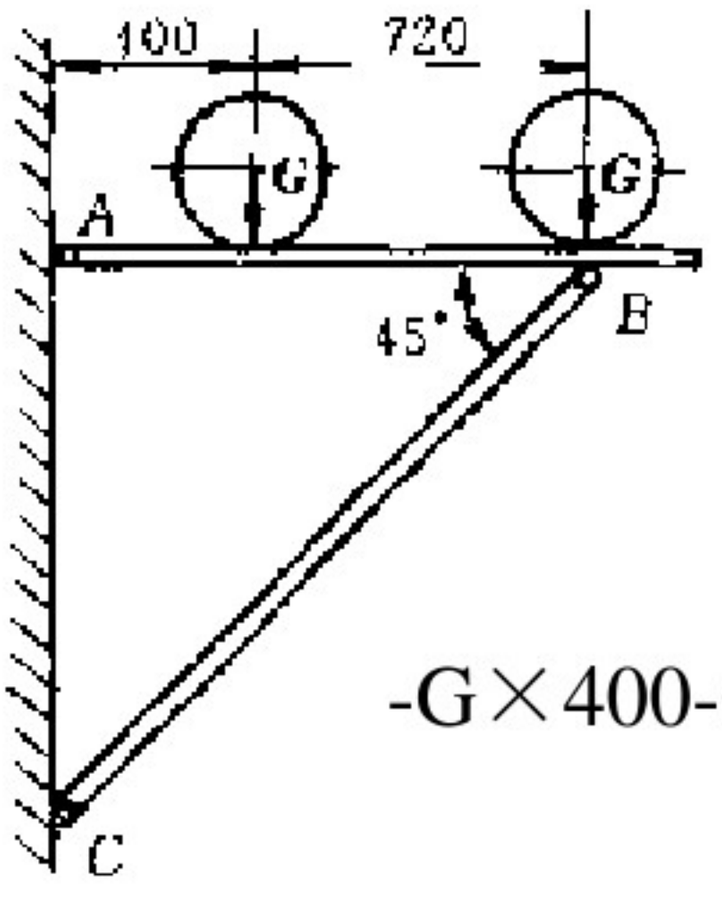
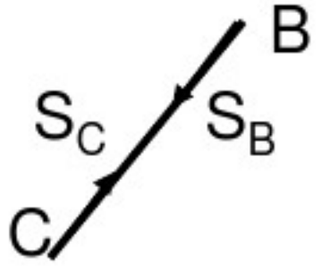


图 1-42　题 1-10 图

解 取支架BC为研究对象

画受力图如图

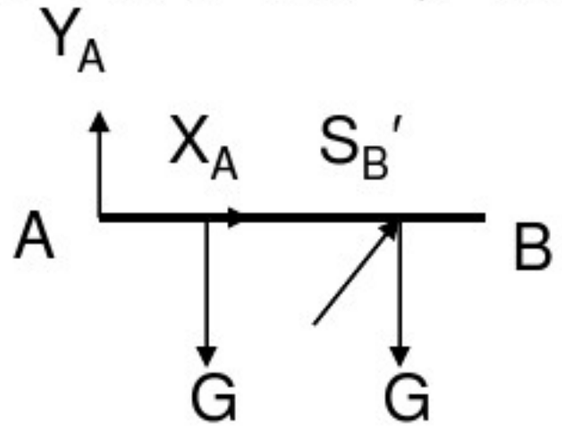


$$\sum M_A = 0$$

$$-G\times 400-G\times (400+720)+S_B\times (400+720)\sin 45^\circ = 0$$

$$S_B = 8.64\text{kN}$$

取支架AB为研究对象 ,画受力图如图（2）



$$\sum X=0,\ X_A+S_B\times\cos45^\circ=0,\quad X_A=-6.11\text{kN}$$

$$\sum Y=0,\ Y_A+S_B'\times\sin45^\circ-2G=0,\quad Y_A=2.89\text{kN}$$

1-13如图1-45所示结构，B、E、C处均为铰接。已知P=1KN，试求的A处反力以及杆EF和杆CG所受的力。

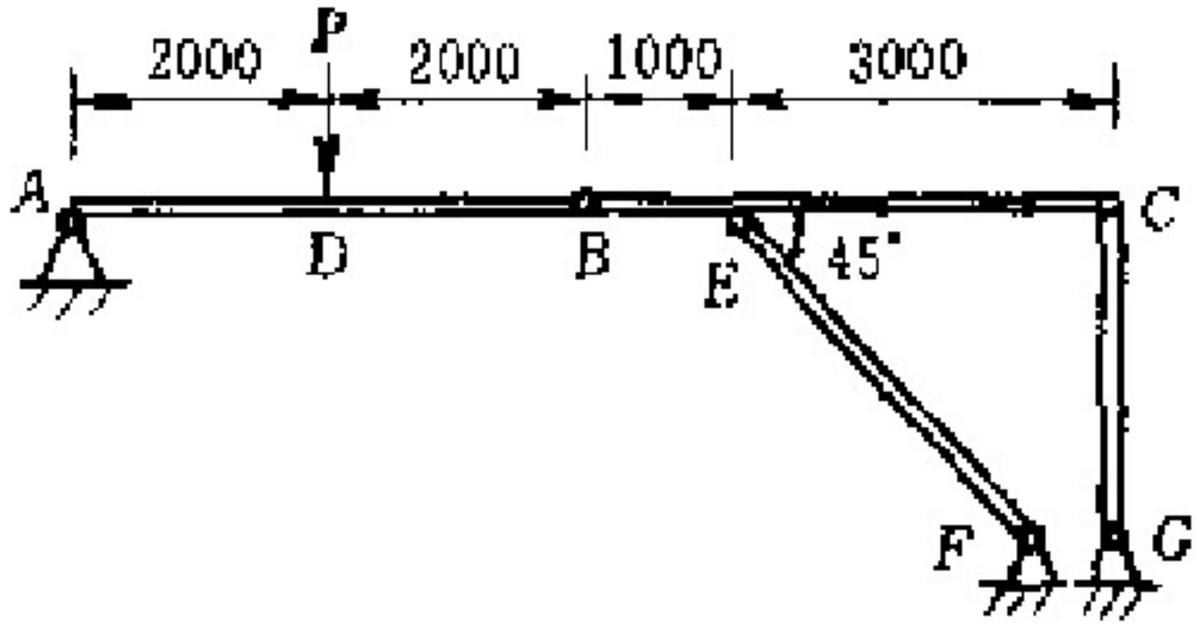
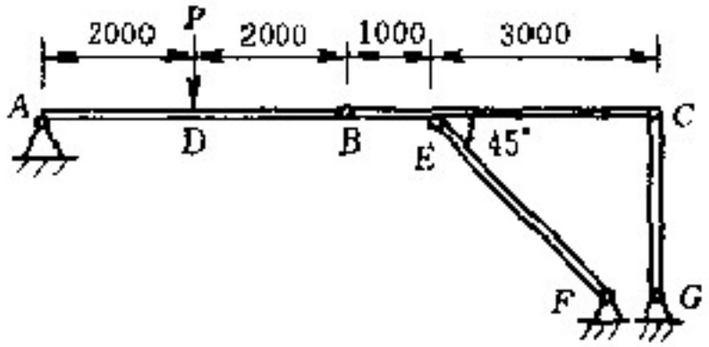
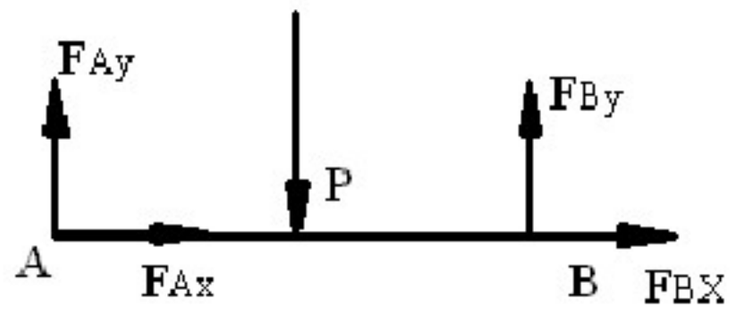


图 1-45　题 1-13 图



解：取AB为研究对象，受力如图



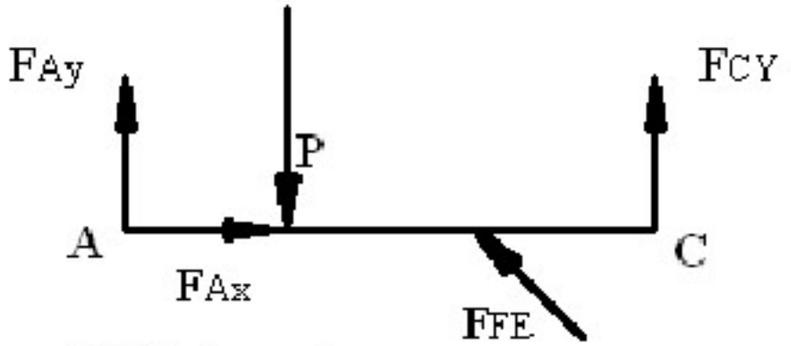
$\sum Y = 0$ $F_{AY}+F_{BY}=P$

$\sum X = 0$ $F_{AX}+F_{BX}=0$

$\sum M_A = 0$ $-P\cdot 2000+F_{BY}\cdot 4000=0$

联立得 $F_{AY}=F_{BY}=1/2P=1/2KN$

取AC为研究对象，受力如图



$$\sum M_C = 0$$

$$P\cdot 6000 - F_{AY}\cdot 8000 - F_{FE}\cdot 3000\cdot \sin 45^\circ = 0$$

$$F_{FE}= \frac{2\sqrt{2}}{3} KN$$

$$\sum Y = 0$$

$$F_{AY}+F_{CY}+F_{FE}\sin 45^\circ - P=0$$

$$F_{CY}= \frac{1}{6} KN$$

$$\sum X = 0$$

$$F_{AX} - F_{FE}\sin 45^\circ = 0$$

$$F_{AX} = \frac{2}{3} KN$$

1-14 求图1-46所示桁架中各杆所受的力

解：以节点A为研究对象，受力如图

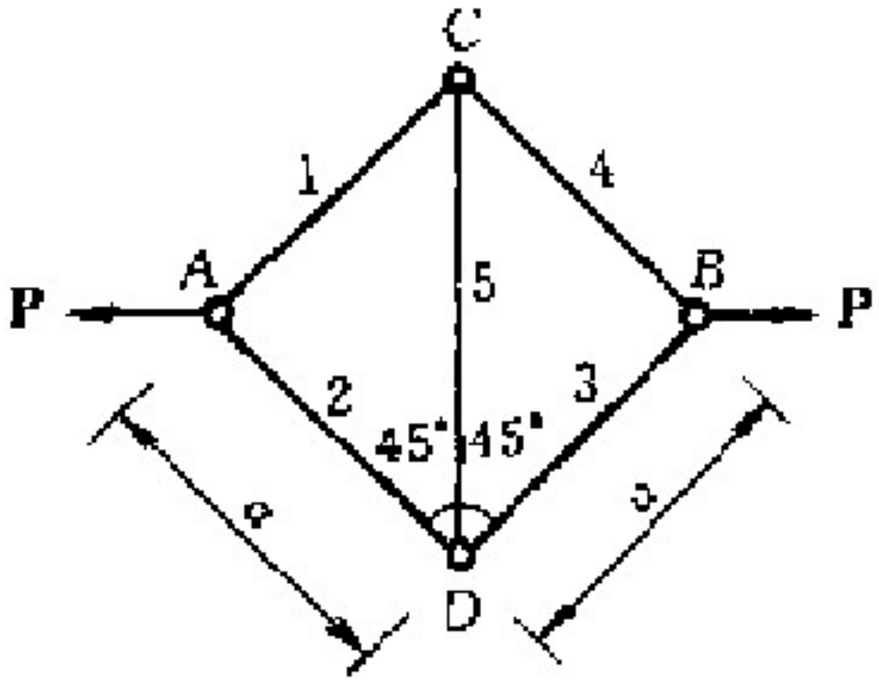
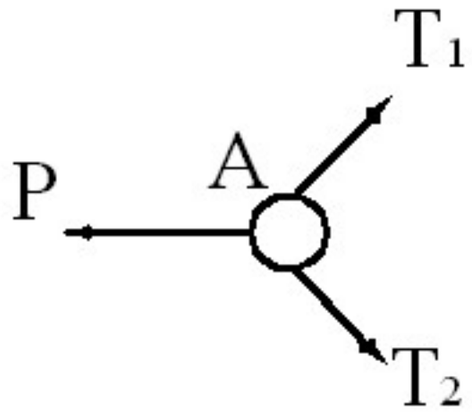


图 I-46　题 1-14 图



$$\sum X = 0$$

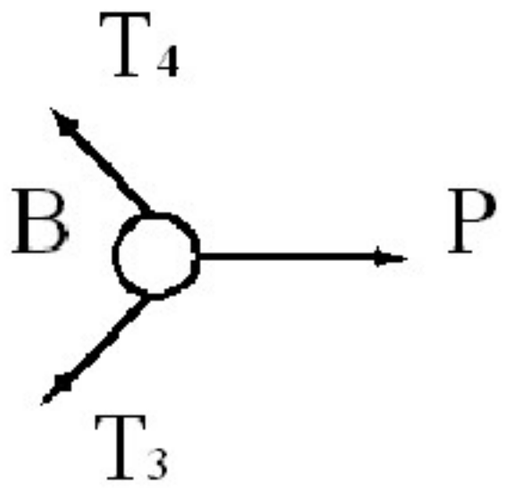
$$-P + T_1 \sin 45° + T_2 \sin 45° = 0$$

$$\sum Y = 0$$

$$T_1 \cos 45° - T_2 \cos 45° = 0$$

得　$T_1 = T_2 = \dfrac{\sqrt{2}}{2} P$

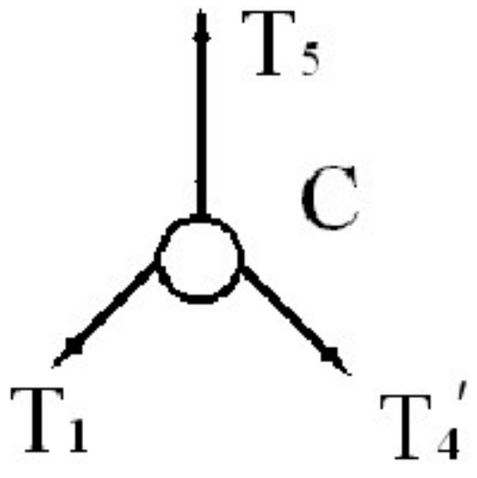
以B节点为研究对象，受力如图



同理可得

$$T_3 = T_4 = \frac{\sqrt{2}}{2} P$$

以C节点为研究对象，受力如图



$$T_5 - T_1 \cos 45° - T_4^{'} \cos 45° = 0$$

$$T_5 = P$$

1-16 图1-48所示水平传动轴上装有两个皮带轮C和D,半径分别为r_1=200mm和r_2=250mm，C轮上皮带是水平的，两边张力为$T_1=2t_2$=5KN，D轮上皮带与铅直线夹角 $\alpha=30°$，两张力为$T_2=2t_2$。当传动轴均匀传动时，试求皮带张力T_2、t_2和轴承A、B的反力。

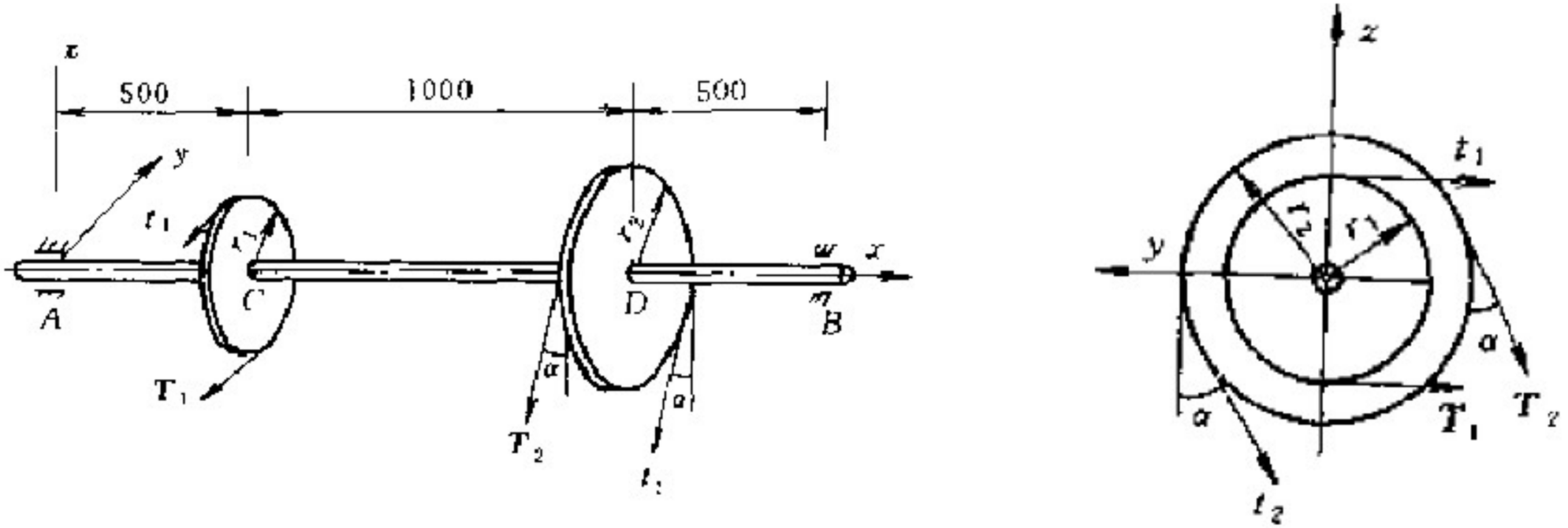
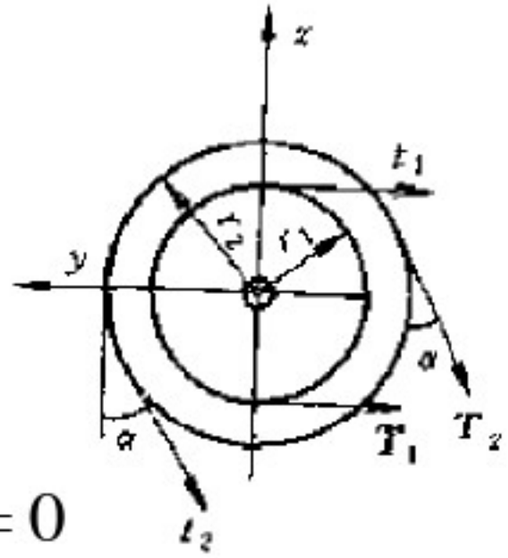
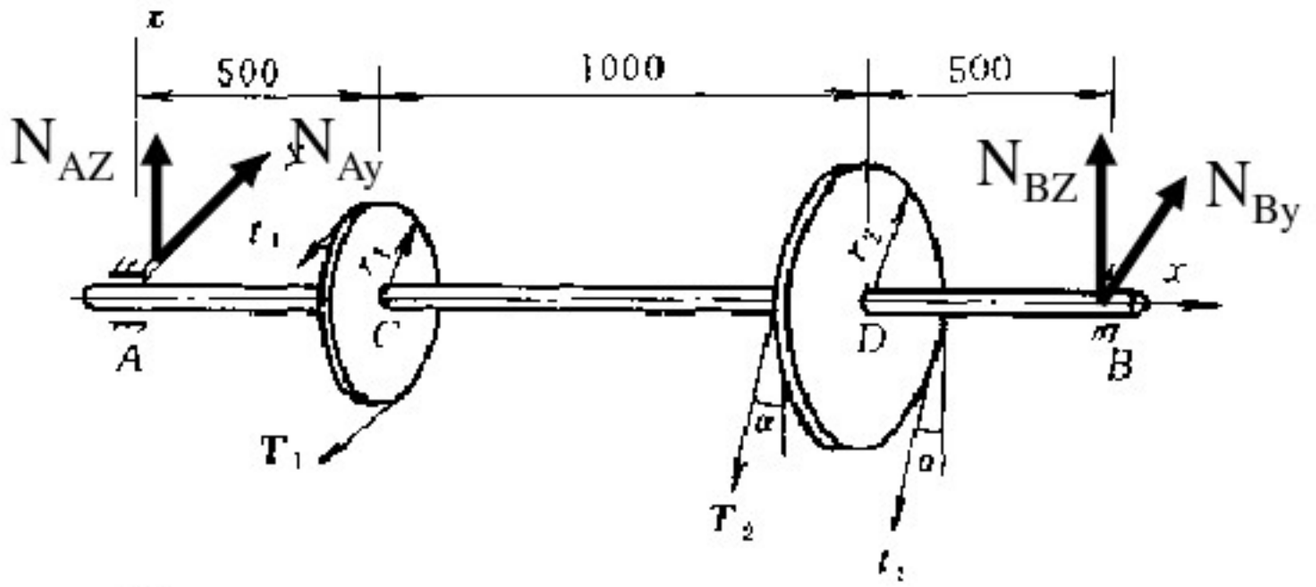


图 1-48　题 1 16 图





解：　$\sum M_X = 0$

$$(-T_1 + t_1)r_1 + (T_2 - t_2)r_2 = 0$$

得：$T_2 - t_2 = 2KN$　　∵ $T_2 = 2t_2$

$T_2 = 4KN$　　$t_2 = 2KN$

$$\sum M_Z = 0$$

$$-[(T_1 + t_1) \times 500 + (T_2 + t_2)\sin 30° \times 1500)] + N_{BY} \times 2000 = 0$$

$N_{BY} = 4.125KN$

$$\sum M_Y = 0$$

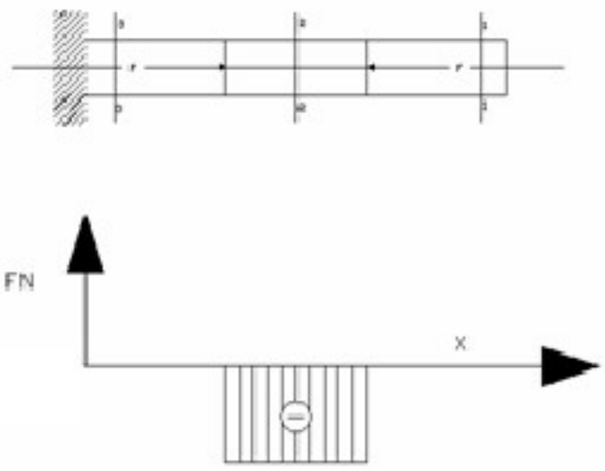
$$(T_2 + t_2)\sin 30° \times 1500 - N_{BZ} \times 2000 = 0$$

$$N_{BZ} = 3.897$$

2-1 试求出图2-34所示各杆1-1，2-2，及3-3截面上的轴力，并作轴力图。

$$\sum Z = 0$$

$$N_{AZ} + N_{BZ}$$



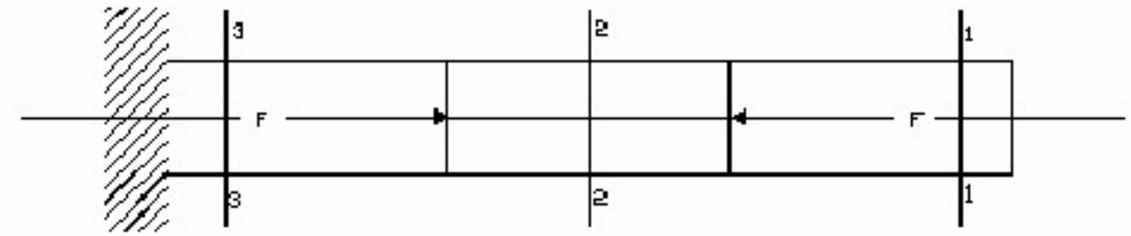
$$N_{AZ} = 1.29$$

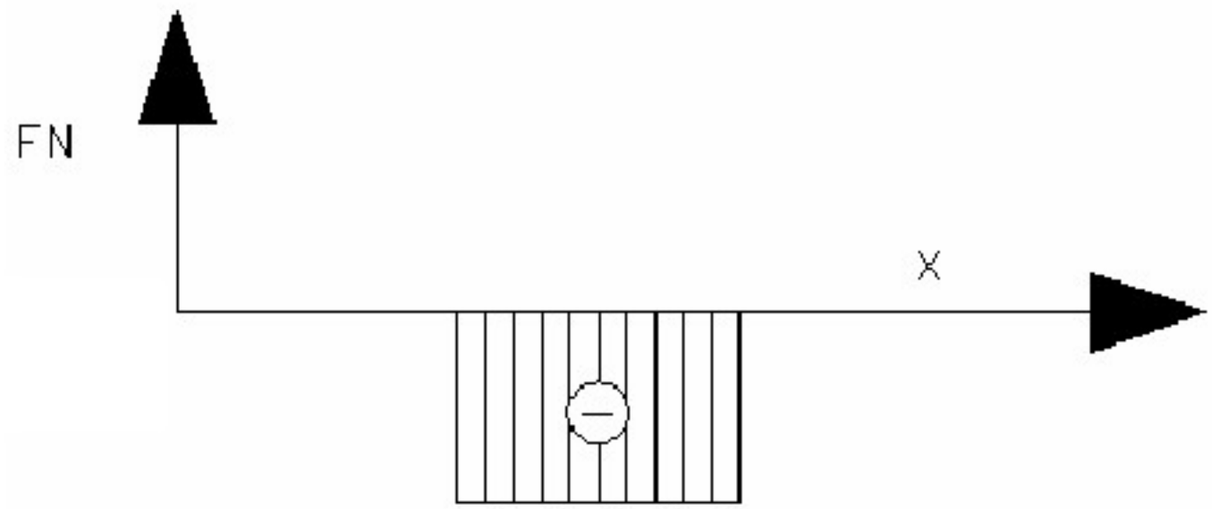
$$\sum Y = 0$$

$$N_{Ay} + N_{By} -$$

$$N_{Ay} = 6.375KN$$

2-1 试求出图2-34所示各杆1-1，2-2，及3-3截面上的轴力，并作轴力图。

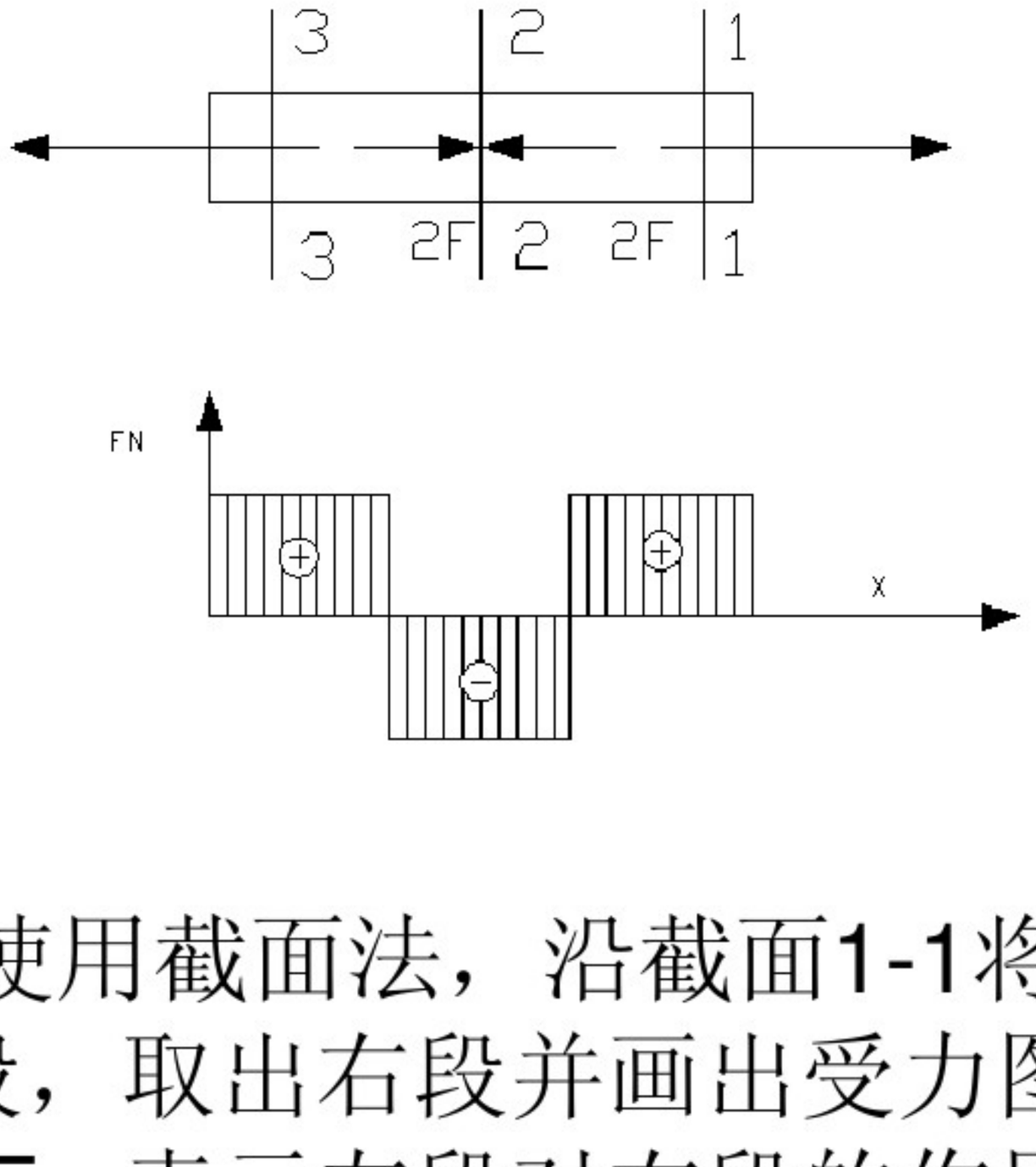




解b　使用截面法，沿截面1-1将杆分成两段，取出右段并画出受力图（b）用F_{N1}表示左段对右段的作用，由平衡方程∑Fx=0，得F_{N1}=F（拉）

同理，可以计算横截面2-2上的轴力FN2，由截面2-2右段图（c）的平衡方程∑Fx=0，得FN2= F（压）

同理，可以计算横截面3-3上的轴力FN3，由截面3-3右段图（d）的平衡方程∑Fx=0，得FN3= 0



解b 使用截面法，沿截面1-1将杆分成两段，取出右段并画出受力图（b）用F_{N1}表示左段对右段的作用，由平衡方程$\sum Fx=0$，得$F_{N1}=F$（拉）

同理，可以计算横截面2-2上的轴力F_{N2}，由截面2-2右段图（c）的平衡方程$F_x=0\sum$，得FN2= F（压）

同理，可以计算横截面3-3上的轴力F_{N3}，由截面3-3左段图（d）的平衡方程$\sum Fx=0$，得$F_{N3}=F$（拉）

2-2 试求图2-35所示钢杆各段内横截面上的应力和杆的总变形。钢的弹性模量E=200GPa。

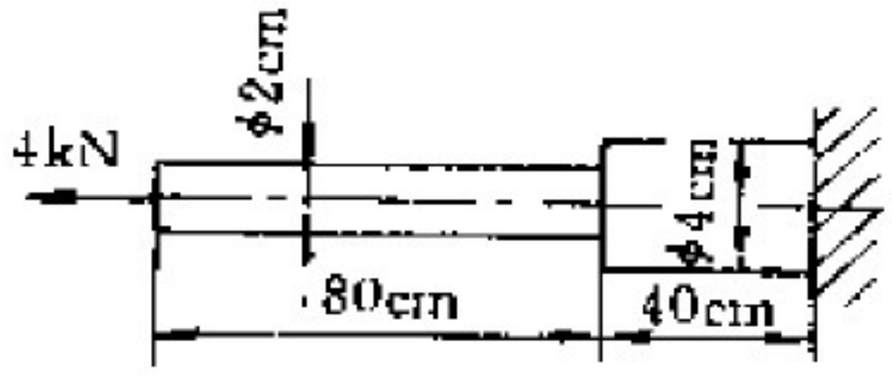


图 2-35　题 2-2 图

解 1、内力计算

用截面法分别计算左段和右段的内力并作杆的轴力图（b）

得 F左 =4kN（拉）

F右 =4kN（拉）

左段：

$$\Delta L_{左}=\frac{F_{左}L_{左}}{EA_{左}}=\frac{4\times10^{3}\times80\times10^{-2}}{200\times10^{9}\times\frac{\pi}{4}\times4\times10^{-4}}=5.0955\times10^{-5}(\mathrm{m})$$

右段：

$$\Delta L_{右}=\frac{F_{右}L_{右}}{EA_{右}}=\frac{4\times10^{3}\times40\times10^{-2}}{200\times10^{9}\times\frac{\pi}{4}\times16\times10^{-4}}=0.637\times10^{-5}(\mathrm{m})$$

$$\delta_{左} = \frac{F_{左}}{A_{左}} = \frac{4 \times 10^3}{\frac{\pi}{4} \times 2^2 \times 10^{-4}} = 12.73\text{MPa}$$

$$\delta_{右} = \frac{F_{右}}{A_{右}} = \frac{4 \times 10^3}{\frac{\pi}{4} \times 4^2 \times 10^{-4}} = 3.18\text{MPa}$$

2、各段变形的计算左、右两段的轴力为$F_{左}$ $F_{右}$，横截面面积$A_{左}$、$A_{右}$，长度$L_{左}$，$L_{右}$均不相同，变力计算应力分别进行。

3、总变形计算

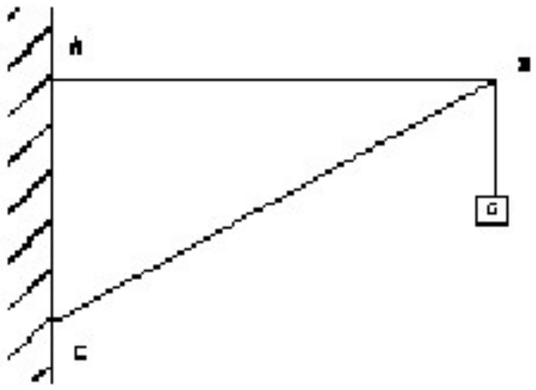
$$\Delta L = \Delta L_{左} + \Delta L_{右}$$

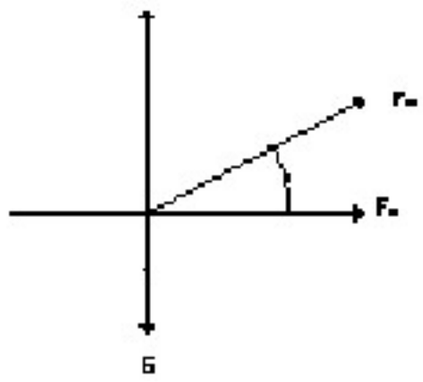
$$= 5.0955 \times 10^{-5} + 0.637 \times 10^{-5}$$

$$= 5.73 \times 10^{-5} (\mathrm{m})$$

4、计算结果表明，左段伸长5.0955x 10^{-5}m，右段伸长0.637x 10-5m，全杆伸长5.73 x10-5m。

2-3 图2-36所示三角形支架，杆AB及BC都是圆截面的。杆AB直径d_1＝20mm，杆BC直径d_2＝40mm，两杆材料均为Q235钢。设重物的重量G＝20kN。问此支架是否安全。





解　选取B为研究对象

1、如图所示，由平衡方程得

$\sum Fx=0 \quad G-F_{BC}Sin30^{o}=0$

$\sum Fy=0 \quad F_{AB}-F_{BC}Cos30^{o}=0$

解得　　$F_{BC}=G$

$$F_{AB}=\sqrt{3}G$$

2、正应力为

$$\sigma_{AB}=\frac{F_{AB}}{A_{AB}}=\frac{\sqrt{3}\times 20\times 10^{3}}{\frac{\pi}{4}\times 400\times 10^{-6}}=110.3\text{MPa}<[\sigma]=160\text{MPa}$$

$$\sigma_{BC}=\frac{F_{BC}}{A_{BC}}=\frac{2\times 20\times 10^{3}}{\frac{\pi}{4}\times 1600\times 10^{-6}}=31.85\text{MPa}<[\sigma]=160\text{MPa}$$

2-4 蒸汽机的汽缸如图2-37所示，汽缸的内径D＝400mm，工作压力P＝1.2MPa。汽缸盖和汽缸用直径为18mm的螺栓连接。若活塞杆材料的许用应力为50MPa，螺栓材料的许用应力为40MPa，试求活塞杆的直径及螺栓的个数。

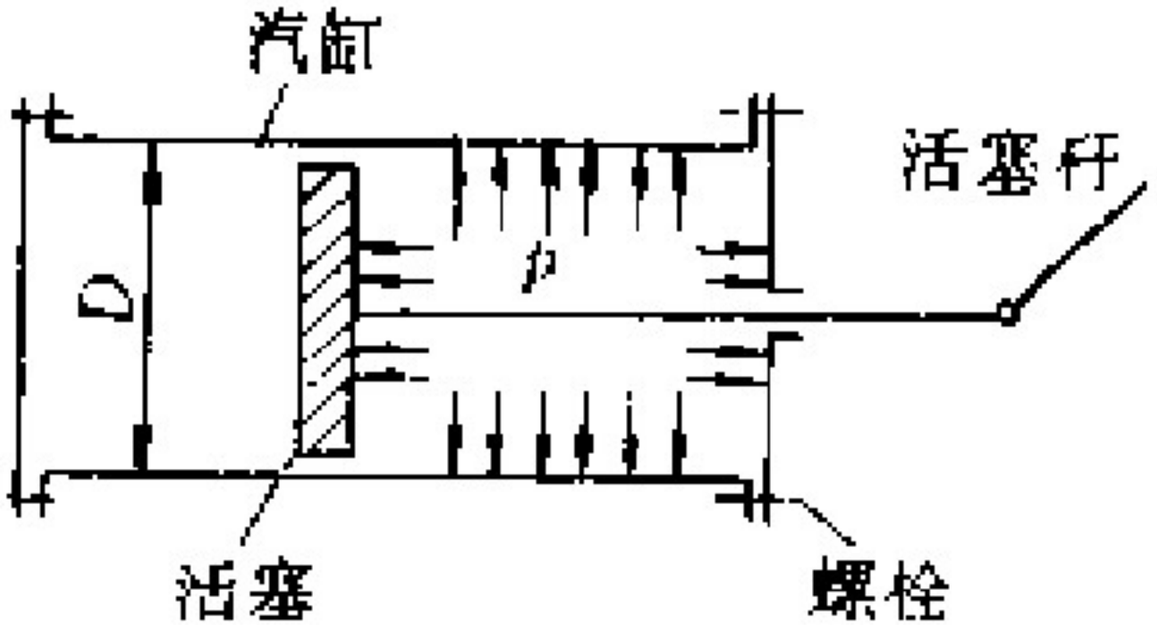


图 2-37　题 2-4 图

解：作用在活塞杆截面的工作应力

$$\sigma = \frac{F_N}{A} = \frac{P\dfrac{\pi D^2}{4}}{\dfrac{\pi d^2}{4}} = P\frac{D^2}{d^2}$$

由强度条件有 $\sigma = [\sigma]$

即 $P\dfrac{D^2}{d^2} \leq [\sigma]$

所以 $$d^2 \geq P\frac{D^2}{[\sigma]} = \frac{1.2\times10^6}{50\times10^6}\times400^2 = 3840mm$$

即活塞杆的直径 $$d \geq 62mm$$

由强度条件式得

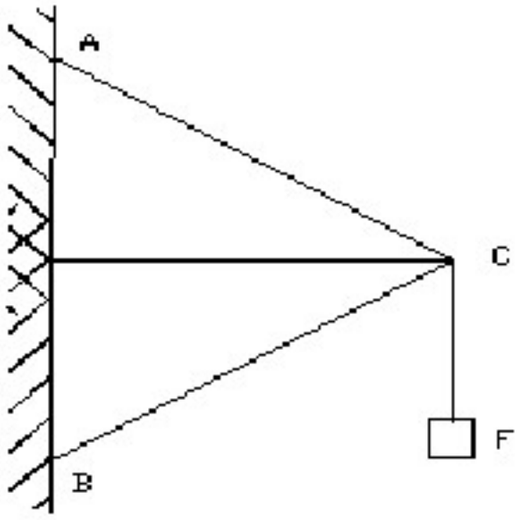
$$\sigma' = \frac{F_N}{nA'} = \frac{P'\frac{\pi D^2}{4}}{n\frac{\pi D'^2}{4}} \leq [\sigma']$$

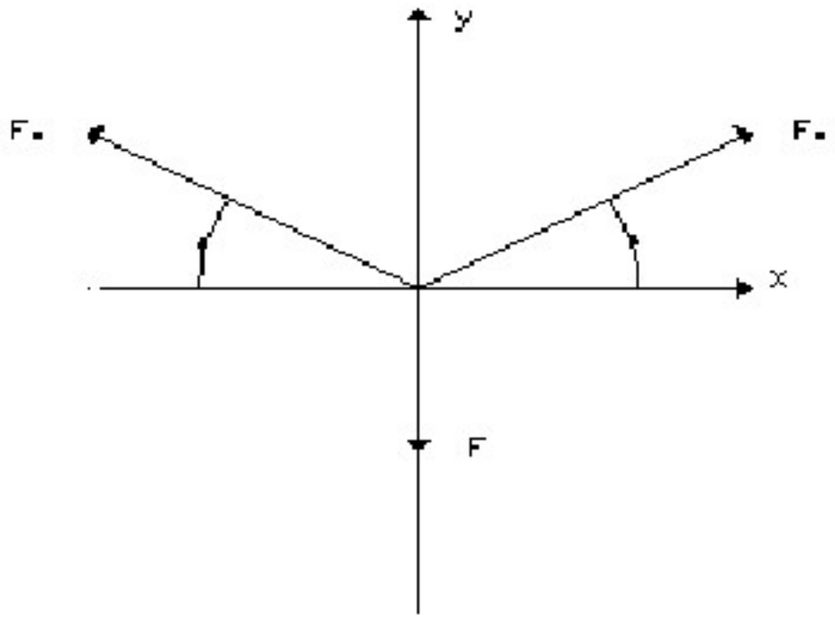
整理得

$$n \geq \frac{\mathrm{PD}^2}{\mathrm{D}'^2[\sigma']} = \frac{1.2\times10^6\times400^2}{40\times10^6\times18^2} = 14.8$$

螺栓应至少为16个

2-5 三角形支架ABC如图2-38所示，在C点受到载荷F的作用。已知，杆AC由两根10号槽钢所组成，$[\sigma]_{AC}$=160MPa；杆BC是20a号工字钢所组成，$[\sigma]_{BC}$=100MPa。试求最大许可载荷F。





解　选取C为研究对象

1、如图所示，由平衡方程得

$\sum Fx=0 \quad F_{AC}\,Cos30^{o}-F_{BC}Cos30^{o}=0$

$\sum Fy=0 \quad F_{AC}\,Sin30^{o}-F+F_{BC}Sin30^{o}=0$

解得　$F_{AC}=F_{BC}=F$

2、许用应力为

杆AC的承载极限：

$$F_1 \le [\sigma]_{AC} \times A_{AC} \le 160 \times 10^6 \times 2 \times 12.74 \times 10^{-4} = 406.78KN$$

杆BC的承载极限：

$$F_2 \le [\sigma]_{BC} \times A_{BC} \le 100 \times 10^6 \times 35.58 \times 10^{-4} = 355.8KN$$

由 $F_{AC} = F_{BC} = F \le 355.8KN$

得 $F = 355.8KN$

2-6 图2-39所示结构中梁AB的变形及重量可忽略不计。杆1为钢制圆杆，直径d_1=20mm，E_1=200GPa；杆2为铜制圆杆，直径d_2=25mm，E_2=100GPa。试问：(1)载荷F加在何处，才能使梁AB受力后仍保持水平？ (2)若此时F=30kN，求两拉杆内横截面上的正应力。

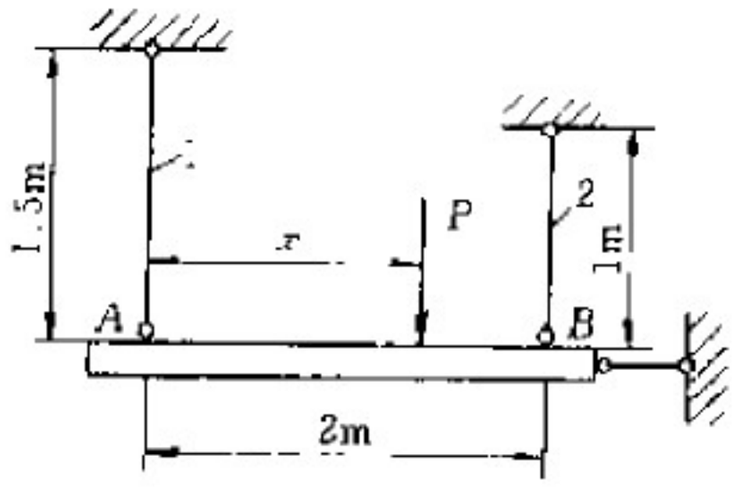


图 2-39 题 2-6 图

解　有平衡方程得

$$\sum M_A=0,\ \ F_X=F_B\times 2\Rightarrow F_B=\frac{FX}{2}$$

$$\sum M_B=0,\ \ F\ (2-X)=F_A\times 2\Rightarrow F_A=\frac{F\ (2-X)}{2}$$

由FA引起的变形等于FB引起的变形

$$\Delta\ell=\frac{F_A\ell_A}{E_AA_A}=\frac{F_B\ell_B}{E_BA_B}$$

即有

$$\frac{\frac{F\ (2-X)}{2}\times 1.5}{200\times 10^9\times\frac{\pi}{4}\times 20^2\times 10^{-6}}=\frac{\frac{FX}{2}\times 1}{100\times 10^9\times\frac{\pi}{4}\times 25^2\times 10^{-6}}$$

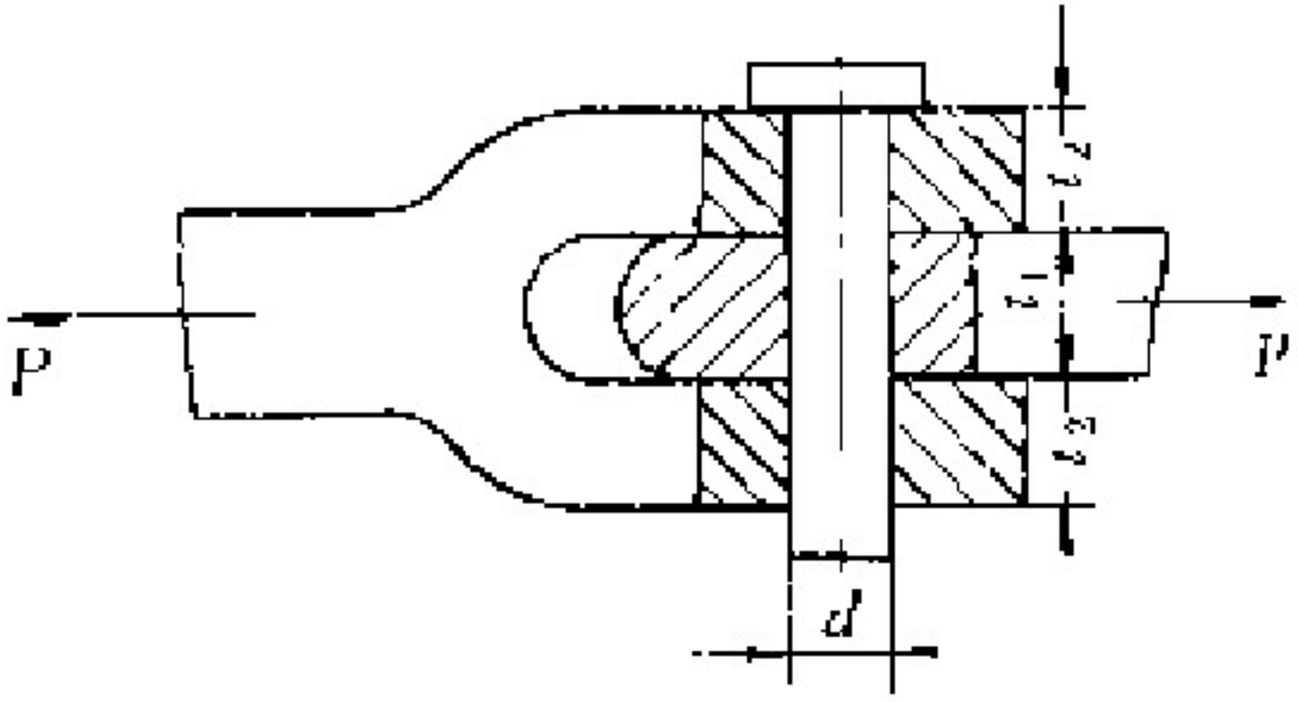
解得X=1.08m，当F=30KN时，

$$F_A = 13.8KN \qquad F_B = 16.2KN$$

正应力 $$\sigma_A = \frac{F_A}{A_A} = \frac{13.8\times10^3}{\frac{\pi}{4}\times20^2\times10^{-6}} = 43.97MPa$$

正应力 $$\sigma_B = \frac{F_B}{A_B} = \frac{16.2\times10^3}{\frac{\pi}{4}\times25^2\times10^{-6}} = 32.97MPa$$

2-7　目2-40所示销钉连接、已知F＝18kN．板厚t_1＝8mm，t_2＝5mm．销钉与板的材料相同，许用切应力[τ]＝60MPa．许用挤压应力[σp]＝200MPa。试设计销钉直径d。



解　许用剪应力

$$\tau = \frac{p}{2A_1} = \frac{p}{2\times\frac{\pi}{4}\times d^2} = \frac{18\times 2\times 10^3}{\pi d^2} \le [\tau] = 60\text{MPa}$$

得　$d \ge 14\text{mm}$

挤压应力　$\sigma_{p_1} = \frac{p}{A_2} = \frac{p}{t_1\times d} = \frac{18\times 10^3}{8\times 10^{-3}\times d} \le [\sigma_p] = 200\text{MPa}$

得 $d \ge 11\text{mm}$

挤压应力

$$\sigma_{p_1}=\frac{p}{2A_1}=\frac{p}{t_2\times 2\times d}=\frac{18\times 10^3}{5\times 10^{-3}\times 2\times d}\le\left[\sigma_p\right]=200\text{MPa}$$

得 $d\ge 9\text{mm}$

综上可知 $d\ge 14\text{mm}$

2- 8 如图2-41所示，齿轮与轴用平键连接，已知轴直径d=70mm，键的尺寸 $b\times h\times \ell = 20mm\times 12mm\times 100mm$ ，传递的力偶矩 $M = 2KN\bullet m$ ；键材料的许用应力 $[\tau] = 80MPa$ $[\sigma_p] = 200MPa$，试校核键的强度。

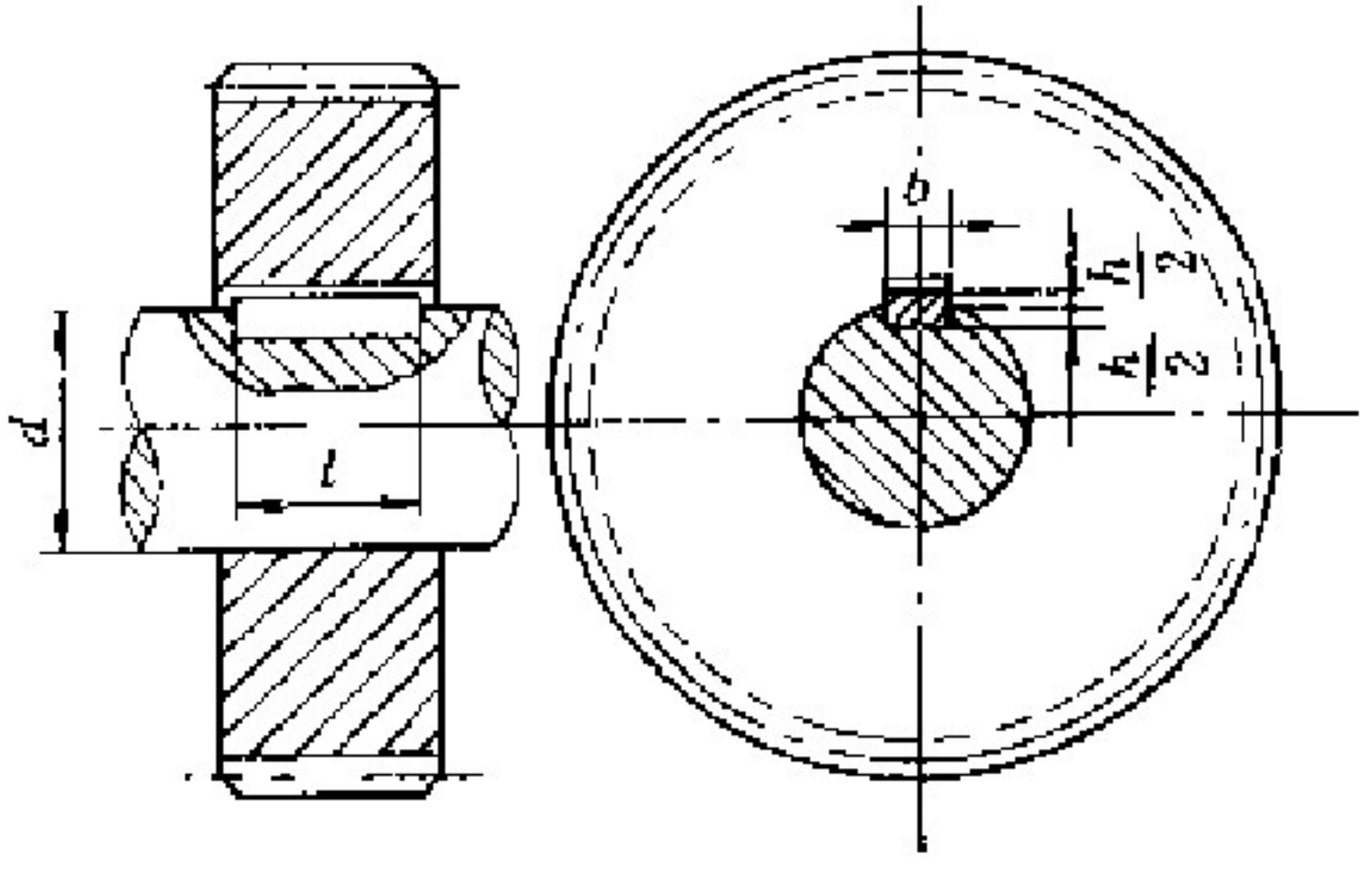


图 2-41　题 2-8 图

解　对轴心取矩，由平衡方程得：

力偶　$$M = F_s \times \frac{d}{2} = \frac{2 \times 2}{0.07} = 57.1KN$$

剪切面积为

$$A = b \times \ell = 20 \times 100 = 2000 mm^2$$

切应力

$$\tau = \frac{F_s}{A_1} = \frac{57.1 \times 10^3}{2000 \times 10^{-6}} = 28.55 MPa < [\tau] = 80 MPa$$

校核键的挤压强度，挤压面积

$$A_2 = \frac{h \times \ell}{2} = \frac{12 \times 100}{2} = 600 mm^2$$

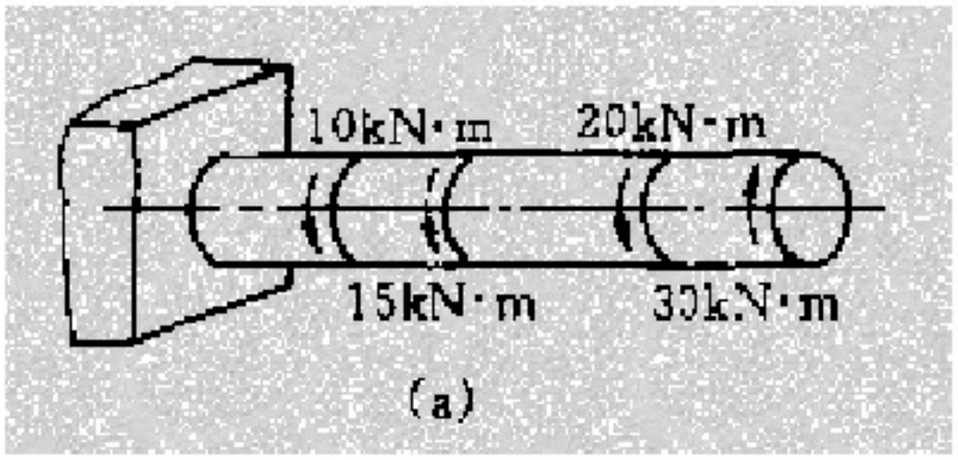
挤压应力

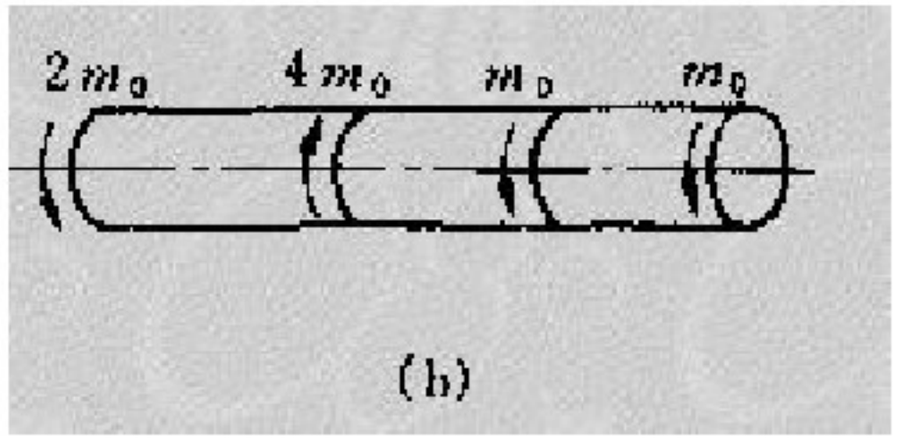
$$\sigma_p = \frac{F_s}{A_2} = \frac{57.1\times10^3}{600\times10^{-6}} = 95.17\text{MPa} < [\sigma_p] = 200\text{MPa}$$

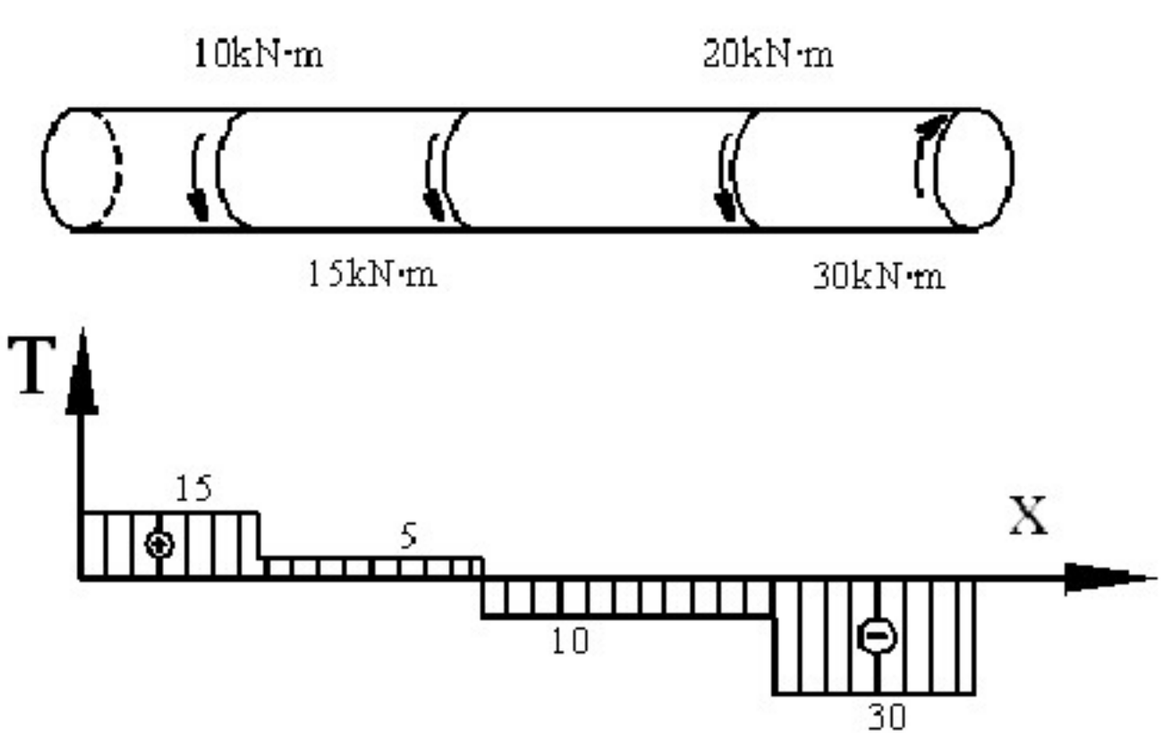
第三章 扭转

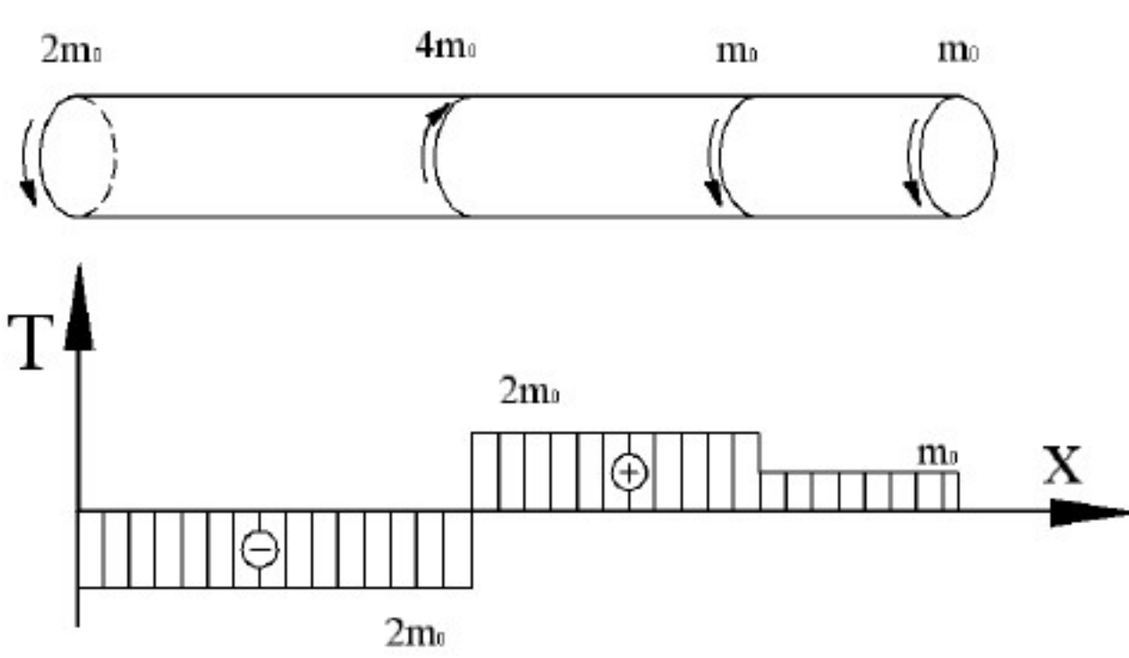
习题解答

3-1 作出图3-15所示各轴的扭矩图。









3-4一根钢轴，直径为20mm，如$[\tau]$=100MPa,求此轴能承受的扭矩。如转速为100r/min,求此轴能传递多少kW的功率。

解：　钢轴抗扭截面模量　$W_t = \dfrac{\pi d^3}{16}$

根据扭转的刚度条件　$\tau_{\max} = \dfrac{T}{W_t} \le [\tau]$

代入数值得：m=157 N·m

根据　$m = 9550\dfrac{N}{n}$

解得：N=1.64 kW

3-5 如图3-17所示，在一直经为75mm的等截面圆轴上，作用者外力偶矩：m_1=1kN·m, m_2=0.6kN·m, m_3=0.2kN·m, m_4=0.2kN·m。

（1）求作轴的扭矩图。

（2）求出每段内 的最大剪力。

（3）求出轴的总扭转角。设材料的剪切模量 G=80GPa。

（4）若m_1和m_2的位置互换，问在材料方面有何增减。

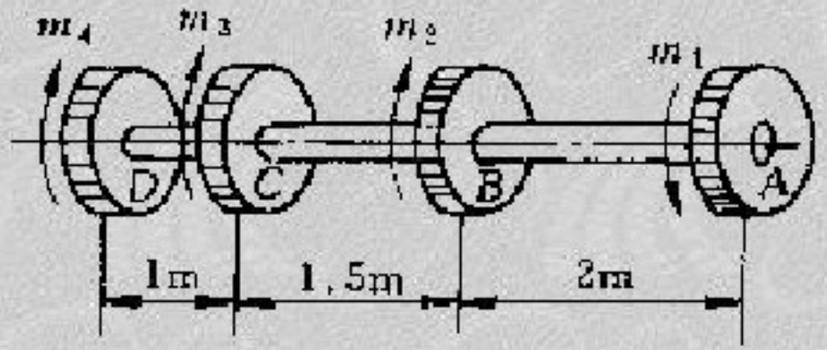
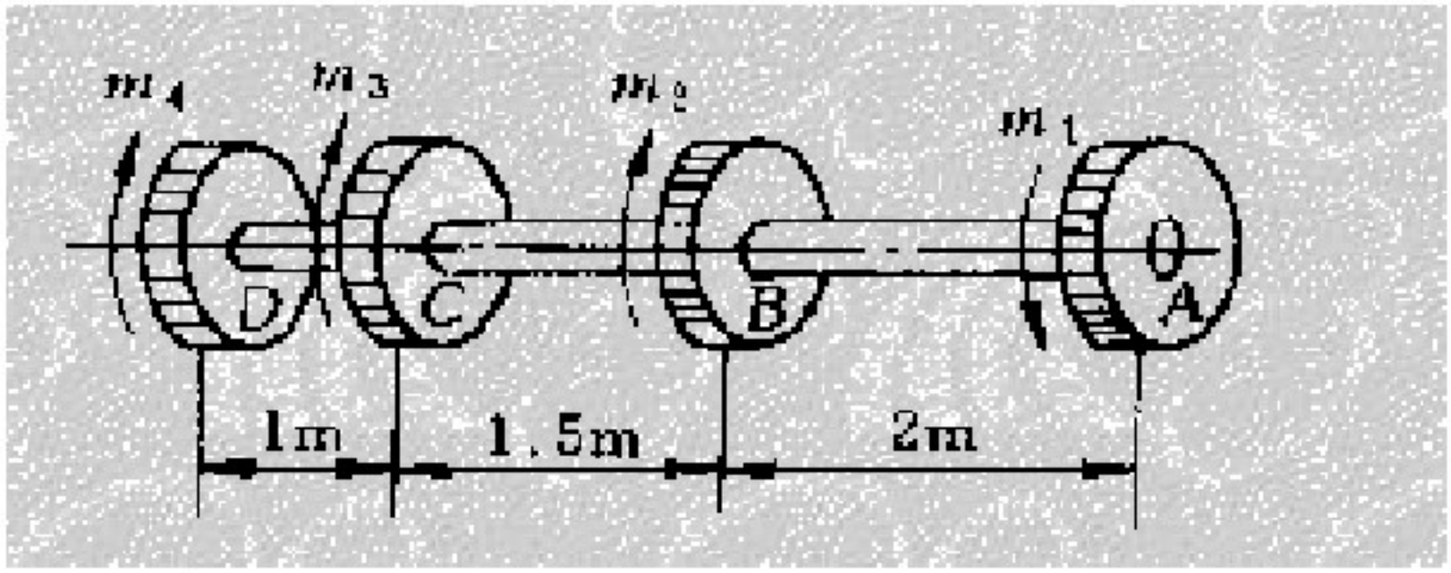
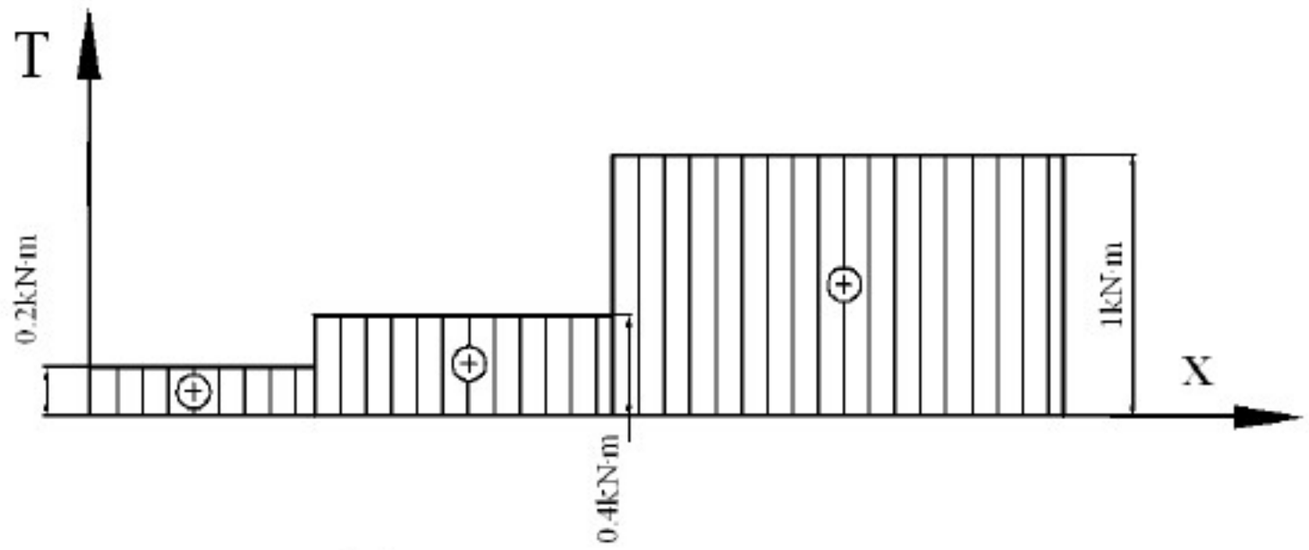


图 3-17　题 3-5 图



（1）



（2）

$$\tau_{1\max}=\frac{T_1}{W_t}=\frac{1}{\frac{\pi(0.075)^2}{16}}=12.08\ \text{MPa}$$

$$\tau_{2\max}=\frac{T_2}{W_t}=\frac{0.4}{\frac{\pi(0.075)^2}{16}}=4.83\ \text{MPa}$$

$$\tau_{3\max}=\frac{T_3}{W_t}=\frac{0.2}{\frac{\pi(0.075)^2}{16}}=2.42\text{MPa}$$

（3）根据 $\varphi=\dfrac{Tl}{GI_P}$

$$\varphi_1=\frac{T_1l_1}{GI_P}=\frac{1\times10^3\times2\times32}{80\times10^9\times\pi\times0.075^4}=8\times10^{-4}\ \text{rad}\qquad \varphi_2=\frac{T_2l_2}{GI_P}=2.41\times10^{-3}\ \text{rad}$$

$$\varphi_3=\frac{T_3l_3}{GI_P}=8\times10^{-3}\ \text{rad}\qquad \varphi=\varphi_1+\varphi_2+\varphi_3\qquad \varphi=0.0112=0.642^\circ$$

（4）若m_1和m_2的位置互换，

$T_1 = -0.6\,\text{kN·m}$　　$T_2 = 0.4\ \text{kN·m}$　　$T_3 = 0.2\ \text{kN·m}$

最大剪力减少，所以用料将降低

3-6 如图3-18所示汽车方向盘外径 ϕ=500mm，驾驶员每只手加在方向盘上的里P=300N，方向盘轴为空心圆轴，其外径之比$\alpha = d/D = 0.8$，材料的许用应力$[\tau]$=600MPa。试求方向盘轴的内外直径。

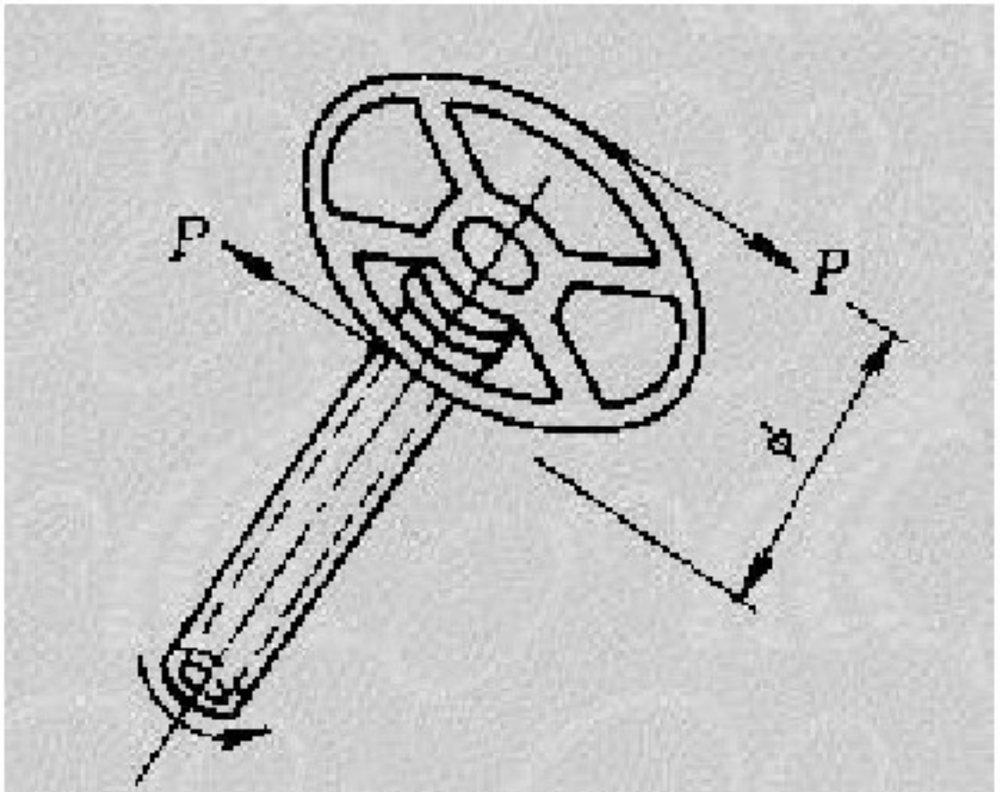


图 3-18　题 3-6 图

解：　$T = P \times \phi = 300 \times 0.5 = 150\,\mathrm{N\cdot m}$

由

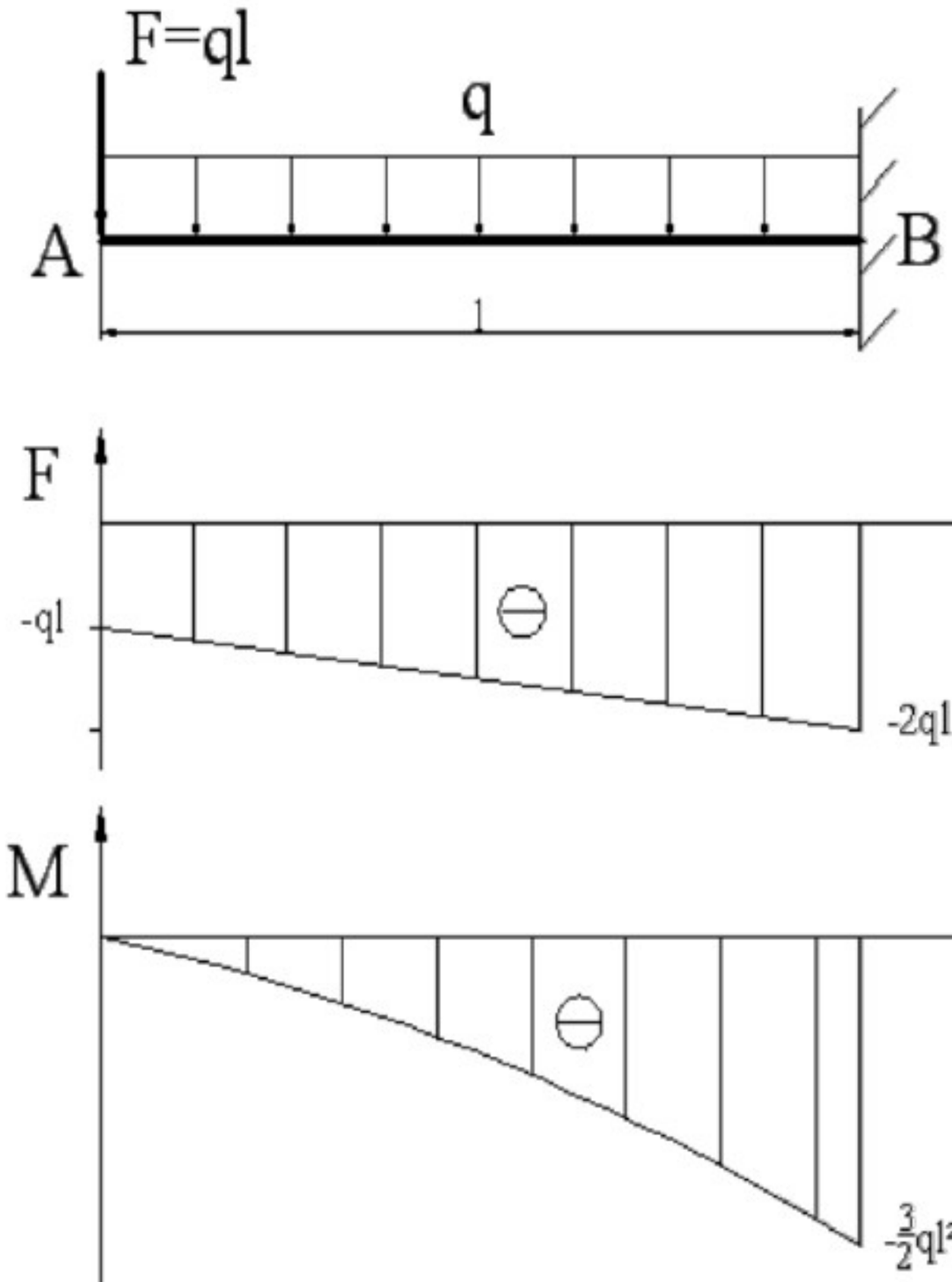
$$\tau = \frac{T}{W_t} = \frac{T}{\frac{\pi D^3}{16}\left(1-\alpha^4\right)} \le 60\times 10^6$$

解得：　$D \ge 2.78\times 10^{-2}\ \mathrm{m}$

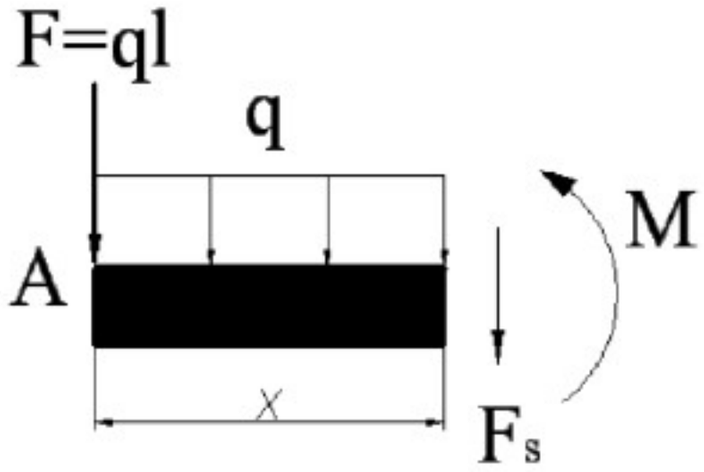
$d \ge 2.23\times 10^{-2}\,\mathrm{m}$

4-1　试列出各梁的的剪力和弯矩方程，并作剪力图和弯矩图，求出$F_{S}max$和$Mmax$

(a)



(1) 列剪力方程和弯矩方程

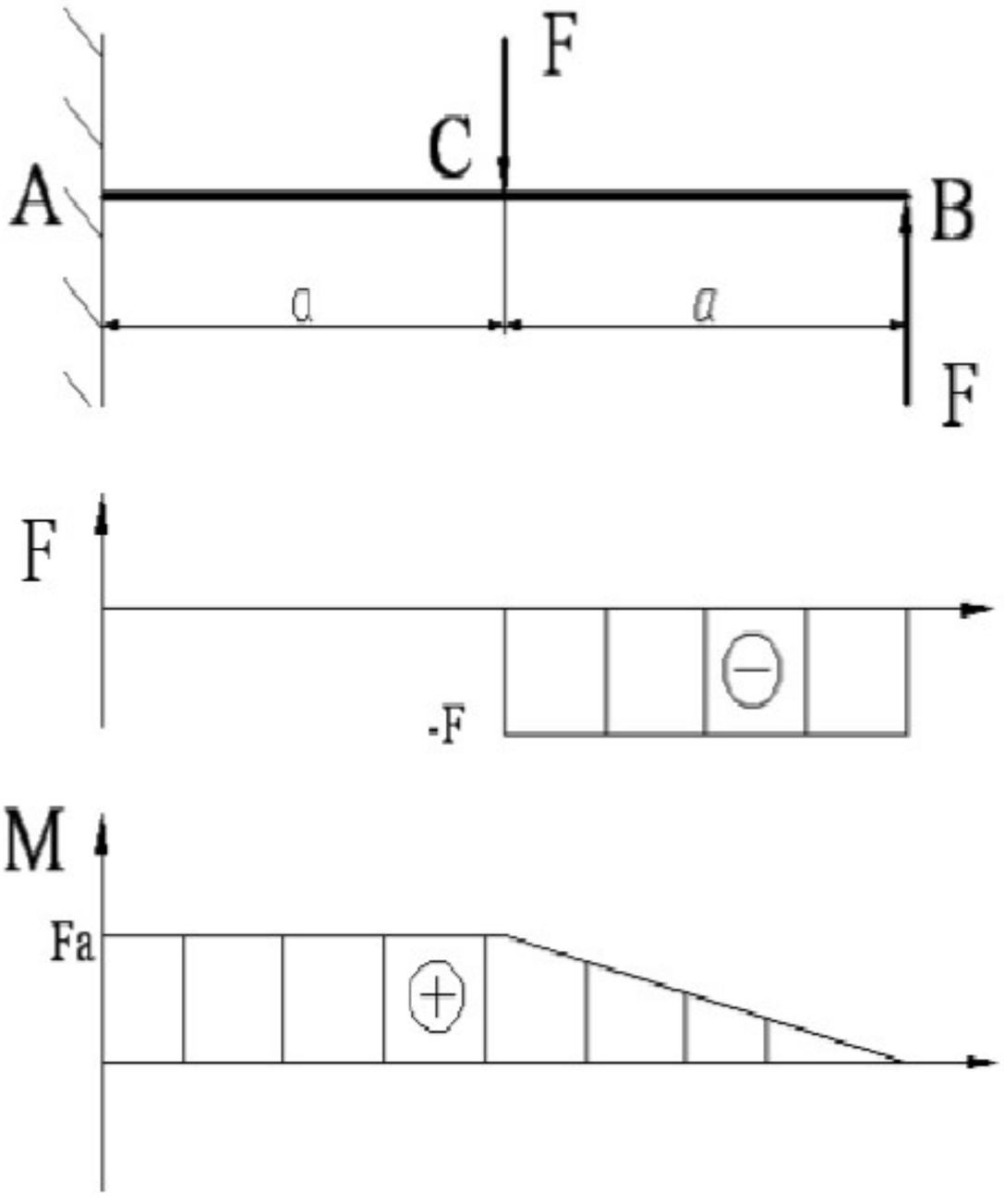


$$F_S(x)=-ql-qx \quad (0<x<l)$$

$$M(x)=-qxl-\frac{qx^2}{2} \quad (0\le x<l)$$

(2) 作剪力图和弯矩图

(c)



$F_S(x) = -F \quad (a < x < 2a)$

$M(x) = M_1 - F(x-a) \quad (a \le x < 2a)$

（1）求支反力

由平衡方程$\sum M_A = 0$,

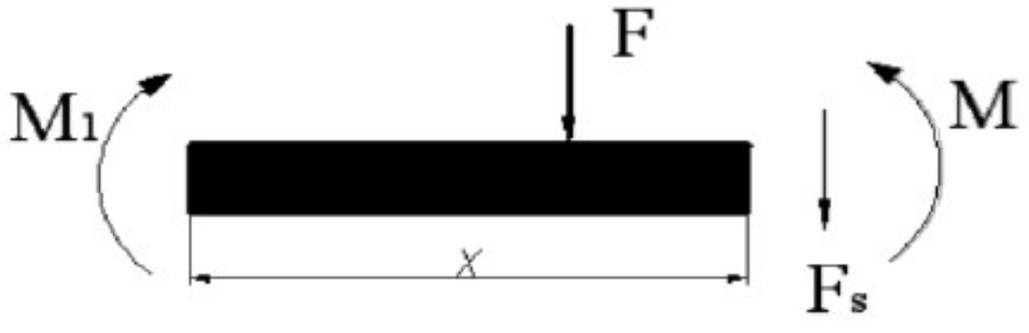
$2Fa - M_1 - Fa = 0$, 求得 $M_1 = Fa$

（2）列剪力方程和弯矩方程



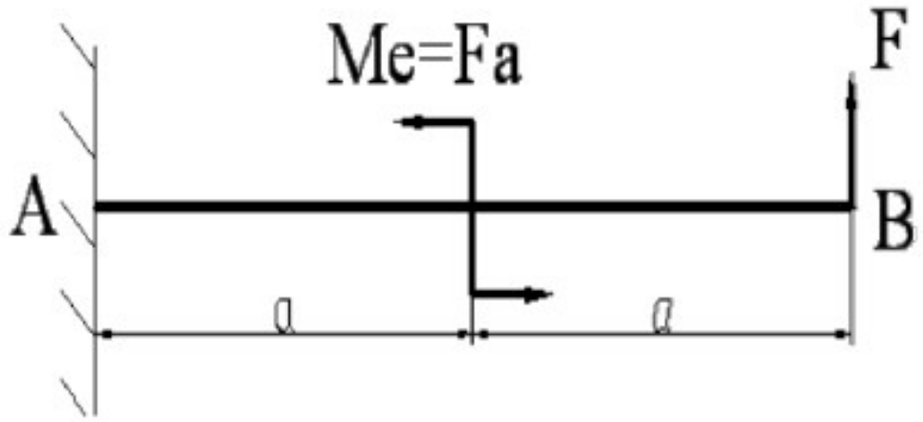
$F_S(x) = 0 \quad (0 < x < a)$

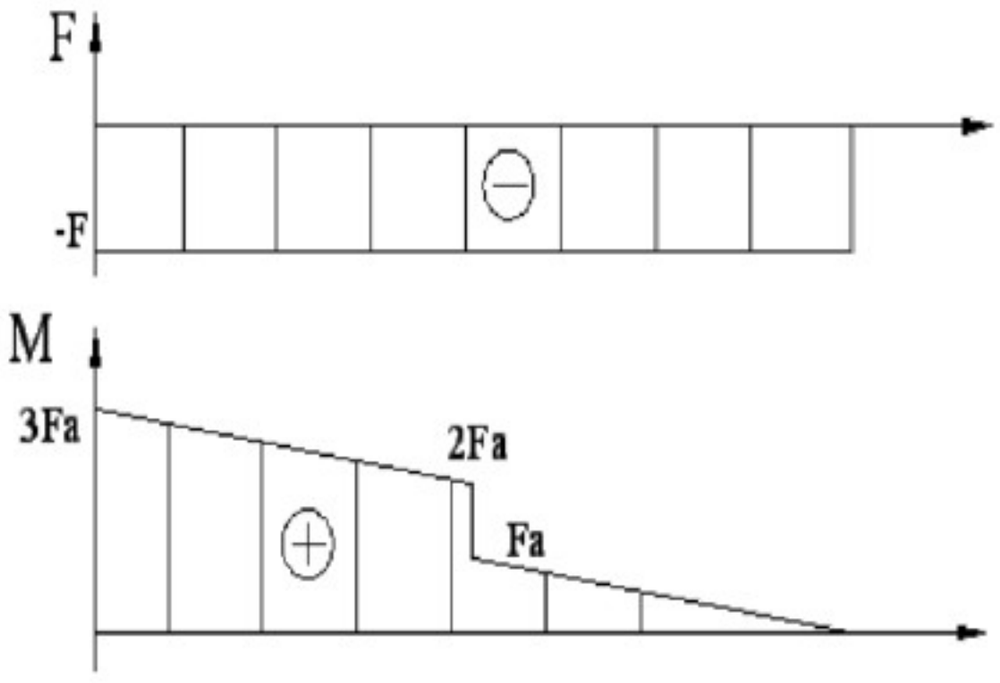
$M(x) = Fa \quad (0 \le x < a)$

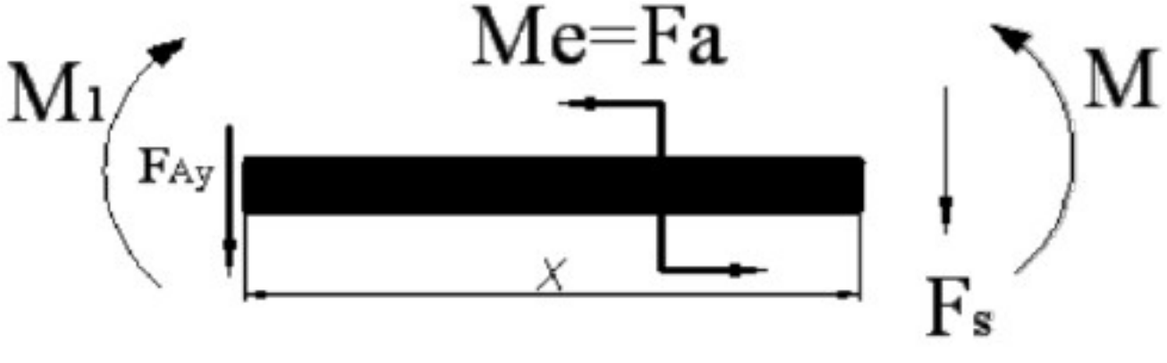


（3）作剪力图和弯矩图

(e)





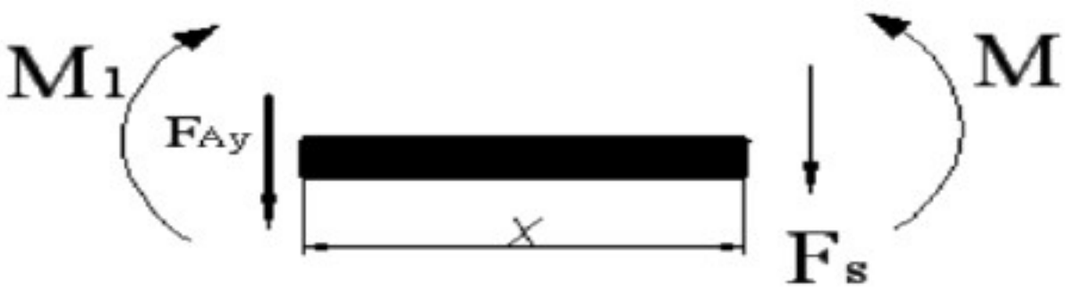


（1）求支反力　由平衡方程$\sum M_A=0$,

$$2Fa-M_1+Me=0$$

$$F_{Ay}=F \quad M_1=3Fa$$

（2）列剪力方程和弯矩方程



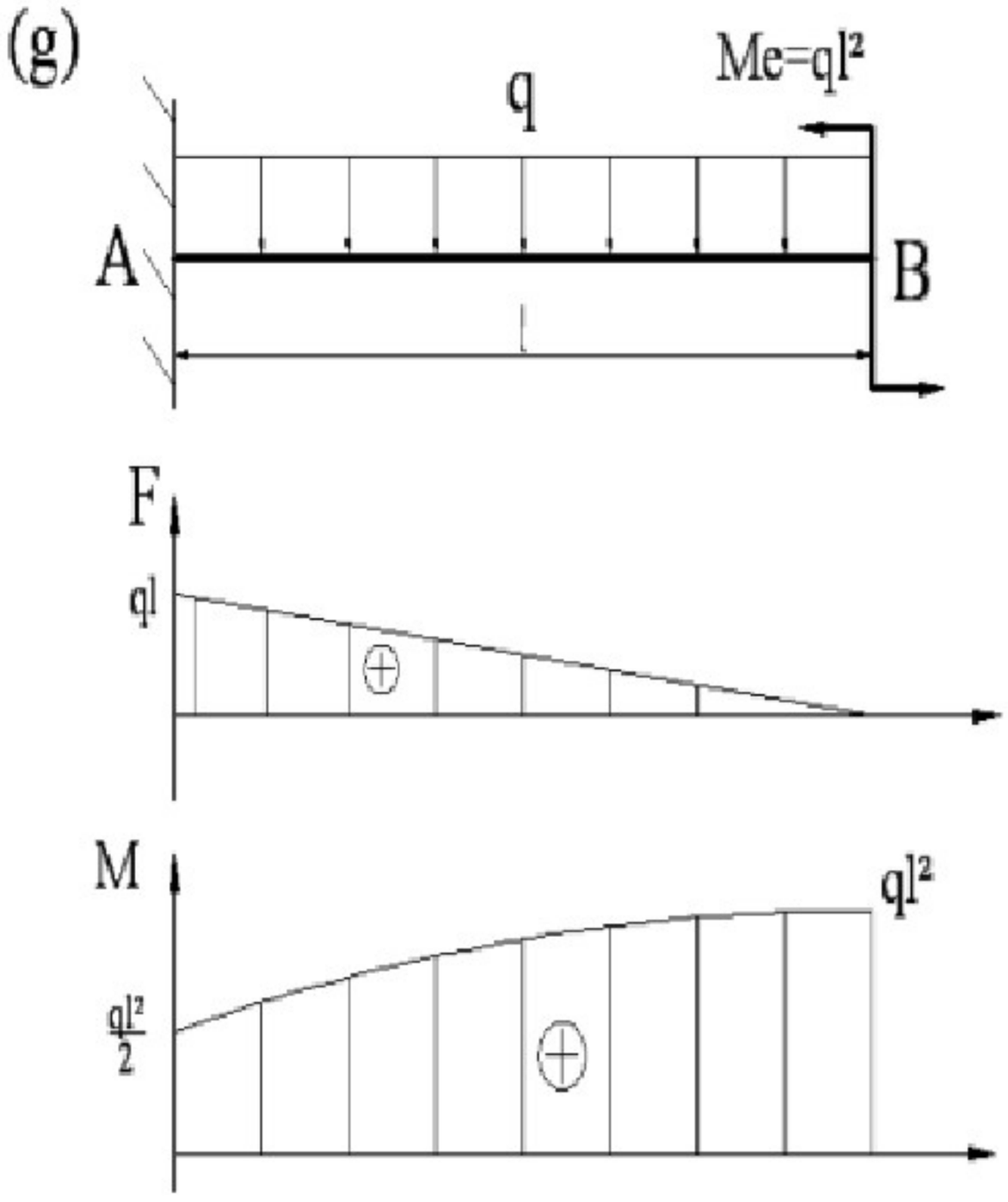
$$F_S(x)=-F \quad (0<x<a)$$

$$M(x)=M_1-Fx \quad (0\le x<a)$$

$$F_S(x)=-F \quad (a<x<2a)$$

$$M(x)=M_1-Fx-Me \quad (a\le x<2a)$$

（3）作剪力图和弯矩图



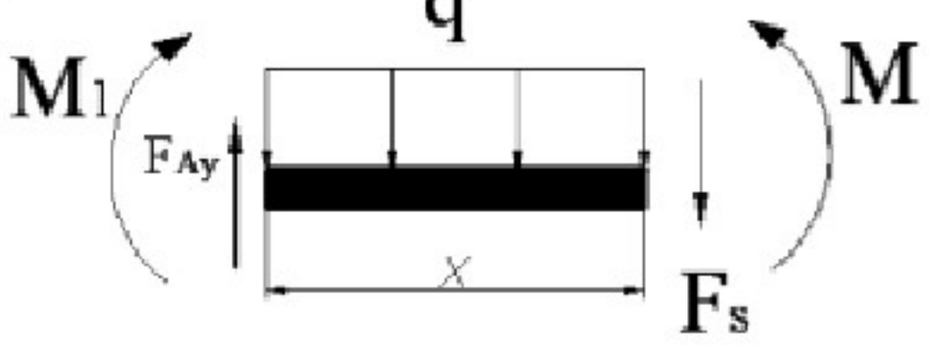
（1）求支反力

由平衡方程∑F=0 ,∑M_A=0

$$Me - M_1 - \frac{ql^2}{2} = 0$$

$$F_{Ay} = ql \qquad M_1 = \frac{ql^2}{2}$$

（2）列剪力方程和弯矩方程



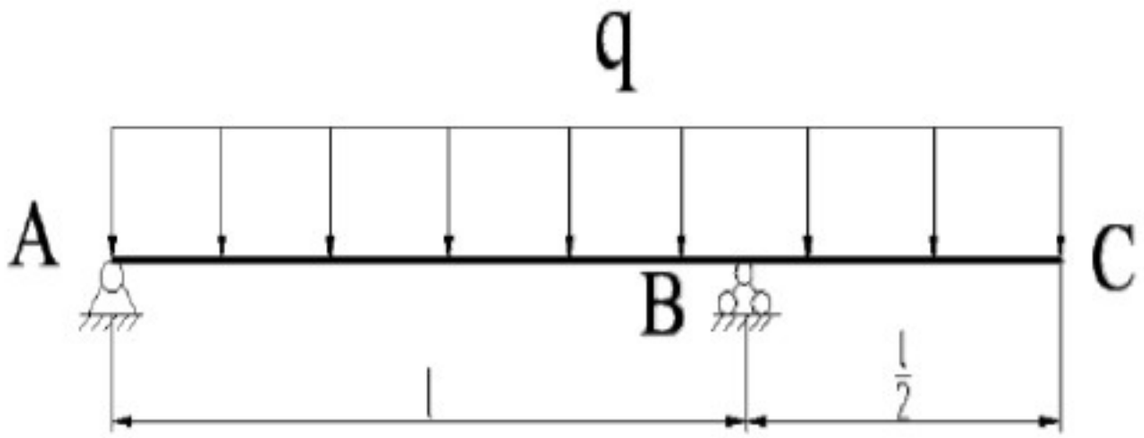
$$F_S(x) = ql - qx \quad (0 < x < l)$$

$$M(x) = M_1 - \frac{qx^2}{2} + F_{Ay}x \quad (0 \le x < l)$$

（3）作剪力图和弯矩图

（1）求支反力　由平衡方程$\sum F=0$,$\sum M_A=0$

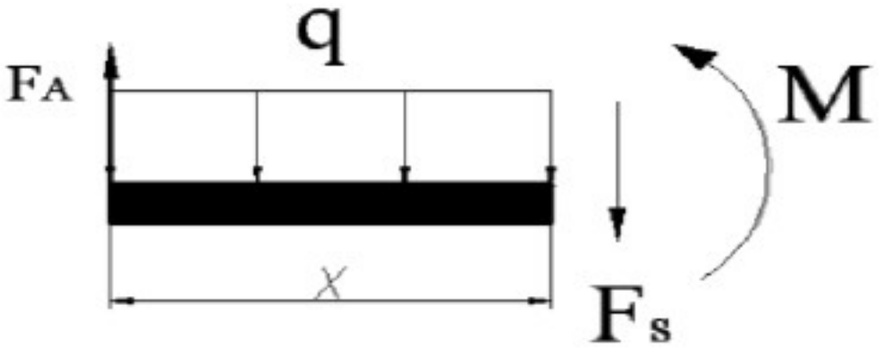
(i)



$$\begin{cases} F_A + F_B - \dfrac{3ql}{2} = 0 \\ F_B l - \dfrac{3ql}{2} \cdot \dfrac{3l}{4} = 0 \end{cases}$$

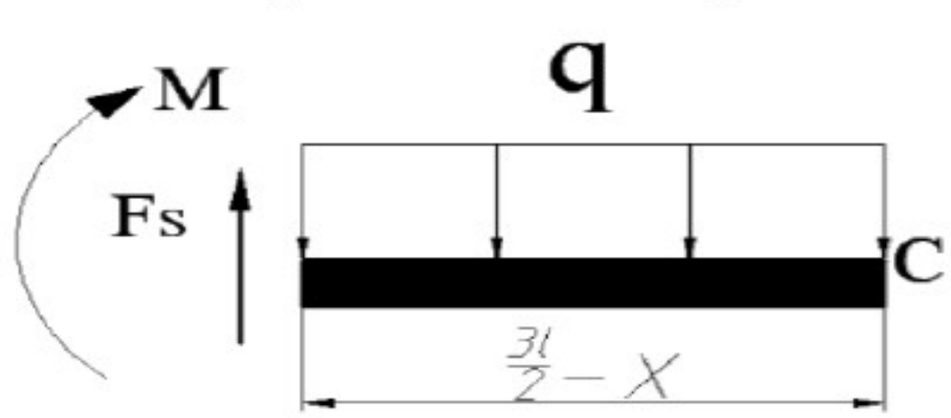
$$F_A = \frac{3ql}{8} \qquad F_B = \frac{9ql}{8}$$

（2）列剪力方程和弯矩方程



$$F_S(x) = \frac{3ql}{8} - qx \quad (0 < x < l)$$

$$M(x) = \frac{3qlx}{8} - \frac{qx^2}{2} \quad (0 \le x < l)$$

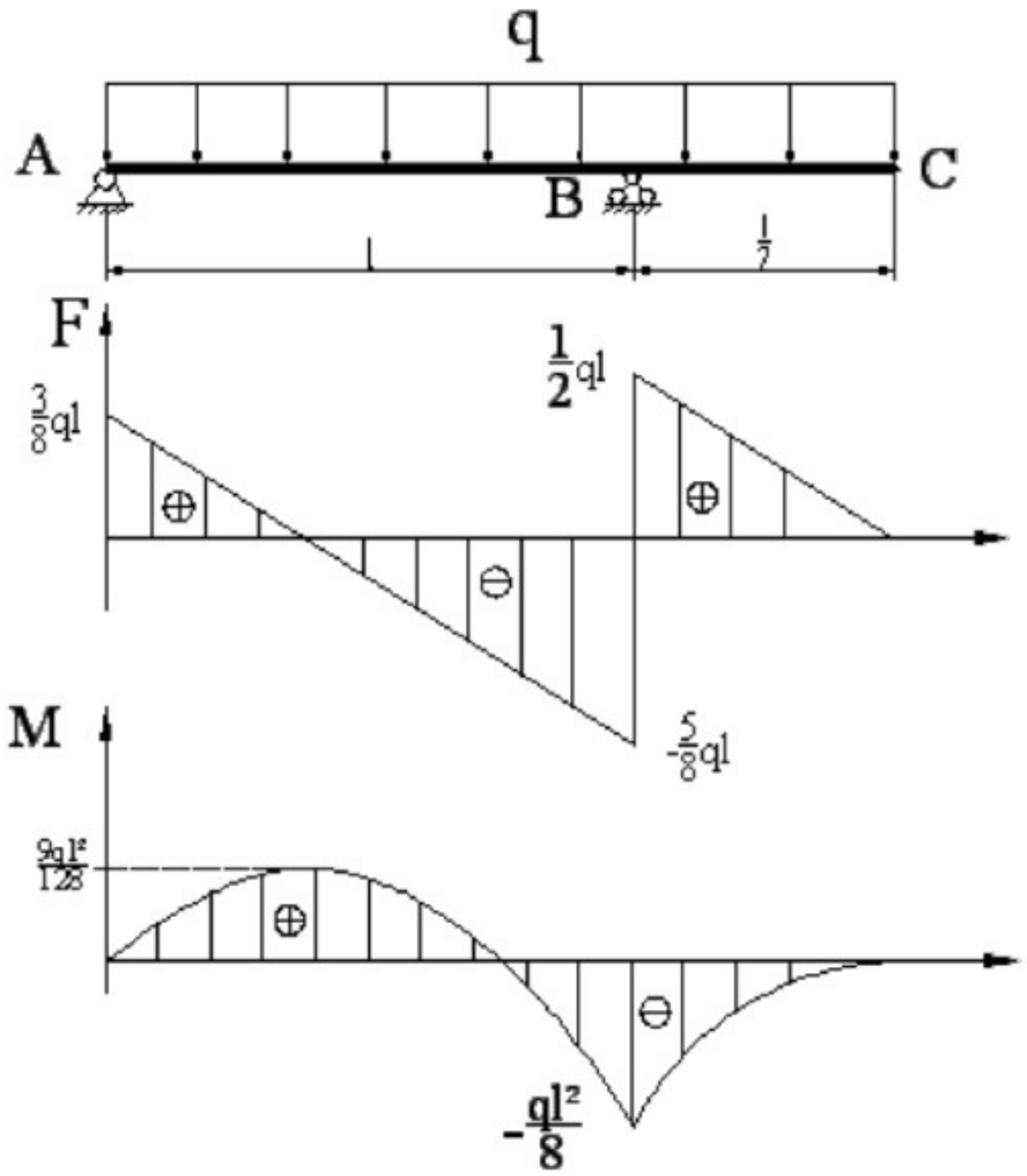


$$F_S(x) = -q(1.5l - x) \quad (l < x \le \frac{3l}{2})$$

$$M(x) = -\frac{q\left(\frac{3l}{2} - x\right)^2}{2} \quad (l \le x \le \frac{3l}{2})$$

（3）作剪力图和弯矩图

(i)



当 $x=\frac{3}{8}l$ 时，

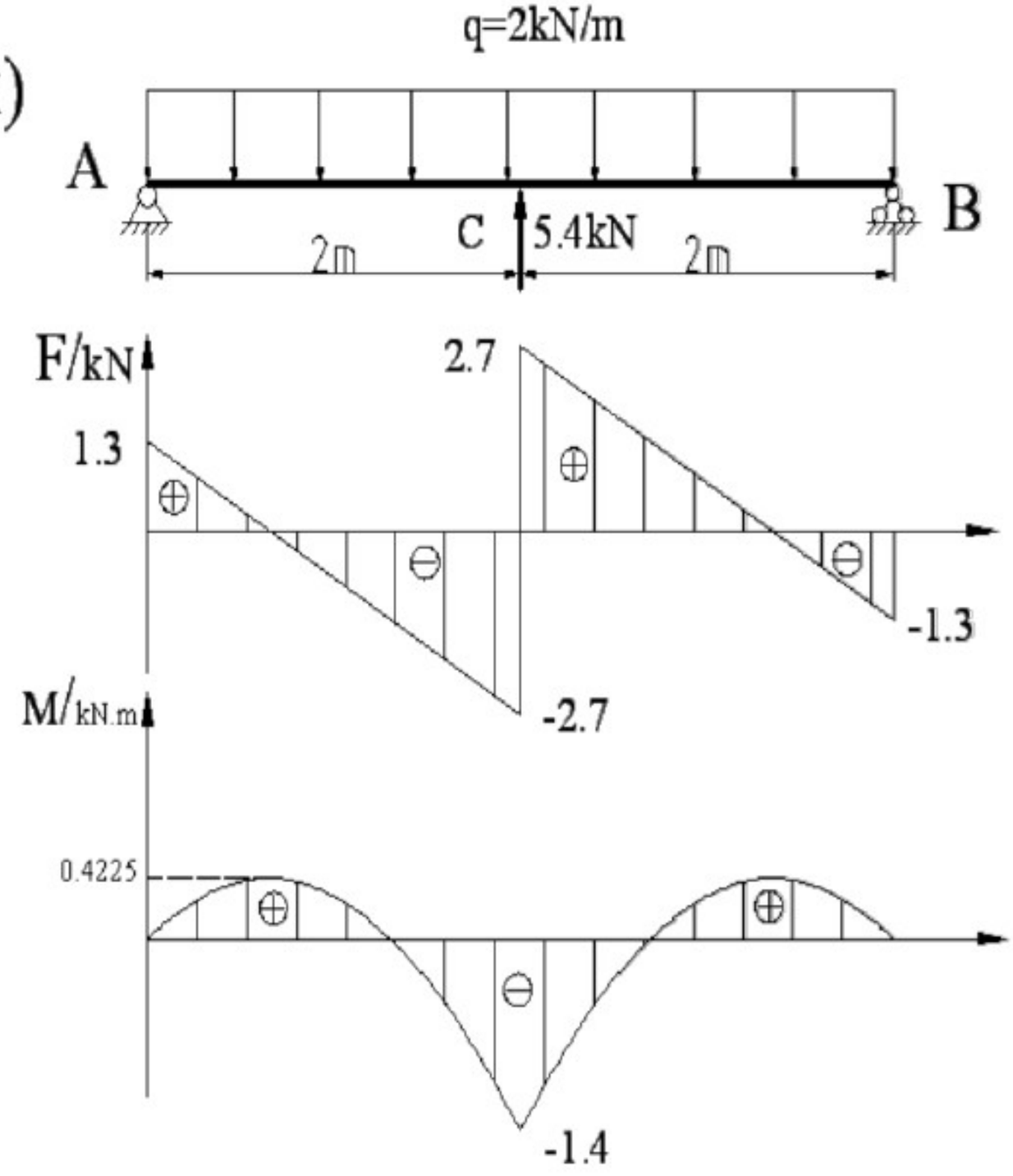
$F_S=0$ $M=\frac{9ql^2}{128}$

当 $x=l$ 时

$M=\frac{ql^2}{8}$

$|F|_{max}=\frac{5ql}{8}$ $|M|_{max}=\frac{ql^2}{8}$

(k)



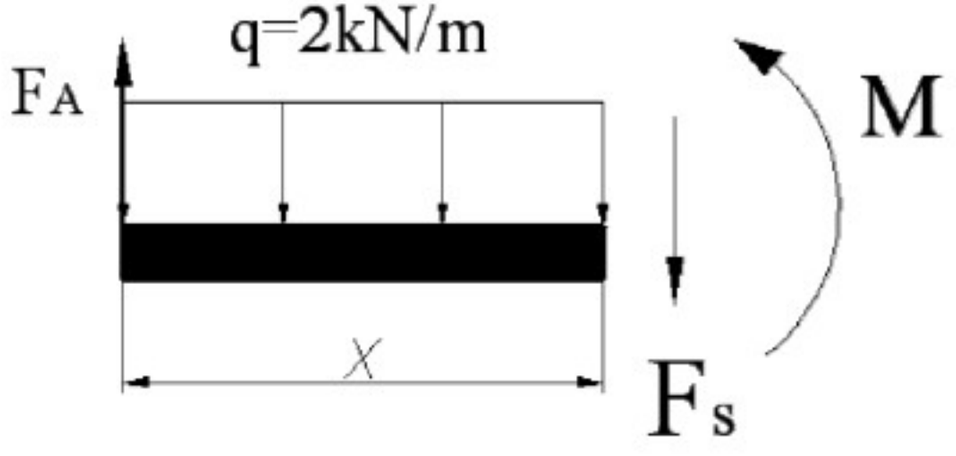
（1）求支反力

由平衡方程∑F=0 ,∑M_A=0

$$\begin{cases} F_A + F_B + F_C = 4q \\ 4F_B + 2F_C - 4q \times 2 = 0 \end{cases}$$

$$F_A = F_B = 1.3\ \text{kN}$$

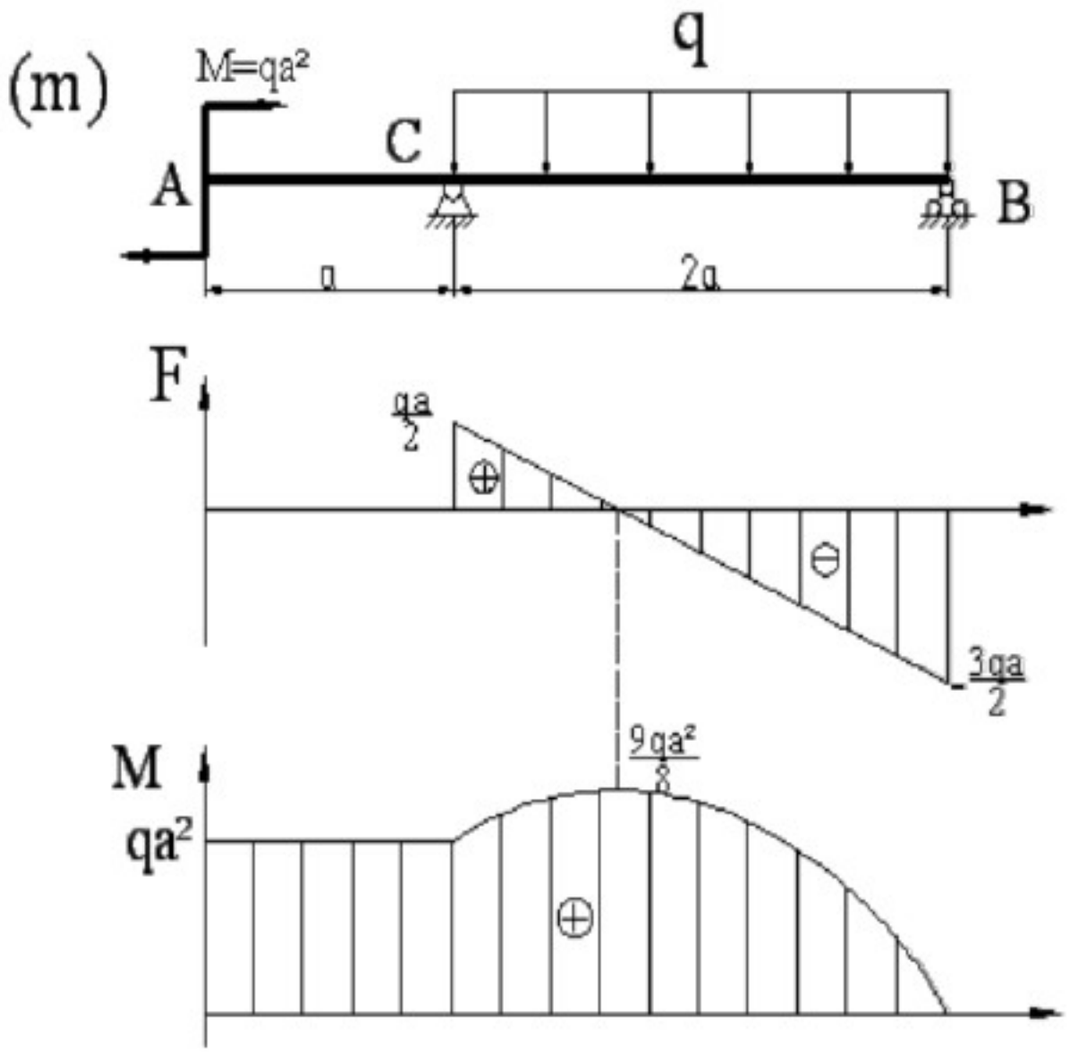
（2）列剪力方程和弯矩方程



$$F_S(x) = F_A - qx \quad (0 < x < 2)$$

$$M(x) = F_A x - \frac{qx^2}{2} \quad (0 \le x < 2)$$

（3）作剪力图和弯矩图



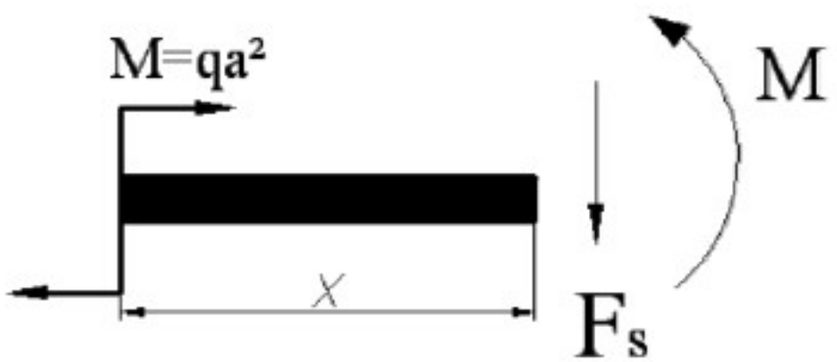
（1）求支反力

由平衡方程∑F=0 ,∑M_C=0

$$\begin{cases} F_B + F_C = 2aq \\ 2aF_B - qa^2 - 2qa^2 = 0 \end{cases}$$

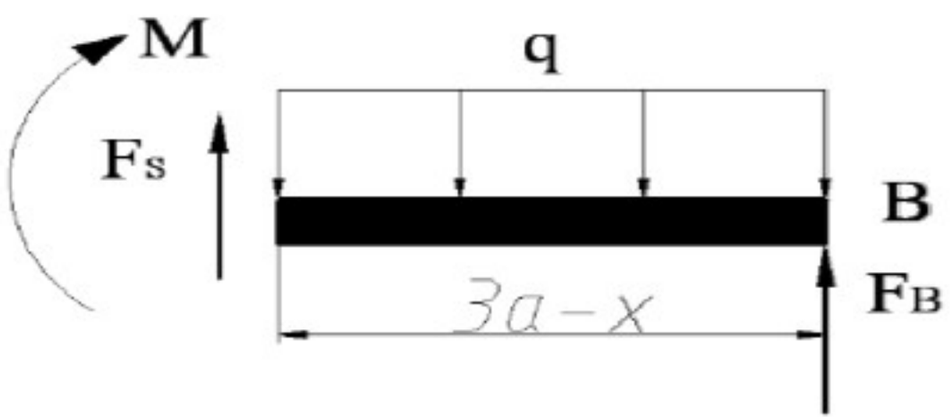
$$F_B = \frac{3}{2}qa \qquad F_C = \frac{aq}{2}$$

（2）列剪力方程和弯矩方程

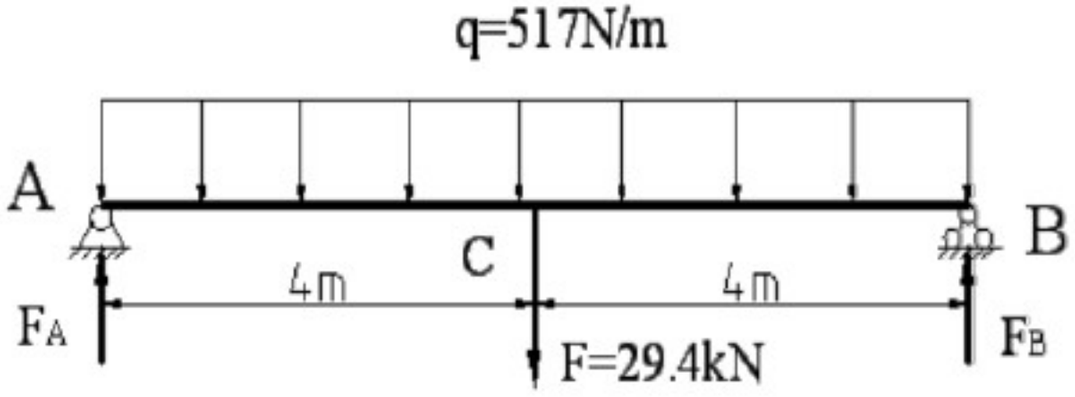


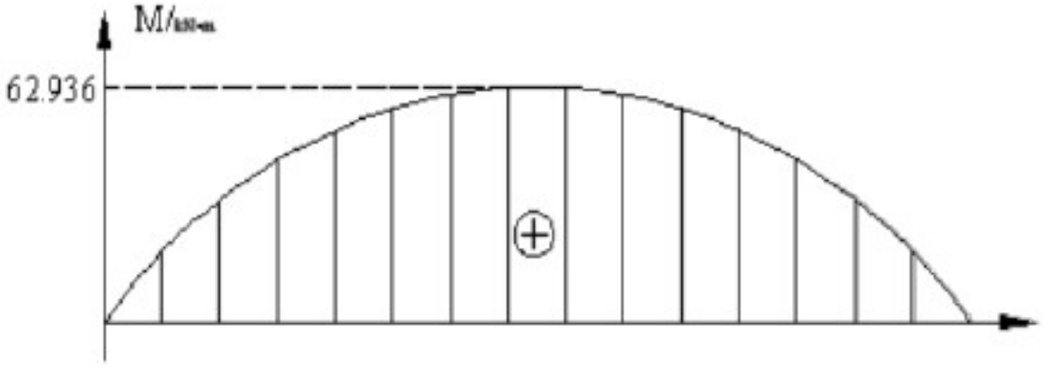
$$\begin{cases} F_S(x) = 0 & (0 < x < a) \\ M(x) = qa^2 & (0 \le x < a) \end{cases}$$

$$\begin{cases} F_S(x) = qx - F_B & (a < x \le 3a) \\ M(x) = F_B(3a - x) - \dfrac{q(3a-x)^2}{2} & (a \le x \le 3a) \end{cases}$$



4-2





（1）求支座反力

$$\sum M_A = 8\times F_B - 0.517\times 8\times 4 - 29.4\times 4 = 0$$

$$F_A = F_B = \frac{8\times 517\times 10^{-3} + 29.4}{2} = 16.768\,\text{kN}$$

(2) 列弯矩方程，求M_{max}

$$\begin{cases} M(x) = F_A x - \dfrac{qx^2}{2} & (0 \le x \le 4) \\ M(x) = F_A x - F(x-4) - \dfrac{qx^2}{2} & (4 \le x \le 8) \end{cases}$$

当 $x = 4$ 时，有 $M_{\max} = 62.936\text{kN}\cdot\text{m}$

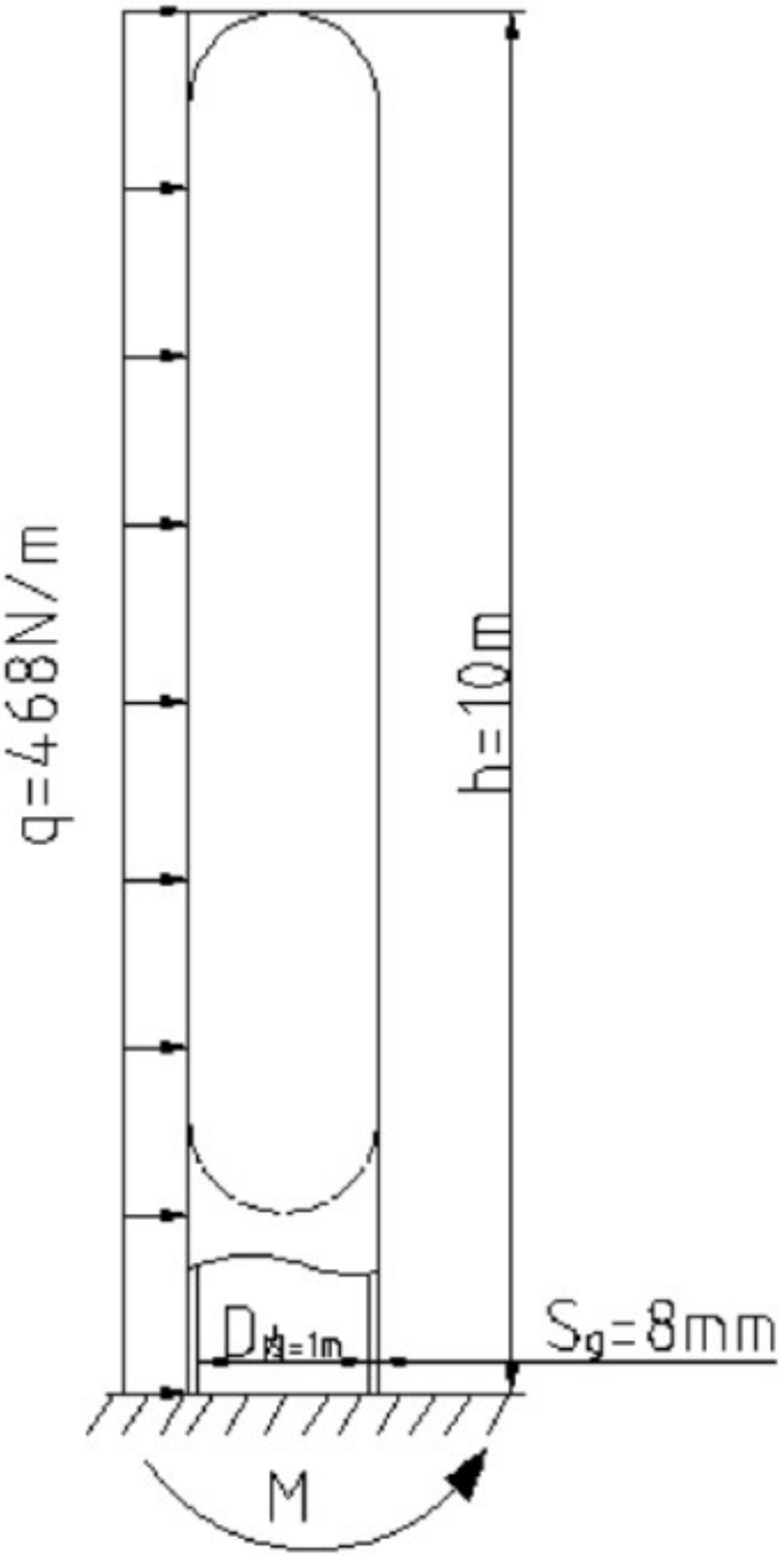
查表4-2，32a工字钢的抗弯截面系数$W_Z=692\times10^{-6}m^3$

$$\sigma_{\max} = \frac{M_{\max}}{W_Z} = \frac{62.936\times10^3}{692\times10^{-6}}$$

$$= 90.95\text{MPa} < [\sigma] = 120\text{MPa}$$

满足要求

4-3



（1）求支座底部弯矩M

$$M = M_{\max} = \frac{qh^2}{2} = 23.4\text{kN}\cdot\text{m}$$

环形截面的抗弯截面系数W

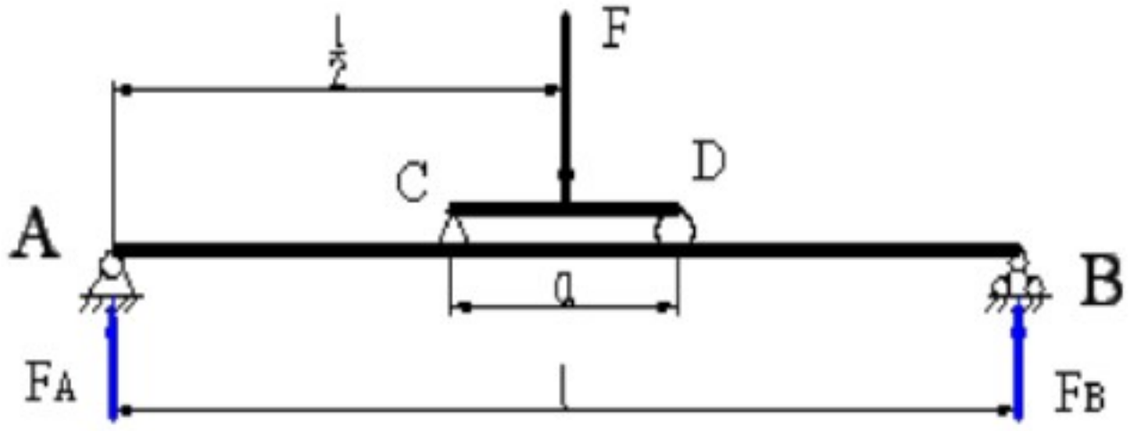
$$\alpha = \frac{D_{内}}{D_{外}} = \frac{1}{1+0.008\times 2}$$

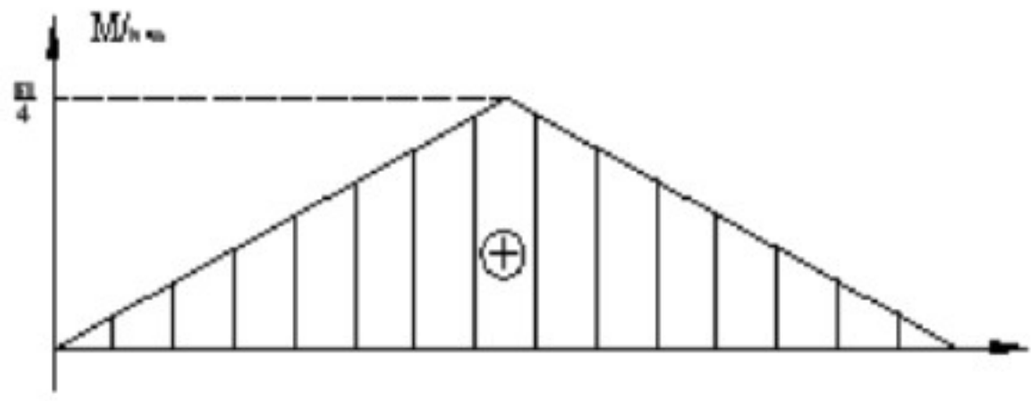
$$\text{W} = \frac{\pi D^3(1-\alpha^4)}{32} = 6.334\times 10^{-3} m^3$$

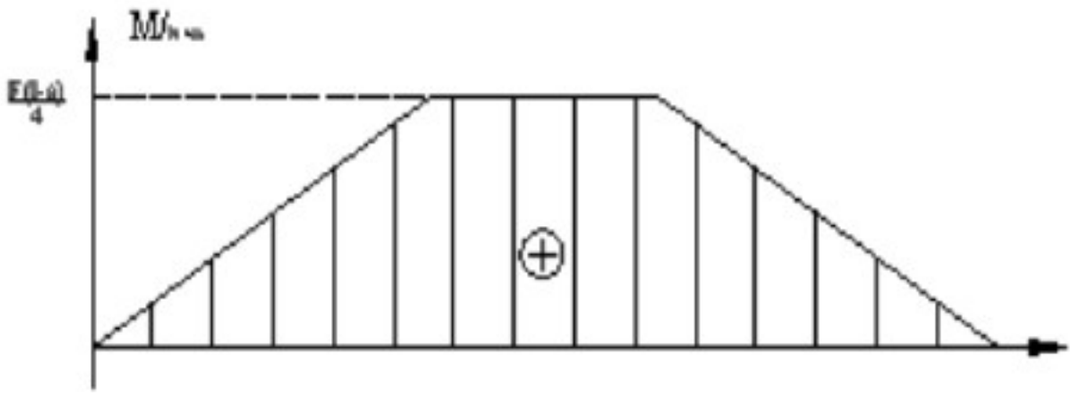
$$\sigma_{\max} = \frac{M_{\max}}{W} = \frac{23.4\times 10^3}{6.334\times 10^{-3}}$$

$$= 3.694\text{MPa}$$

4-5







（1）当P作用在AB梁的中点时

$F_A = F_B = 0.5F$

$$M_{\max} = \frac{F}{2} \times \frac{l}{2} = \frac{Fl}{4}$$

$$\sigma_{\max} = \frac{M_{\max}}{W_Z} = 1.3[\sigma]$$

(2)当P作用在AB梁的中点时

$F_C = F_D = 0.5F$

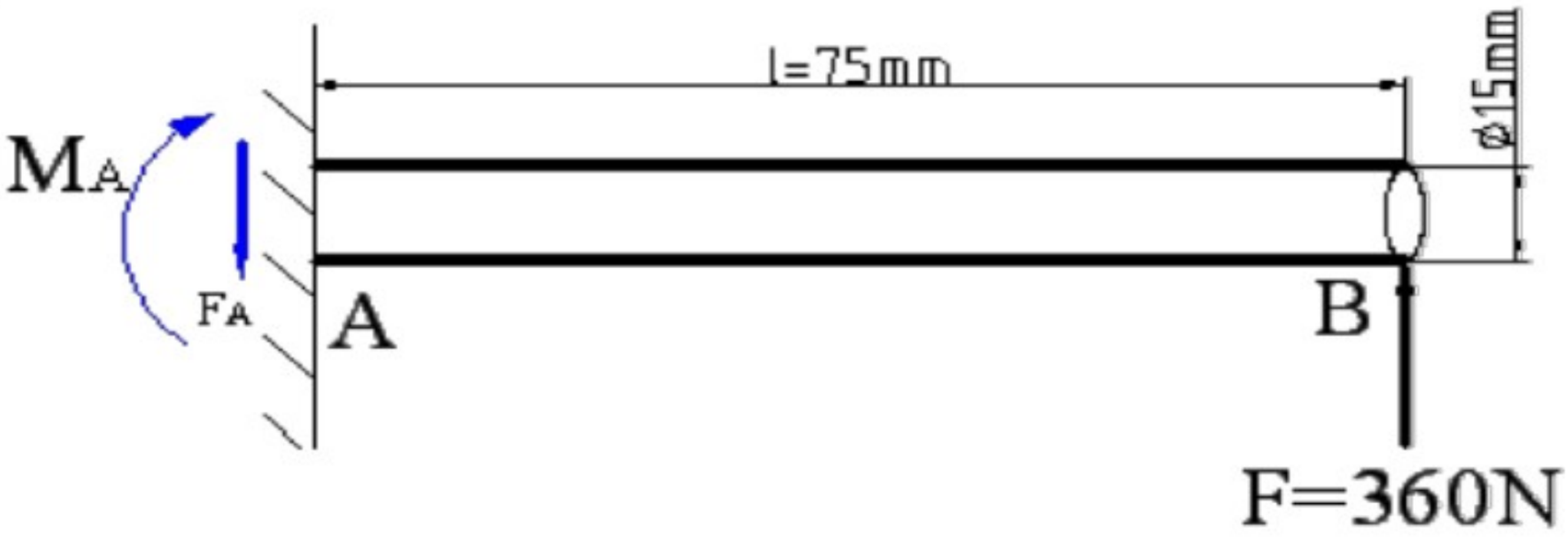
$$M(x)=\begin{cases}\dfrac{\mathrm{F}x}{2} & (0\le x\le 3-\dfrac{a}{2})\\ \dfrac{\mathrm{F}x}{2}-\dfrac{\mathrm{F}}{2}(x-\dfrac{l-a}{2}) & (\dfrac{l-a}{2}\le x\le\dfrac{l+a}{2})\\ \dfrac{\mathrm{F}}{2}(l-x) & (\dfrac{l+a}{2}\le x\le l)\end{cases}$$

$$M'_{\max}=\frac{F(l-a)}{4},\qquad \sigma'_{\max}=[\sigma]$$

从而有 $$\frac{Fl}{4}=1.3\times\frac{F(l-a)}{4}$$

解得 $$a=1.38\mathrm{m}$$

4-7



（1）弯矩方程

$$M(x)=\mathrm{M}_A-F_Ax=F(0.075-x)$$

$$EI\upsilon''=M(x)=F(0.075-x)$$

（2）代入挠曲线近似微分方程并积分

$$EI\upsilon'=0.075Fx-\frac{Fx^2}{2}+C$$

$$EI\upsilon=0.0375Fx^2-\frac{Fx^3}{6}+Cx+D$$

(3)确定积分常数，边界条件为

$x=0,\ \ \nu=0,\ \ \nu'=\theta=0$ 代入上式得

C=0,　D=0

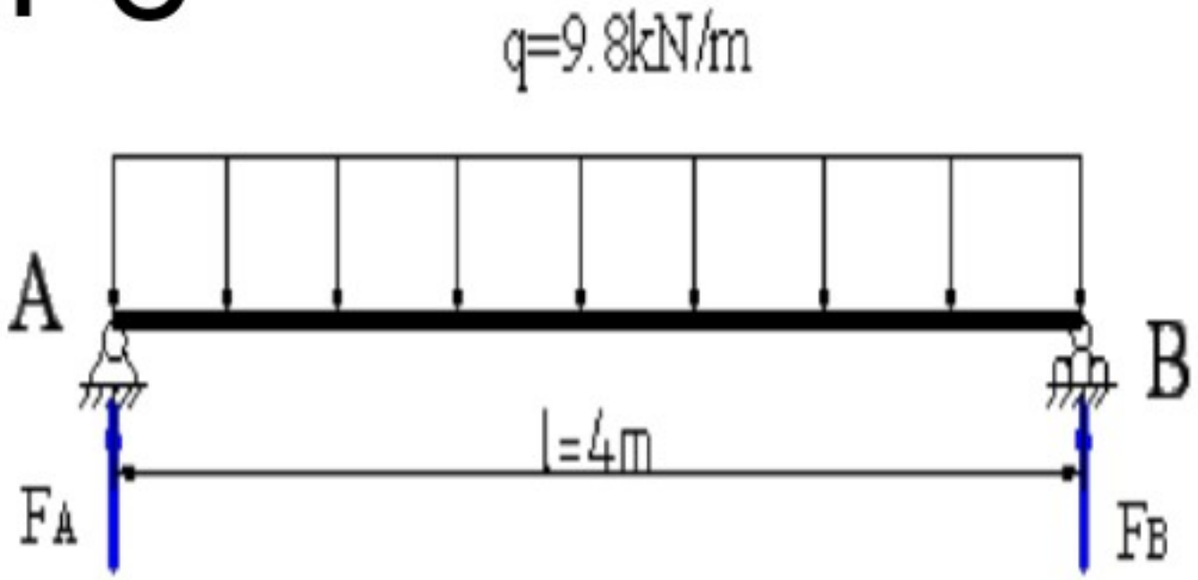
当x=0.075时，挠度最大

$$\nu_{\max}=\frac{0.0375\times360\times0.075^2-60\times0.075^3}{200\times10^9\times\dfrac{\pi\times0.015^4}{64}}=1.02\times10^{-4}\,\mathrm{m}$$

或查表4-3有

$$\nu_{\max}=\frac{Fl^3}{3EI}=\frac{360\times0.075^3}{3\times200\times10^9\times\dfrac{\pi\times0.015^4}{64}}=1.02\times10^{-4}\,\mathrm{m}$$

4-8



（1）求工字钢的抗弯截面系数

$$\sigma_{\max}=\frac{M_{\max}}{W_Z}=\frac{\frac{ql^2}{8}}{W_Z}\le[\sigma]$$

$$\Rightarrow W_Z\ge\frac{ql^2}{8[\sigma]}=\frac{9.8\times10^3\times4^2}{8\times100\times10^6}=196cm^3$$

(2)求工字钢的惯性矩 I_Z

查表4-3

$$[y]=\frac{l}{1000}=|v_{\max}|=\frac{5ql^4}{384EI}$$

$$\Rightarrow I=\frac{5000ql^3}{384E}=3964.4cm^4$$

查表4-2可知，同时满足以上两个条件的工字钢型号为25a，其抗弯截面系数为 $W_Z=402cm^3$

25a工字钢的理论重量为38.105kg/m,考虑自重时，作用在AB梁上的均布载荷为

$$q' = 9.8 + 38.105 \times 9.8 \times 10^{-3} = 10.174\text{kN/m}$$

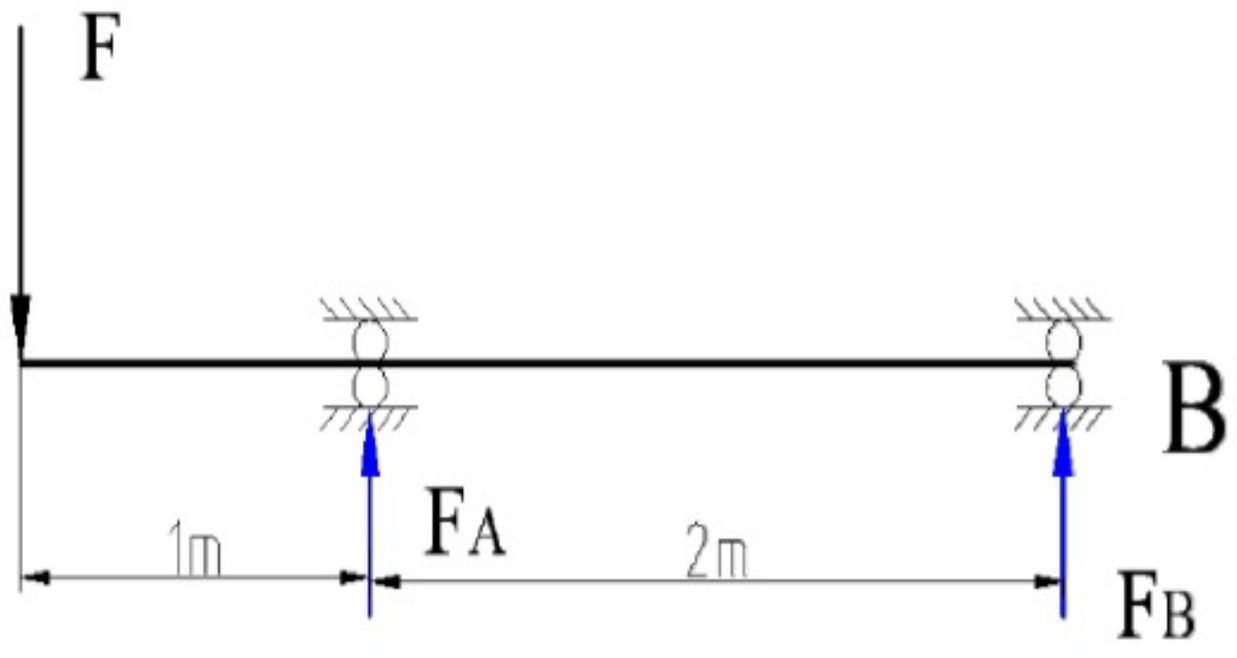
校核应力和挠度

$$\sigma_{\max} = \frac{M_{\max}}{W_Z} = \frac{\frac{q'l^2}{8}}{W_Z} = \frac{10.174 \times 4^2}{8 \times 402 \times 10^{-6}} = 50.61\text{MPa} \le [\sigma]$$

$$\nu_{\max} = \frac{5ql^4}{384EI} = \frac{5 \times 10.174 \times 10^3 \times 4^4}{384 \times 206 \times 10^9 \times 5020 \times 10^{-8}}$$

$$= 3.279 \times 10^{-3}\text{m} \le [y] = \frac{4}{1000}$$

4-9



由$\sum M_A=0$，解得

$F_A=30\text{kN}$，$F_B=-10\text{kN}$

$$M(x)=-Fx \quad (0\le x\le 1)$$

$$M(x)=F_A(x-1)-Fx \quad (1\le x\le 3)$$

当 $0\le x\le 1$ 时

$$EIv''=-20x$$

$$EIv'=-10x^2+C_1$$

$$EIv=-\frac{10}{3}x^3+C_1x+D_1$$

当 $1\le x\le 3$ 时

$$EIv''=10x-30$$

$$EIv'=5x^2-30x+C_2$$

$$EIv=\frac{5}{3}x^3-15x^2+C_2x+D_2$$

代入边界条件：

$$在x=1,x=3处，\quad \nu=0$$

由连续性方程可知，在 x=1处的转角 $\nu'_{左}=\nu'_{右}$

$$-10x^2+C_1=EI\nu'=5x^2-30x+C_2$$

联立求解得：

$$C_1=\frac{70}{3},\mathrm{D}_1=-20,\quad C_2=\frac{115}{3},D_2=-25$$

在x=1处，所允许的最大转角为

：

$$\nu'_{\max}=[\theta]=0.5°$$

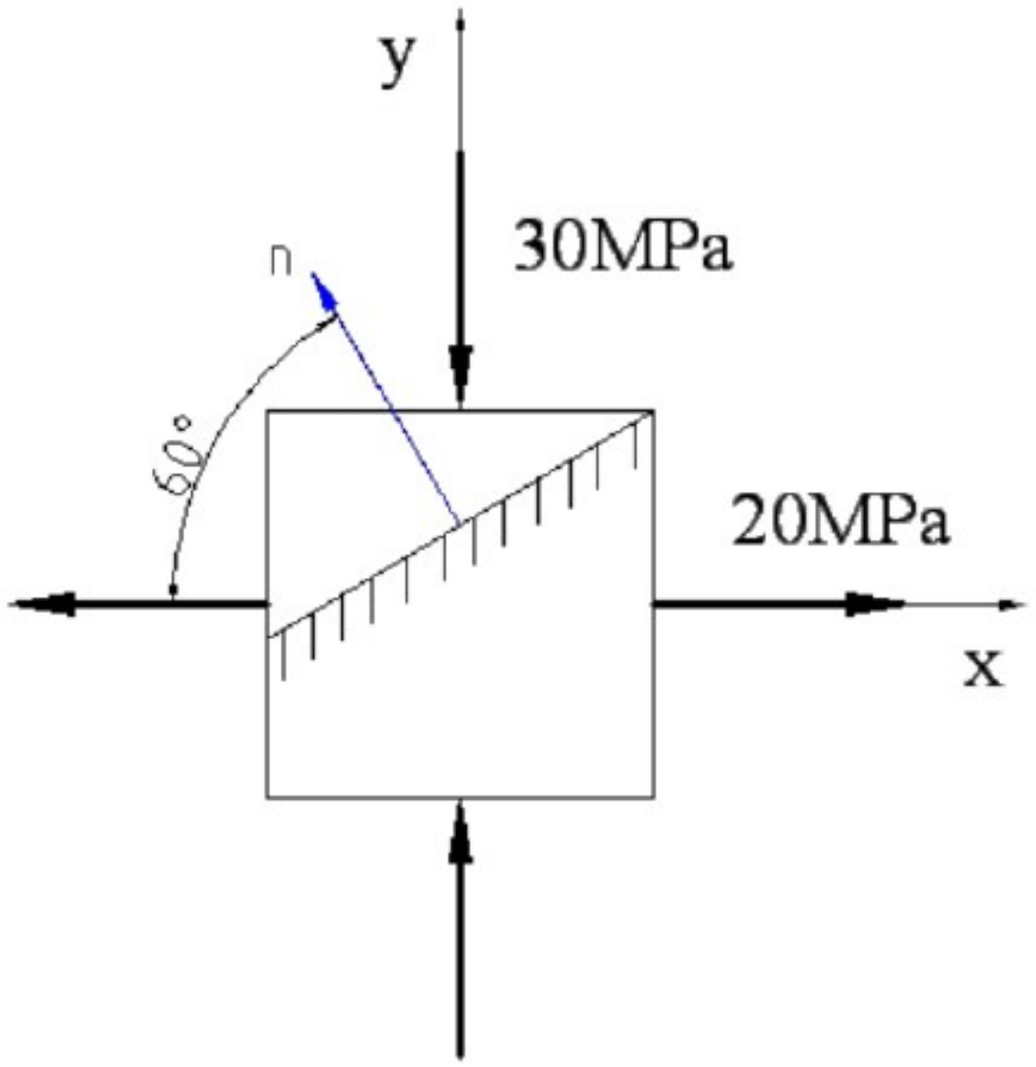
$$EI\nu'_{\max} = -10 + \frac{70}{3} = \frac{40}{3}$$

$$\Rightarrow I = \frac{\pi D^4}{64} = \frac{40\times10^3}{200\times10^9} \times \frac{180}{0.5\pi} = 7.64\times10^{-6}\, m^4$$

$$D \geq \sqrt[4]{\frac{7.64\times10^{-6}\times64}{\pi}} = 0.11169 \text{ m}$$

求解得 $D \geq 111.69$ mm$\approx$**112mm**

5-1、求以下单元体m-m斜截面上的正应力和切应力。



$$(a)\quad \sigma_x = 20\text{MPa},\ \sigma_y = -30\text{MPa}$$

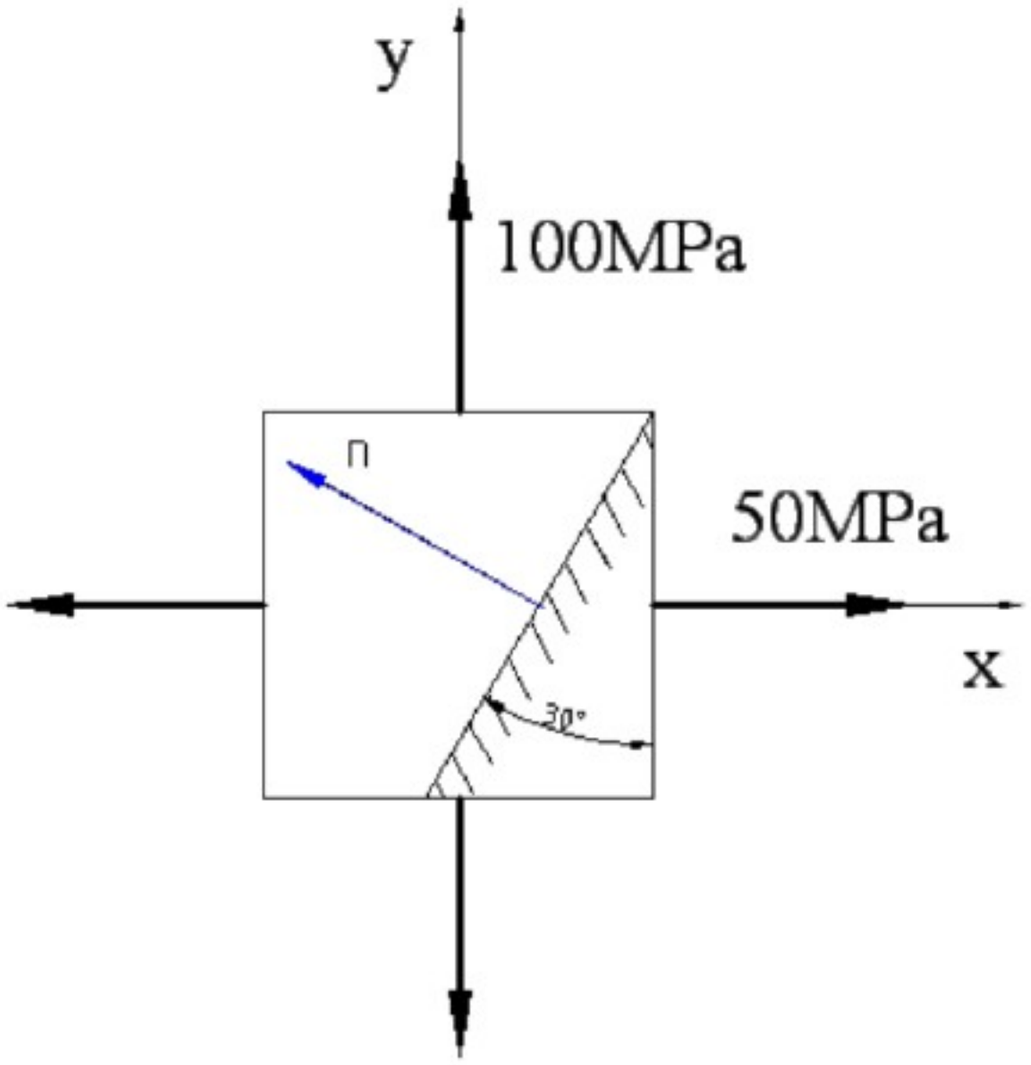
$$\tau_{xy} = 0,\qquad \alpha = 120°$$

代入公式解得：

$$\begin{cases} \sigma_\alpha = \dfrac{\sigma_x + \sigma_y}{2} + \dfrac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha \\ \tau_\alpha = \dfrac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha \end{cases}$$

$$\sigma_\alpha = -17.5\text{MPa}$$

$$\tau_\alpha = -21.65\text{MPa}$$



$$(b)\quad \sigma_x = 50\text{MPa},\ \sigma_y = 100\text{MPa}$$

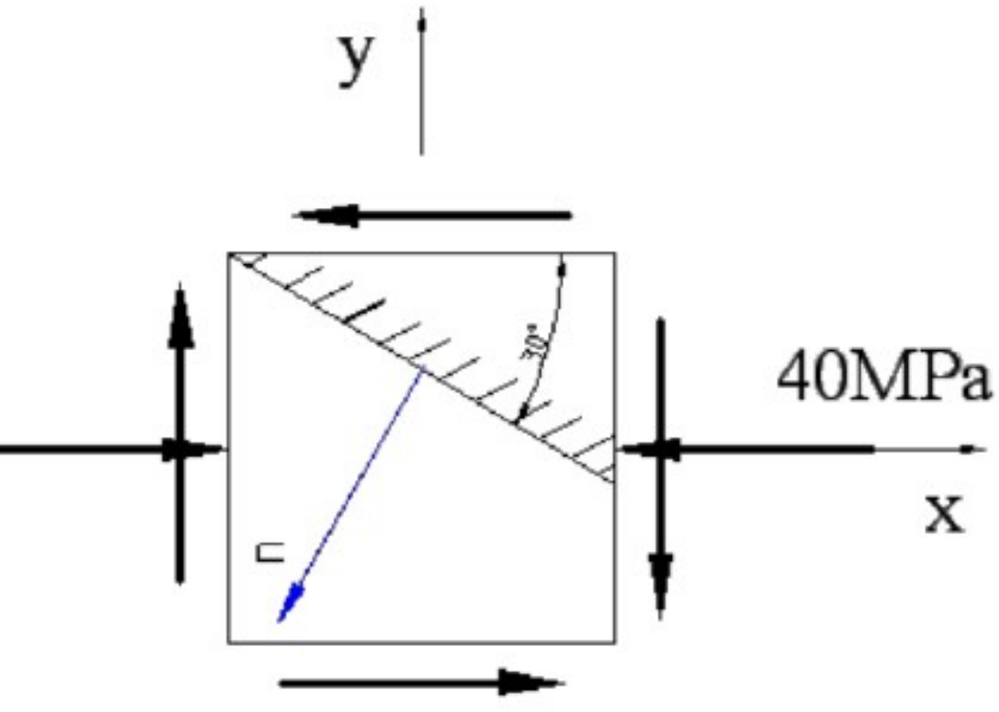
$$\tau_{xy} = 0,\qquad \alpha = 150°$$

代入公式解得：

$$\begin{cases} \sigma_\alpha = \dfrac{\sigma_x + \sigma_y}{2} + \dfrac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha \\ \tau_\alpha = \dfrac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha \end{cases}$$

$$\sigma_\alpha = 62.5\text{MPa}$$

$$\tau_\alpha = 21.65\text{MPa}$$



(c) $\sigma_x = -40\text{MPa},\ \sigma_y = 0\text{MPa}$

$\tau_{xy} = 20\text{MPa},\quad \alpha = -120°$

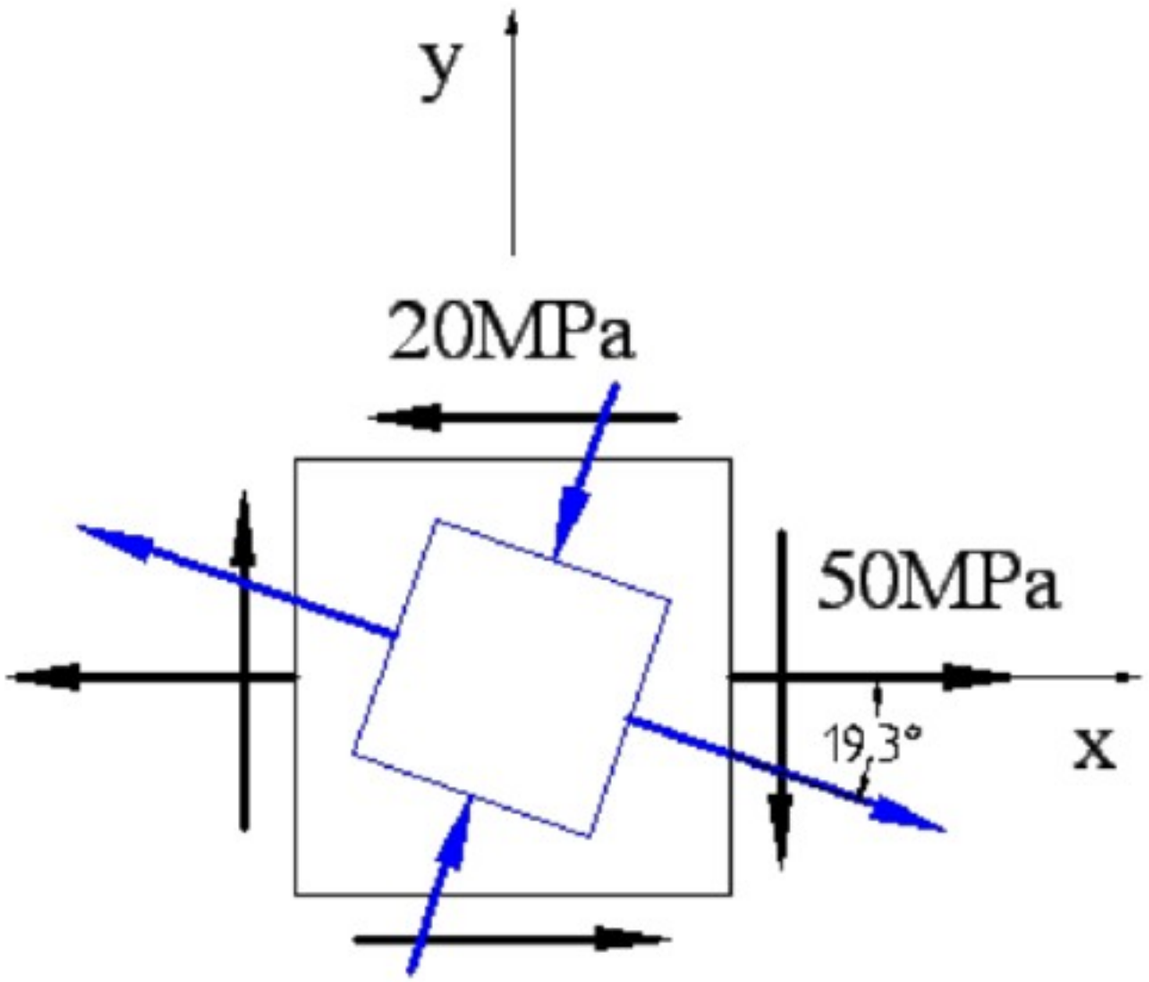
代入公式解得：

$$\begin{cases} \sigma_\alpha = \dfrac{\sigma_x + \sigma_y}{2} + \dfrac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha \\ \tau_\alpha = \dfrac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha \end{cases}$$

$$\sigma_\alpha = -27.32\text{MPa}$$

$$\tau_\alpha = -27.32\text{MPa}$$

5-2、求以下单元体内主应力的大小及方向，并在它们中间绘出仅受主应力作用的单元体。



(a) $\sigma_x = 50\text{MPa},$

$\sigma_y = 0\text{MPa}$

$\tau_{xy} = 20\text{MPa},$

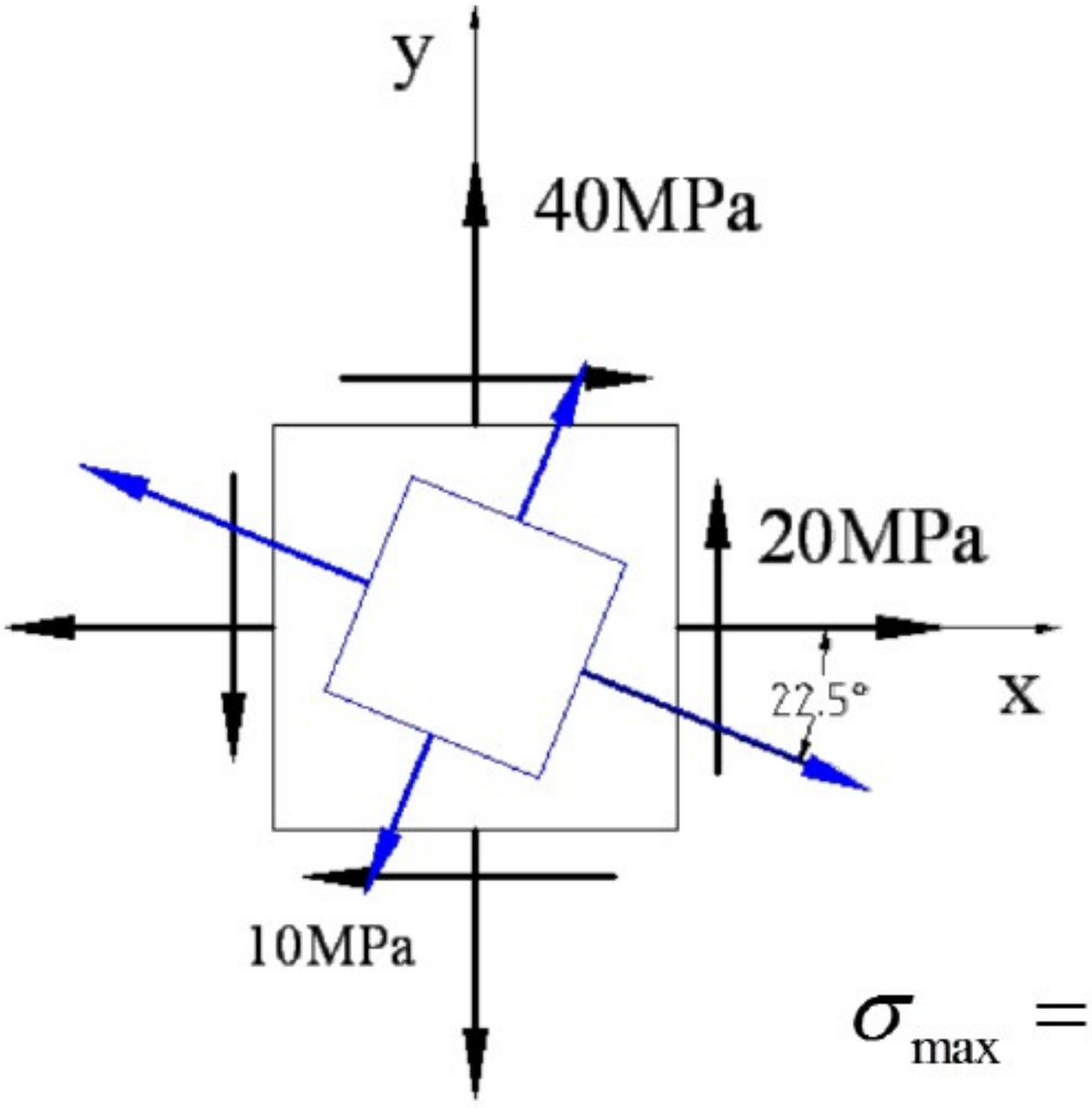
代入公式解得：

$$\begin{cases} \sigma_{\max} = \dfrac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ \sigma_{\min} = \dfrac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \end{cases}$$

$$\sigma_{\max} = 57.02\text{MPa} \quad \sigma_{\min} = -7\text{MPa}$$

$$\tan 2\alpha = -\frac{2\tau_{xy}}{\sigma_x - \sigma_y} = -0.8$$

$$\Rightarrow \alpha_{\sigma_{\max}} = -19.3°, \quad \alpha_{\sigma_{\min}} = 70.7°$$



代入公式解得：

$$\begin{cases}\sigma_{\max}=\dfrac{\sigma_x+\sigma_y}{2}+\sqrt{\left(\dfrac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}\\ \sigma_{\min}=\dfrac{\sigma_x+\sigma_y}{2}-\sqrt{\left(\dfrac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}\end{cases}$$

$\sigma_{\max}=44.14\text{MPa}\quad \sigma_{\min}=15.86\text{MPa}$

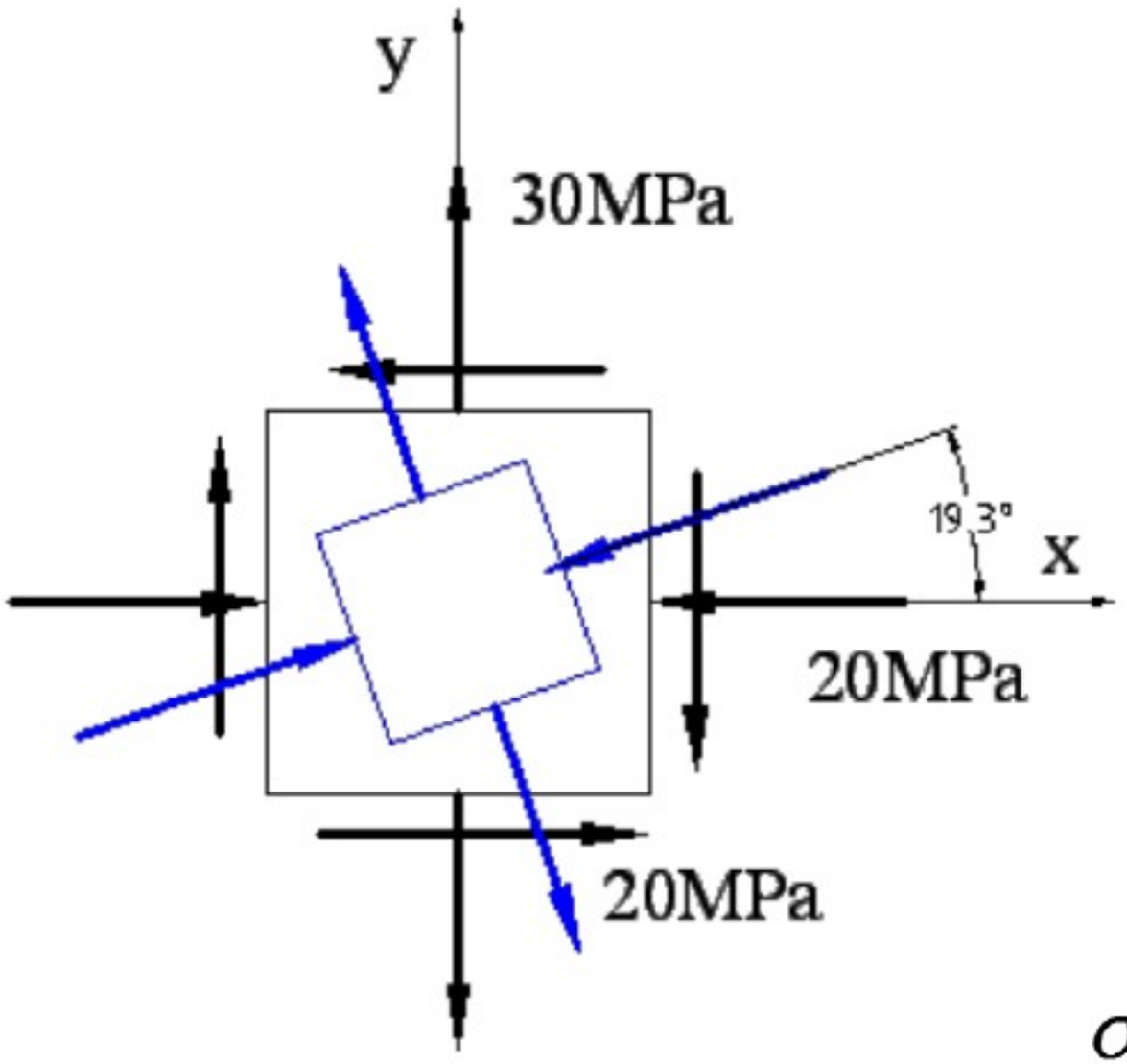
(b) $\sigma_x=20\text{MPa},$

$\sigma_y=40\text{MPa}$

$\tau_{xy}=-10\text{MPa},$

$$\tan 2\alpha=-\frac{2\tau_{xy}}{\sigma_x-\sigma_y}=-1$$

$$\Rightarrow \alpha_{\sigma_{\max}}=67.5°,\ \alpha_{\sigma_{\min}}=-22.5°$$



代入公式解得：

$$\begin{cases} \sigma_{\max} = \dfrac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ \sigma_{\min} = \dfrac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \end{cases}$$

$$\sigma_{\max} = 37\text{MPa} \quad \sigma_{\min} = -27\text{MPa}$$

(c) $\sigma_x = -20\text{MPa},$

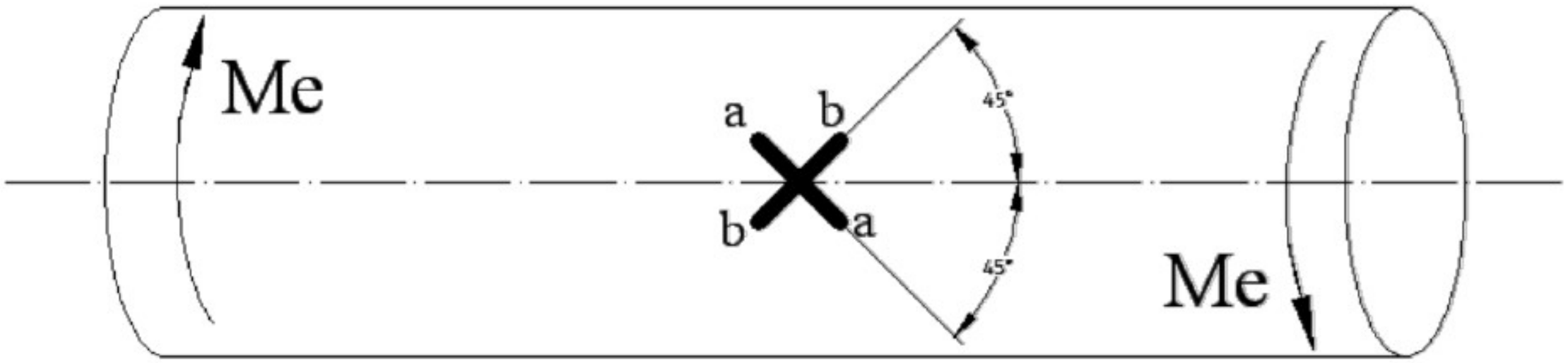
$\sigma_y = 30\text{MPa}$

$\tau_{xy} = 20\text{MPa},$

$$\tan 2\alpha = -\frac{2\tau_{xy}}{\sigma_x - \sigma_y} = 0.8$$

$$\Rightarrow \alpha_{\sigma_{\max}} = -70.66°, \quad \alpha_{\sigma_{\min}} = 19.3°$$

5-3



受扭圆轴的应力状态为纯剪切，其剪切力为：$\tau = \dfrac{\mathrm{Me}}{W_t}$

因此有：　$\sigma_x = 0,\ \ \sigma_y = 0,\ \ \tau_{xy} = \tau$

代入公式求出主应力　$\left.\begin{matrix}\sigma_{\max}\\ \sigma_{\min}\end{matrix}\right\} = \dfrac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$$\sigma_1 = \tau,\ \sigma_2 = 0,\ \sigma_3 = -\tau$$

由主应变和主应力的关系式有

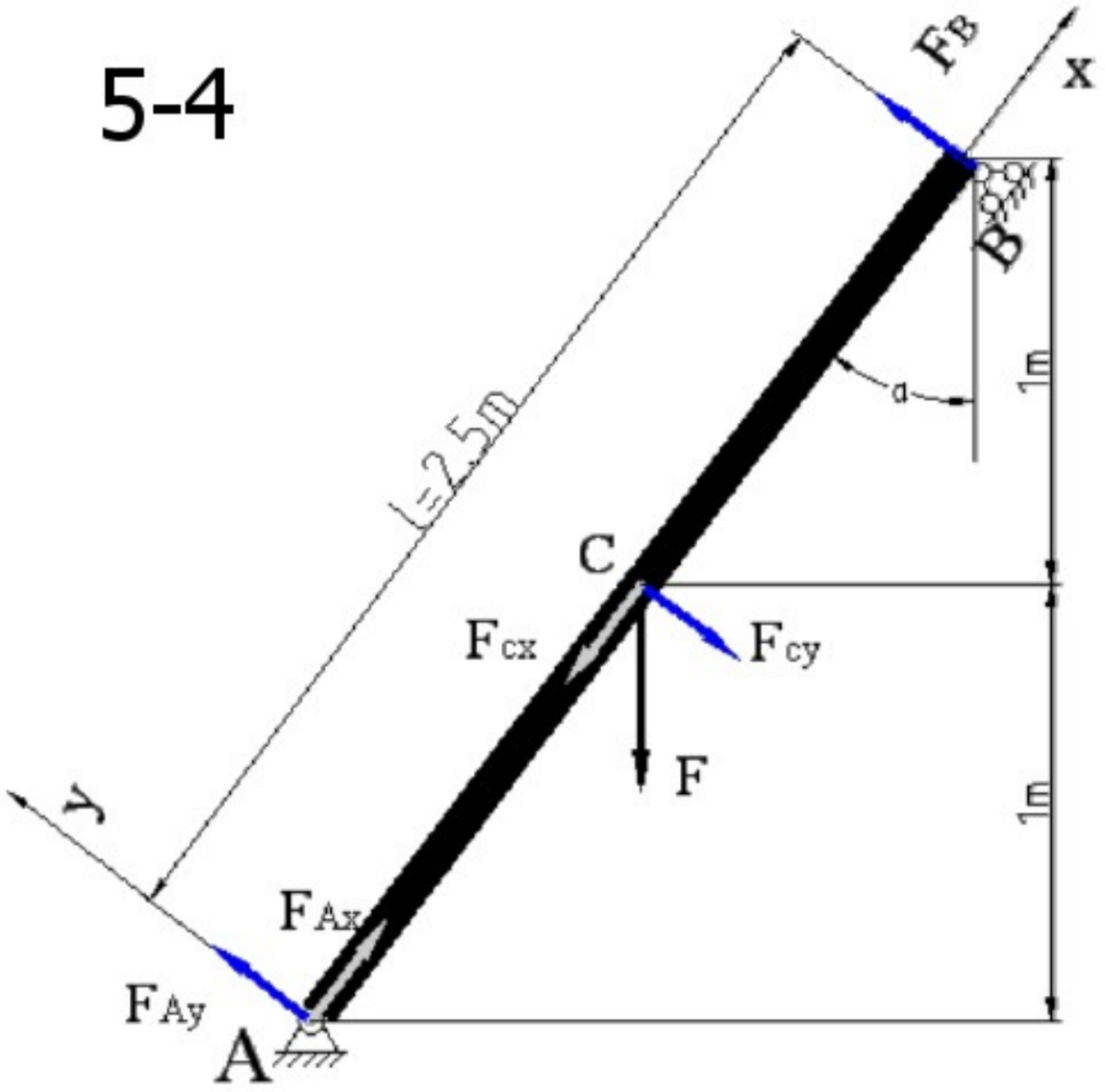
$$\begin{cases} \varepsilon_1 = \dfrac{1}{E}\left[\sigma_1 - \mu\left(\sigma_2 + \sigma_3\right)\right] = 425\times10^{-6} \\ \varepsilon_3 = \dfrac{1}{E}\left[\sigma_3 - \mu\left(\sigma_1 + \sigma_2\right)\right] = -425\times10^{-6} \end{cases}$$

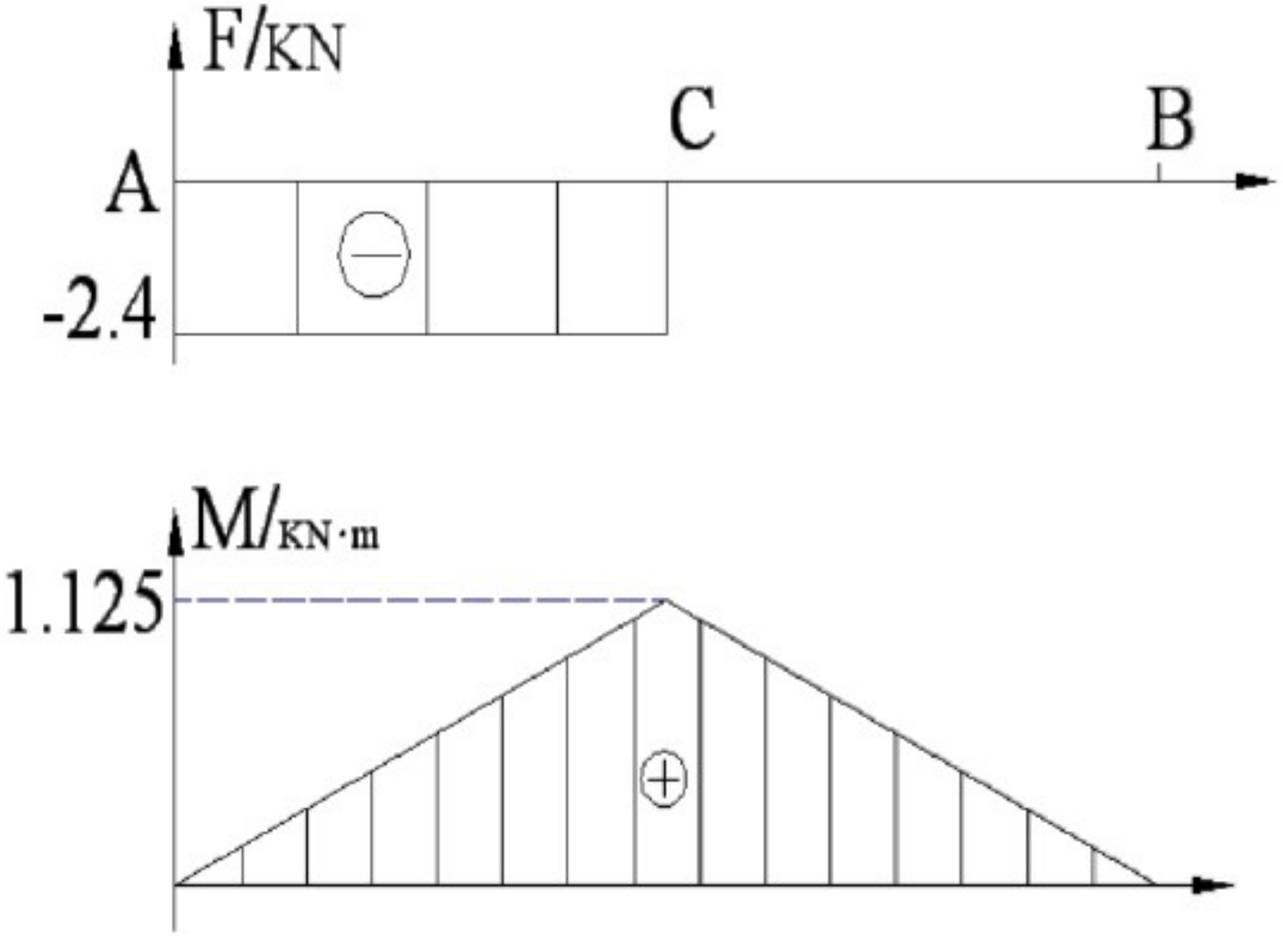
代入已知条件：$\sigma_1 = \tau,\ \sigma_2 = 0,\ \sigma_3 = -\tau,\ \mu$=0.3

解得：$\tau = 67.673\text{MPa}$

$$\text{Me}=\tau \cdot W_t = 67.673\times10^6 \times \frac{\pi \times 0.2^3}{16} = 106\text{KN}\cdot\text{m}$$

5-4





建立x-y坐标轴，如图所示

将力F分解

$$F_{Cx} = F\cos\alpha = 2.4\text{KN}$$

$$F_{Cy} = F\sin\alpha = 1.8\text{KN}$$

作轴力图和弯矩图

由F_{Cx}引起的压应力为

$$\sigma_c' = \frac{F_{Cx}}{A} = \frac{2.4\times10^3}{0.1^2} = 0.24\text{MPa}$$

由F_{cy}引起的拉、压应力为：

$$\left.\begin{matrix}\sigma_t \\ \sigma_c''\end{matrix}\right\} = \pm\frac{M_{\max}}{W} = \pm\frac{0.9\times10^3\times1.25}{\frac{0.1^3}{6}} = \pm 6.75\text{MPa}$$

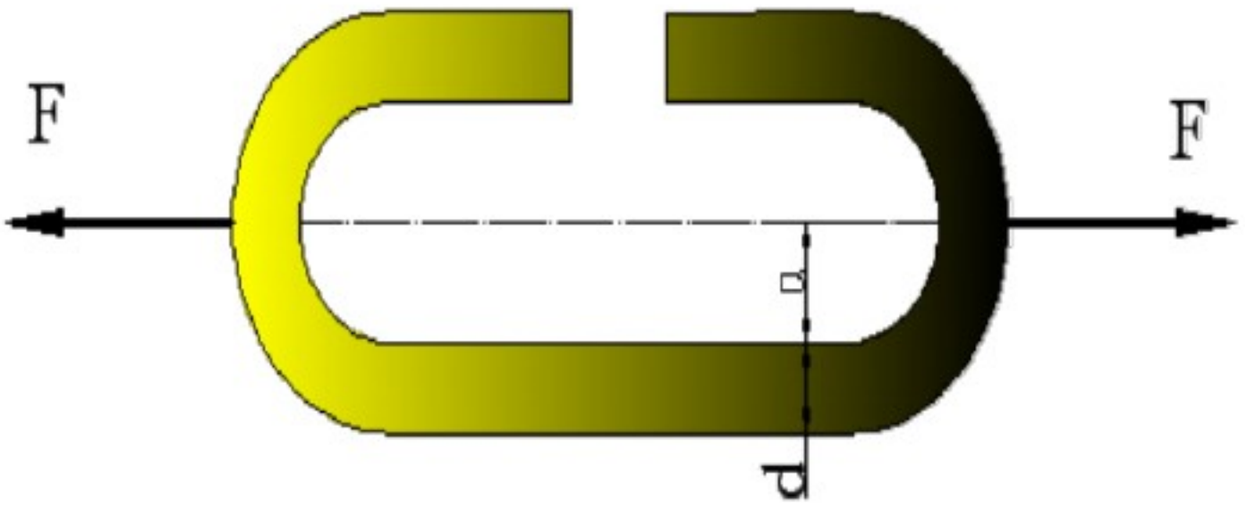
故最大压应力为：

$$\sigma_c = \sigma_c' + \sigma_c'' = 0.24 + 6.75 = 6.99\text{MPa}$$

最大的拉应力为：

$$\sigma_t = 6.75\text{MPa}$$

5-5、已知条件a=60mm，d=50mm，





$$\sigma_{t\max}=\frac{F}{A}+\frac{M_{\max}}{W}=\frac{F}{\frac{\pi \mathrm{d}^2}{4}}+\frac{F\left(a+\frac{d}{2}\right)}{\frac{\pi d^3}{32}}\le[\sigma]=120\mathrm{MPa}$$

$$\Rightarrow\quad F\le\frac{120\times10^6\times\pi\times0.05^3}{32\times0.085+4\times0.05}=16.138\mathrm{KN}$$

5-8、解：

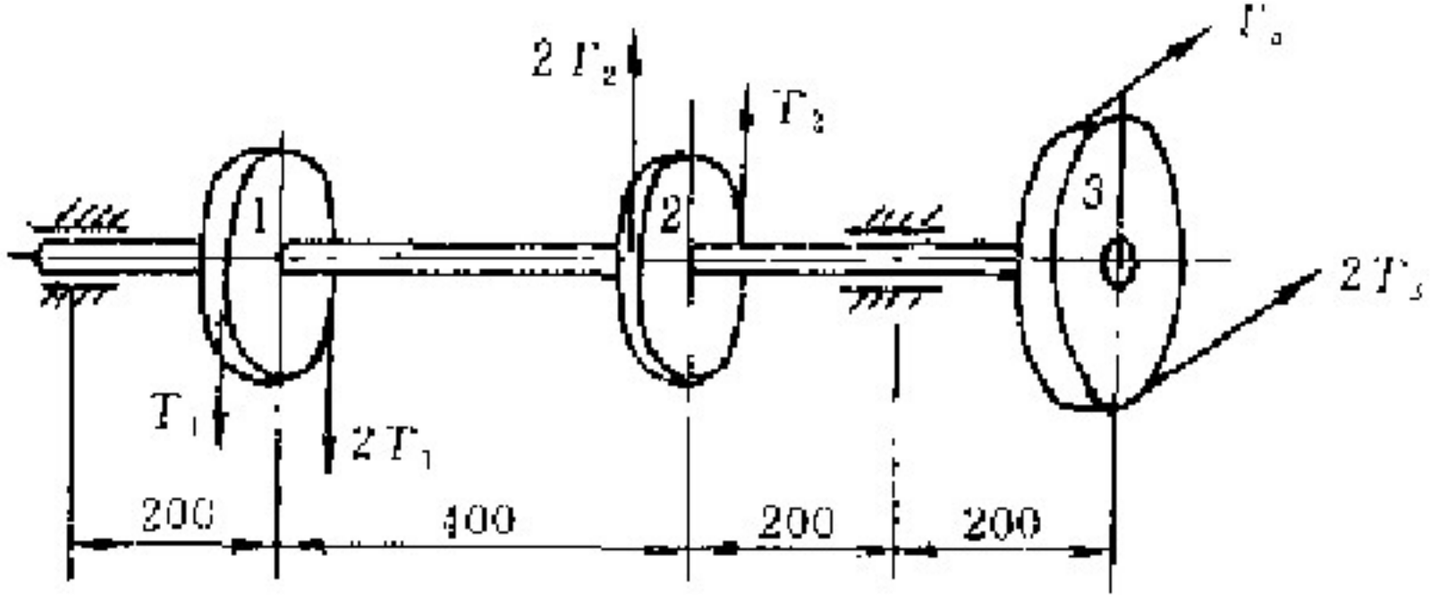
$F_{T1}=F_{T2}=1.5\text{kN}$

$D_{1轮}=D_{2轮}=300\text{mm}$

$D_{3轮}=450\text{mm}$，

$d_{轴}=60\text{mm}$

许用应力$[\sigma]=80\text{MPa}$



由$M_y=0$，$M_z=0$,$M_1+M_2=M_3$求得

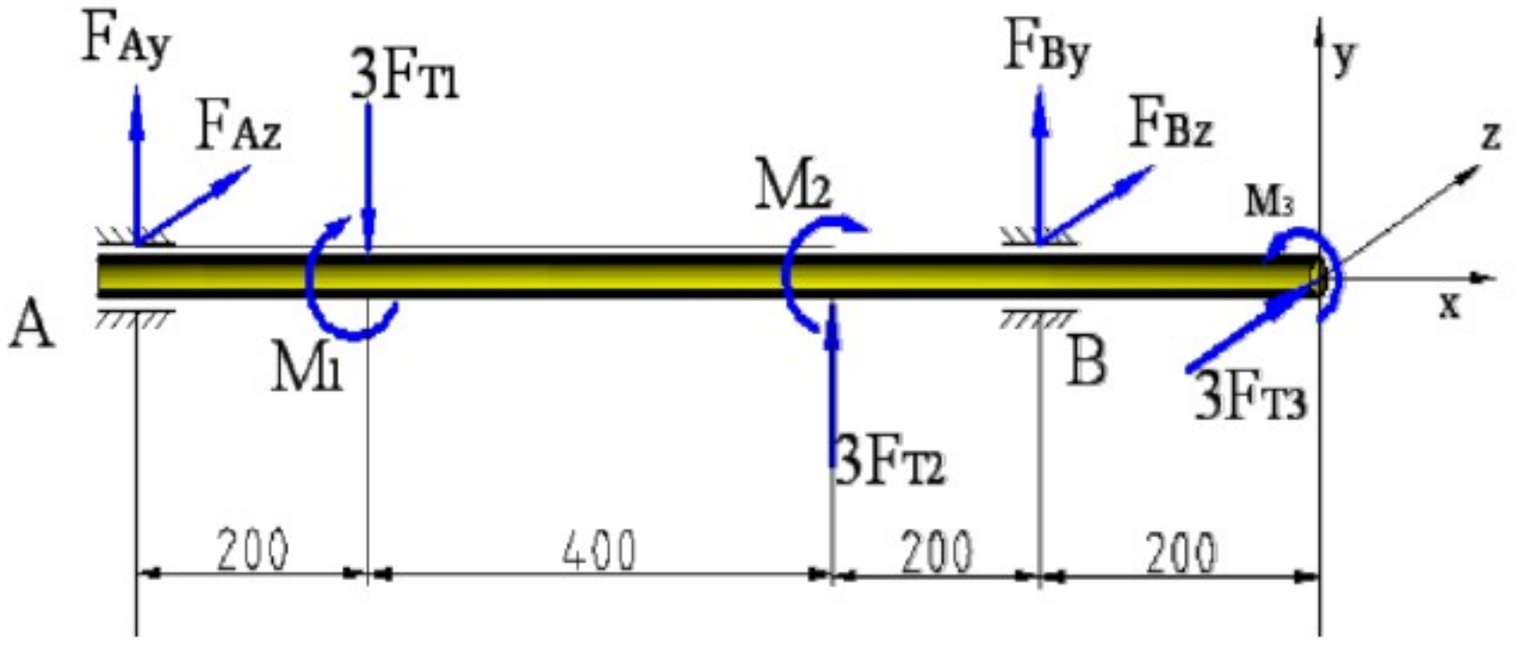
$F_{Ay}=2.25\text{kN}$

$F_{Az}=1.5\text{kN}$

$F_{By}=-2.25\text{kN}$

$F_{Bz}=-7.5\text{kN}$

$F_{T3}=2\text{kN}$，



$M_1=M_2=0.225\text{kN}\cdot\text{m}$，$M_3=0.45\text{kN}\cdot\text{m}$

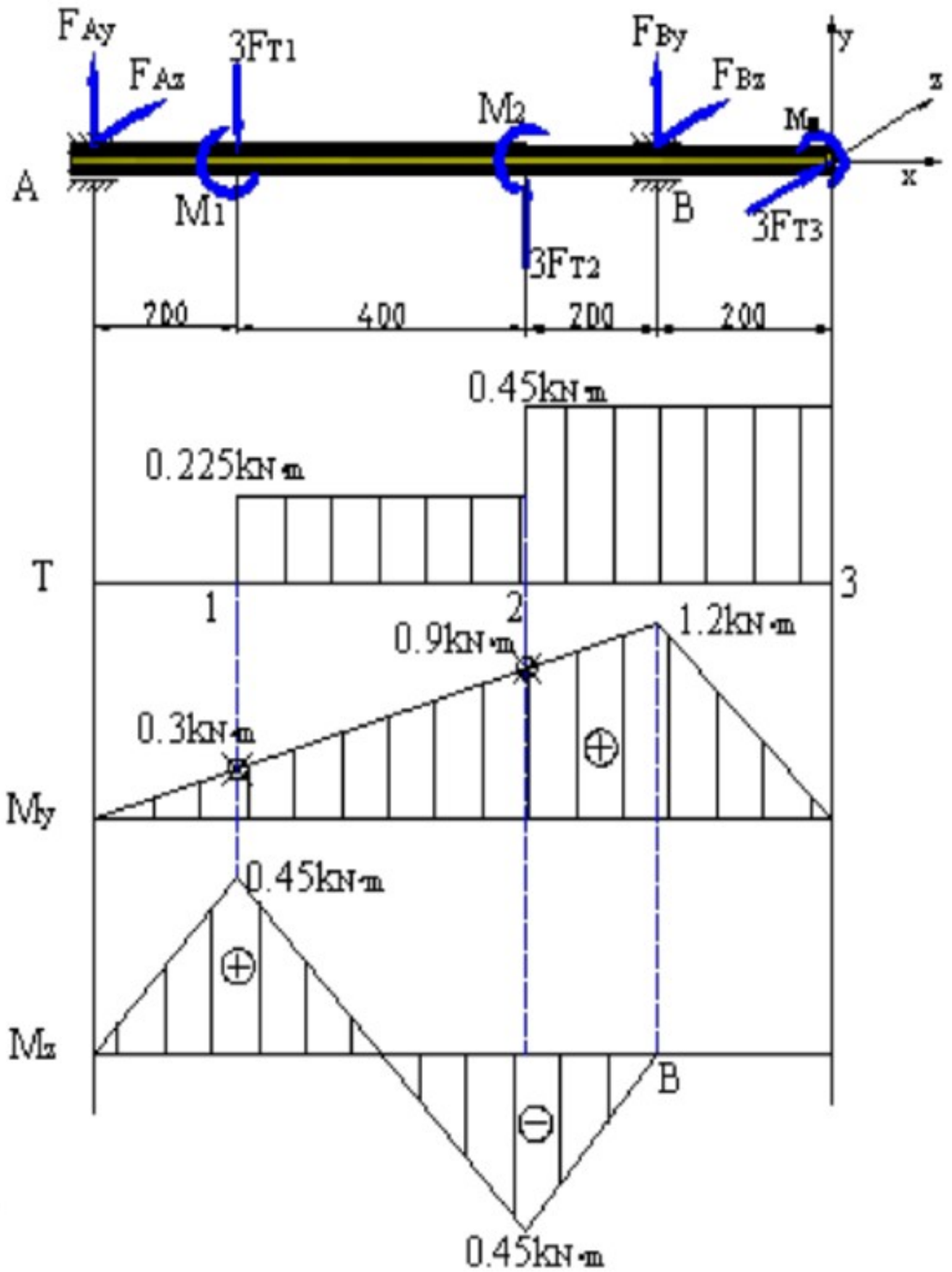
作出各个弯矩图并判断危险截面

1、2、B点处的合力矩为：

$$M_1' = \sqrt{M_{y1}^2 + M_{z1}^2} = 0.54\text{kN}\cdot\text{m}$$

$$M_2' = \sqrt{M_{y2}^2 + M_{z2}^2}$$
$$= 1.006\text{kN}\cdot\text{m}$$

$$M_3' = \sqrt{M_{y3}^2 + M_{z3}^2}$$
$$= 1.2\text{kN}\cdot\text{m}$$



由合力矩可知：B点最危险

圆轴的抗弯截面系数为：

$$W=\frac{\pi d^3}{32}=2.12\times10^{-5}\mathrm{m}^3$$

代入B点处的数值
根据第三强度理论有：

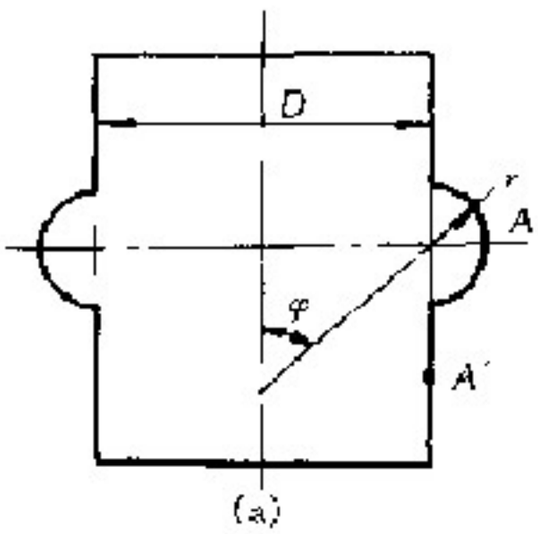
$$\sigma_{r3}=\frac{1}{W}\sqrt{M^2+T^2}=60.45\mathrm{MPa}\le[\sigma]$$

根据第四强度理论有：

$$\sigma_{r4}=\frac{1}{W}\sqrt{M^2+0.75T^2}=59.5\mathrm{MPa}\le[\sigma]$$

故满足强度要求

8-3　试用图8-23(a)、(b)中所注尺寸符号写出各回转壳体中A和A′点的第一曲率半径和第二曲率半径以及平行圆半径。



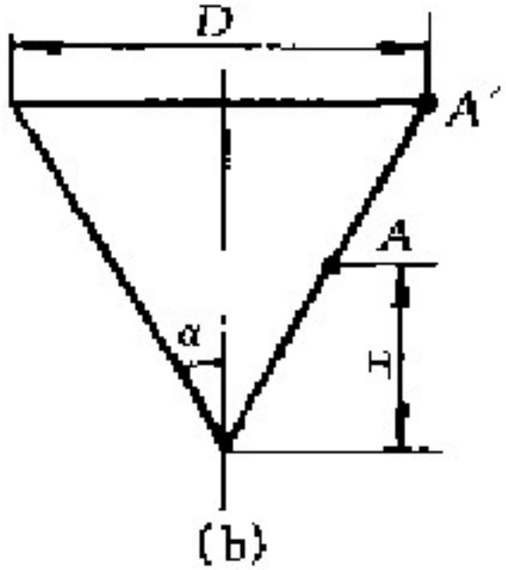
（a）A点：

$$R_1 = r, \quad R_2 = r + \frac{D}{2\sin\varphi}$$

$$R = \frac{D}{2} + r\sin\varphi$$

（a）A′ 点：$R_1 = \infty$

$$R_2 = \frac{D}{2}, \quad R = \frac{D}{2}$$



（b）A点：　$R_1=\infty$，

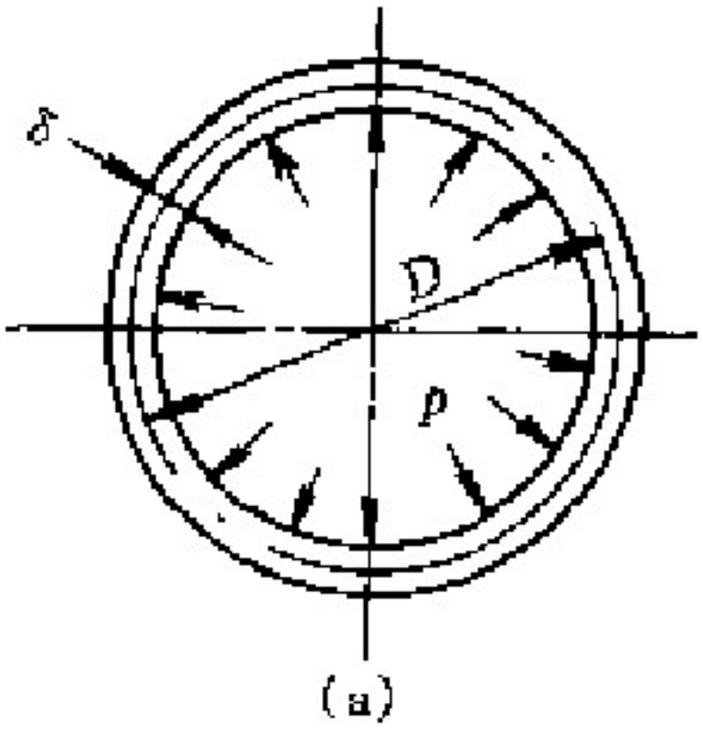
$$R_2=\frac{x\tan\alpha}{\cos\alpha}=\frac{x\sin\alpha}{\cos^2\alpha},\ R=x\tan\alpha$$

（b）A′ 点：　$R_1=\infty$，

$$R_2=\frac{D}{2\cos\alpha},R=\frac{D}{2}$$

8-4 计算图8-24所示各种承受均匀内压作用的薄壁回转壳体上各点的σ_m 和 σ_θ

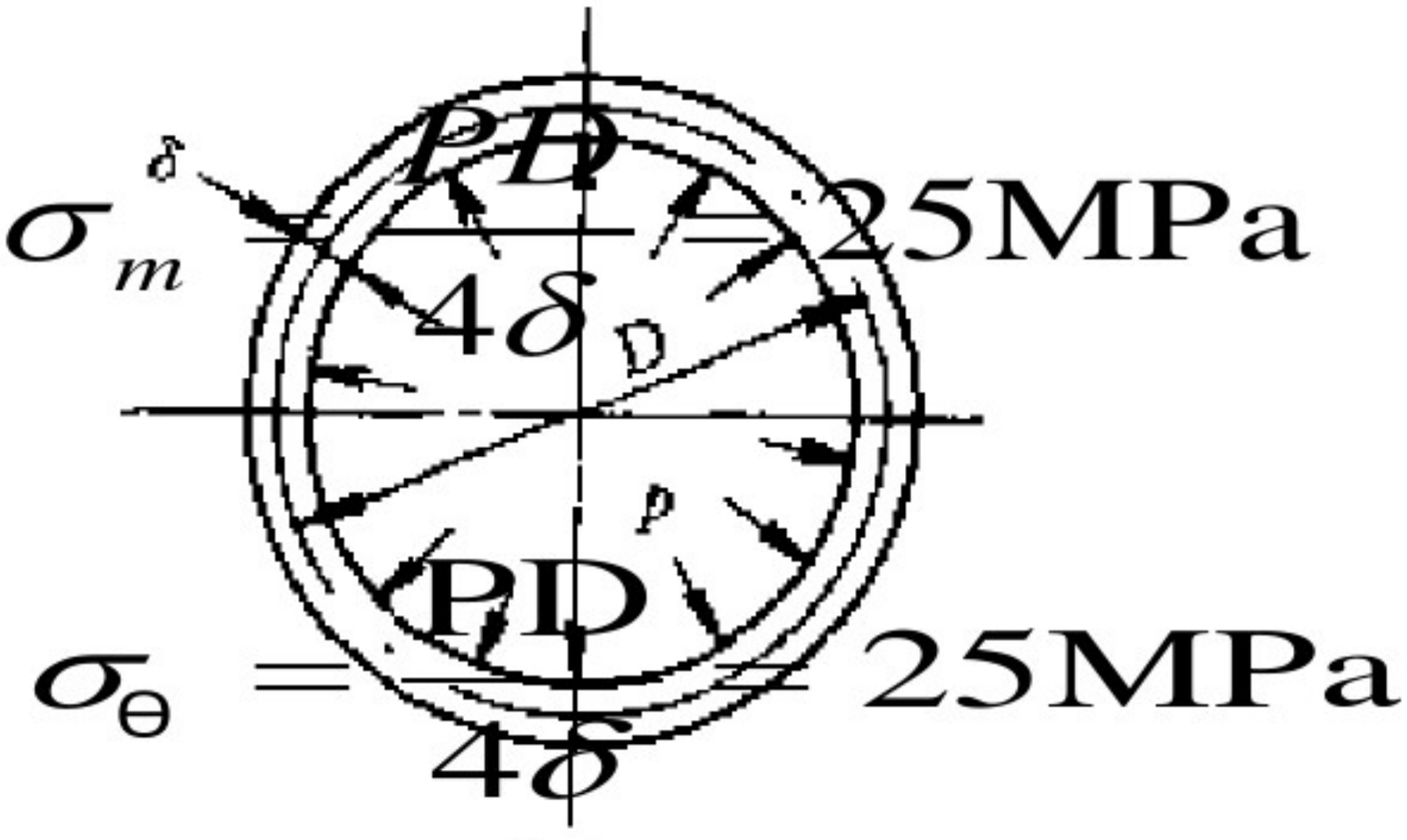
(1) 球壳上任一点。已知：P＝2MPa，D＝1000mm，δ＝20mm。



(a)

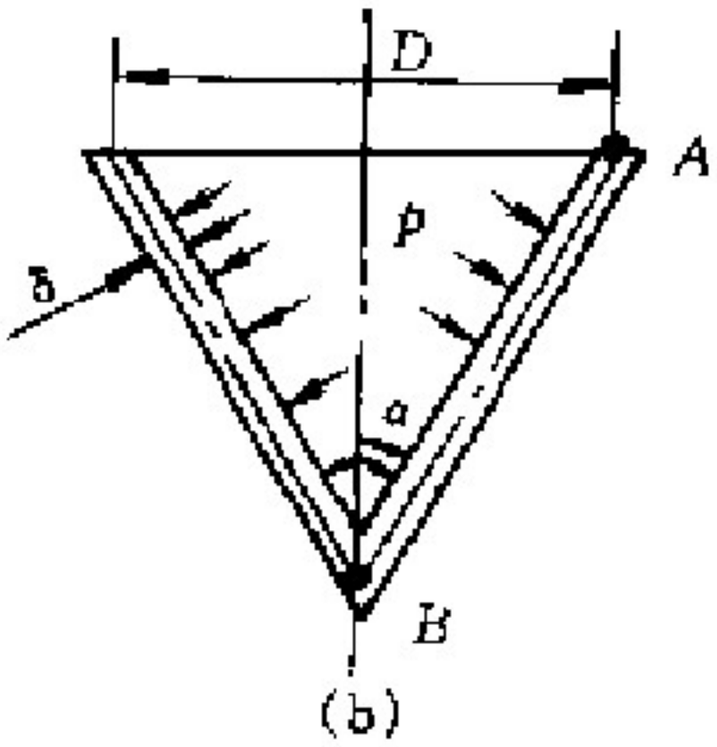
$$\sigma_m = \frac{PD}{4\delta} = 25\text{MPa}$$

$$\sigma_\theta = \frac{PD}{4\delta} = 25\text{MPa}$$



(a)

(2)圆锥壳上A点和B点。已知：P＝0.5MPa，D＝1000mm，δ＝10mm，α=30°。



(b)

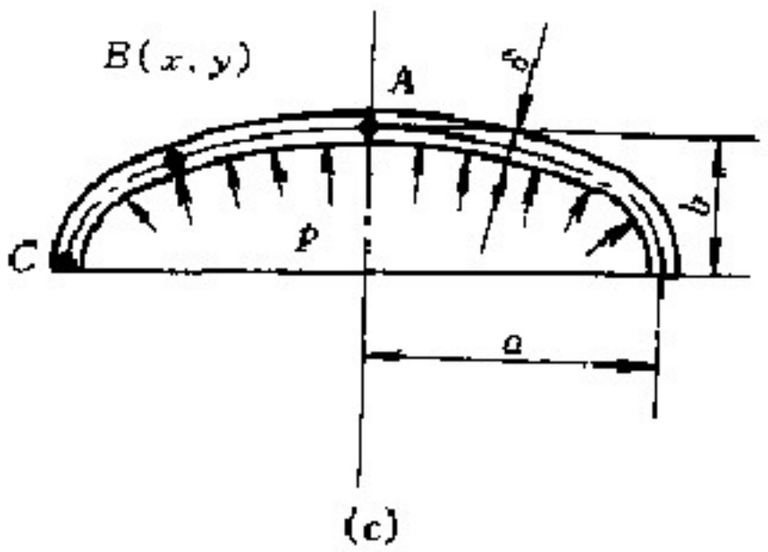
A点：

$$R_1=\infty,\qquad R_2=\frac{D}{2\cos\alpha},$$

$$\sigma_m=\frac{PR_2}{2\delta}=\frac{0.5\times\dfrac{1000}{2\cos 30^0}}{2\times 10}=14.43\text{MPa}$$

$$\sigma_\theta=\left(\frac{P}{\delta}-\frac{\sigma_m}{R_1}\right)\times R_2=(\frac{0.5}{10})\times\frac{1000}{2\cos 30^0}=28.86\text{MPa}$$

B点：$R_1=\infty,\quad R_2=0,\quad \sigma_m=0,\quad \sigma_\theta=0$



（3）椭球壳上A、B、C点。已知：P＝1MPa，a＝1000mm，b＝500mm，δ＝10mm，B点处坐标x＝600mm。

A点：
$$R_1=\frac{1}{a^4\mathrm{b}}\left[a^4-\mathrm{x}^2\left(a^2-\mathrm{b}^2\right)\right]^{\frac{3}{2}}\underline{\underline{\mathrm{x}=0}}\frac{a^2}{\mathrm{b}}$$

$$R_2=\frac{\left(a^4y^2+\mathrm{b}^4\mathrm{x}^2\right)^{\frac{1}{2}}}{\mathrm{b}^2}=a^2$$

$$\sigma_{\mathrm{m}}=\frac{\mathrm{P}}{2\delta\mathrm{b}}\sqrt{a^4-\mathrm{x}^2\left(a^2-\mathrm{b}^2\right)}=\frac{1\times1000^2}{2\times500\times10}=100\mathrm{MPa}$$

$$\sigma_\theta = \frac{\mathrm{P}}{2\delta \mathrm{b}} \sqrt{a^4 - \mathrm{x}^2\left(a^2 - \mathrm{b}^2\right)} \left[2 - \frac{a^4}{a^4 - \mathrm{x}^2\left(a^2 - \mathrm{b}^2\right)} \right]$$

$$= \frac{1 \times 1000^2}{2 \times 10 \times 500} = 100\mathrm{MPa}$$

B点：x=600mm,

$$\sigma_{\mathrm{m}} = \frac{1}{2 \times 10 \times 500} \sqrt{1000^4 - 600^2\left(1000^2 - 500^2\right)} = 85.44\mathrm{MPa}$$

$$\sigma_\theta = \sigma_{\mathrm{m}} \left[2 - \frac{1000^4}{1000^4 - 600^2\left(1000^2 - 500^2\right)} \right] = 53.84\mathrm{MPa}$$

C点：x=1000mm,

$$\sigma_{\mathrm{m}}=\frac{1}{2\times10\times500}\sqrt{1000^4-1000^2\left(1000^2-500^2\right)}=50\mathrm{MPa}$$

$$\sigma_{\theta}=\sigma_{\mathrm{m}}\left[2-\frac{1000^4}{1000^4-1000^2\left(1000^2-500^2\right)}\right]$$

$$=-100\mathrm{MPa}$$

8-7 有一立式圆筒形储油罐，如图所示，罐体中径D=5000mm，厚度δ=10mm，油的液面离罐底高H=18m，油的相对密度为0.7，试求：

（1）当P_0=0时，油罐筒体上M点的应力及最大应力。

（2）当P_0=0.1MPa时，油罐筒体上M点的应力及最大应力。

- 解:依题意，筒壁上任意一点的压力为：

- (1)当P_0=0时, p=ρ gx=ρ gz

$$\sigma_m = 0$$

$$\sigma_\theta = \frac{pD}{2\delta} = \frac{0.7\times10^3\times9.8\times5000/10^6}{2\times10} Z = 1.715Z \text{ MPa}$$

最大环向应力在$x=H=18$m

$$\sigma_{\theta\ \max}=1.715\times18=31.5\ \text{MPa}$$

(2) 当P_0=0.1MPa时

$$\sigma_m=\frac{p_0R}{2\delta}=\frac{p_0D}{4\delta}=\frac{0.1\times5000}{4\times10}=12.5\,\text{MPa}$$

$$\sigma_\theta=\frac{(p_0+\rho gx)R}{\delta}=\frac{(p_0+\rho gx)D}{2\delta}=(25+1.715Z)\,\text{MPa}$$

当z=18m时，有

$$\sigma_{\theta\ \max}=(25+31.5)=56.5\ \text{MPa}$$

9-1　已知：$DN2000,\ \delta_n=22\text{mm},$

$$P=2MPa,\ \phi=0.85,\ C=2mm,$$

筒体的有效厚度：$\delta_e=\delta_n-C=20-2=20\text{mm}$

在设计温度下，筒体的计算应力为：

$$\sigma^t=\frac{p_c(D_i+\delta_e)}{2\delta_e}=\frac{2\times(2000+22-2)}{2\times20}=101\text{MPa}$$

9-3　已知：$D_{\mathrm{i}}=10m,\quad \delta_n=22mm,\quad \phi=1.0,$

$$C=2mm,\quad [\sigma]^t=147\mathrm{MPa}$$

球壳的有效厚度为：

$$\delta_e=\delta_n-C=20-2=20\mathrm{mm}$$

球壳的最大允许工作压力为：

$$[P_w]=\frac{4[\sigma]^t\phi\delta_e}{D_i+\delta_e}=\frac{4\times147\times1.0\times20}{10000+20}=1.172\mathrm{MPa}$$

9-4　已知：$D_i = 1600\text{mm}$，　$\phi = 0.85$，P=1.6MPa，

$C = 2mm$，　$[\sigma]^t = 147\text{MPa}$，

材料 0Cr18Ni10Ti 属于高合金不锈钢，查表9-5，

其在工作温度 t=5~105℃时，$[\sigma]^t = 103\text{MPa}$，

有安全阀时：$P_c = (1.05\sim1.1)P = 1.76\text{MPa}$

计算厚度　$\delta = \dfrac{P_c D_i}{2[\sigma]^t \phi - P_c} = \dfrac{1.76\times1600}{2\times103\times0.85-1.76} = 16.25\text{mm}$

查表9-1，取C_1=0.8mm，C_2=0

名义厚度$\delta_n = \delta + C_1 + C_2 + \Delta = 16.25+0.8+\Delta = 18\text{mm}$

9-5 已知：$D_i = 1600\text{mm},\ \delta_n = 16\text{mm},\ \phi = 0.85,$

$C = 1.5mm,\ [\sigma]^t = 170\text{MPa},\ [\sigma_S] = 345\text{MPa}$

材料为16MnR，$P_c = 2.5\text{MPa}$

解：有效厚度$\delta_e = \delta_n - C = 16 - 1.5 = 14.5\text{mm}$

在设计压力下产生的薄膜应力为：

$$\sigma_T = \frac{P_C(D_i + \delta_e)}{2\delta_e} = \frac{2.5 \times (1600 + 14.5)}{2 \times 14.5} = 139\text{MPa}$$

容器承压时所允许产生的最大应力为：

$\phi[\sigma]^t = 0.85 \times 170 = 144.5\text{MPa} > \sigma_T$，故满足要求。

采用水压试验，试验压力为

$$P_{\mathrm{T}}=1.25P\frac{[\sigma]}{[\sigma]^{t}}=1.25\times2.5\times\frac{170}{170}=3.125\mathrm{MPa}$$

压力试验时的薄膜应力

$$\sigma_{\mathrm{T}}=\frac{P_{\mathrm{T}}(D_i+\delta_{\mathrm{e}})}{2\delta_e}=\frac{3.125\times(1600+14.5)}{2\times14.5}=173.97\mathrm{MPa}$$

16MnR的$\sigma_{\mathrm{S}}=345\mathrm{MPa}$，

水压试验的许用应力为：

$0.9\phi\sigma_{\mathrm{s}}=0.9\times0.85\times345=263.9\mathrm{MPa}\ >\sigma_T=173.9\mathrm{MPa}$

所以满足水压试验要求